



**NATIONAL ACADEMY OF SCIENCE
AND TECHNOLOGY (NAST)**

Department of Civil Engineering
Affiliated with Pokhara University
Dhangadhi, Kailali, Sudurpashchim Province, Nepal

**AN INTERNSHIP REPORT ON
WATER SUPPLY AND SANITATION DIVISION
OFFICE (WSSDO), KAILALI**

*Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Civil
Engineering (BCE), National Academy of Science and Technology (NAST), Affiliated with
Pokhara University*



Submitted by
Tek Raj Joshi
Roll No.: 22040384

Submitted to
Jeewan Kunwar, Internship Coordinator

**Department of Civil Engineering
National Academy of Science and Technology (NAST)**

Affiliated with Pokhara University
Dhangadhi, Kailali, Sudurpashchim Province, Nepal

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INTERNSHIP REPORT

on

**Water Supply and Sanitation Division Office
(WSSDO), Kailali**

Internship Duration:

May 19, 2026 – August 21, 2026

*Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Civil Engineering (BCE)*

Submitted By:

Tek Raj Joshi

Roll No.: 22040384

Submitted To:

Jeewan Kunwar

Internship Coordinator

Department of Civil Engineering, NAST

Date of Submission: August 22, 2026

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Tek Raj Joshi

Roll No.: 22040384

BCE, NAST / Pokhara University

SUPERVISOR'S RECOMMENDATION CERTIFICATE

This is to certify that Mr. Tek Raj Joshi (Roll No. 22040384), a student of the Bachelor of Civil Engineering (BCE) program at the National Academy of Science and Technology (NAST), affiliated to Pokhara University, has successfully completed his internship at the Water Supply and Sanitation Division Office (WSSDO), Kailali, Nepal, from 2083-02-05 to 2083-05-05 BS (May 19, 2026 – August 21, 2026 AD).

During the internship period, Mr. Joshi gained practical exposure to various office and field activities related to water supply and sanitation engineering under the guidance of the Senior Engineer and the technical team of WSSDO Kailali. He observed the planning, design, estimation, construction, supervision, and monitoring aspects of water supply projects and developed an understanding of office procedures and project implementation practices. He demonstrated sincerity, enthusiasm, and professional conduct throughout the internship period.

I recommend this internship report for evaluation by the concerned faculty of NAST, Pokhara University.

Signature: _____

Name: Hansha Bahadur Thapa

Designation: Senior Engineer (Internship Supervisor)

Water Supply and Sanitation Division Office (WSSDO), Kailali

Date: 2083/05/05

FOCAL PERSON'S RECOMMENDATION CERTIFICATE

This is to certify that the internship report titled "Internship Report on Water Supply Projects" at WSSDO Kailali, submitted by Mr. Tek Raj Joshi (Roll No. 22040384) has been prepared under my coordination as the Internship Focal Person and is recommended for evaluation in partial fulfillment of the 3-month internship requirement of the Bachelor of Civil Engineering program at NAST, Pokhara University.

Verified By

Signature: _____

Name: Jeewan Kunwar

Designation: Internship Coordinator / Focal Person

Organization: Department of Civil Engineering, NAST

Affiliated to Pokhara University

Date: _____

EXTERNAL EXAMINER'S APPROVAL CERTIFICATE

This is to certify that the internship report titled "Internship Report on Water Supply Projects at Water Supply and Sanitation Division Office (WSSDO), Kailali" submitted by Mr. Tek Raj Joshi (Roll No. 22040384) has been examined and approved in partial fulfillment of the requirements of the Bachelor of Civil Engineering (BCE) program at the National Academy of Science and Technology (NAST), affiliated to Pokhara University, in accordance with the Pokhara University School of Engineering Internship Program Guidelines, 2082.

Signature: _____

Name: _____

Designation: External Examiner

Organization: School of Engineering, Pokhara University

Date: _____

ABSTRACT

This internship report presents the experience and learning outcomes gained during a 13-week internship at the Water Supply and Sanitation Division Office (WSSDO), Kailali, Sudurpashchim Province, Nepal, from 2083-02-05 to 2083-05-05 BS. The internship was undertaken in partial fulfillment of the requirements for the Bachelor of Civil Engineering (BCE) program at the National Academy of Science and Technology (NAST), affiliated to Pokhara University.

The Water Supply and Sanitation Division Office (WSSDO), Kailali is responsible for the planning, implementation, operation, monitoring, and supervision of water supply and sanitation projects within the kailali and kanchanpur district. During the internship period, practical exposure was gained through various office and field activities under the guidance of the Senior Engineer and the technical team of the organization. The internship involved reviewing technical documents and Detailed Project Reports (DPRs), observing planning and design procedures, studying quantity and cost estimation practices, and developing an understanding of project implementation and management processes.

Field visits, site inspections, and measurement activities at ongoing, completed, and rehabilitated water supply schemes provided valuable insight into construction practices, project monitoring, site supervision, measurement of quantity, quality control of ongoing works, and the operational aspects of water supply infrastructure. The internship also provided exposure to professional engineering practices, technical documentation, office procedures, and the overall workflow involved in the execution of water supply projects. Through interaction with engineers and technical personnel, a better understanding of the practical responsibilities, challenges, and multidisciplinary nature of water supply and sanitation engineering was developed.

Overall, the internship served as an important bridge between academic learning and professional practice, enhancing technical knowledge, practical skills, and professional competence while providing valuable insight into the design, construction, operation, and implementation aspects of water supply and sanitation projects.

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LIST OF ABBREVIATIONS.

BCE - Bachelor of Civil Engineering

BOQ - Bill of Quantities

BS - Bikram Sambat (Nepali Calendar)

DPR - Detailed Project Report

DWSSM - Department of Water Supply and Sanitation Management

GoN - Government of Nepal

HDPE - High-Density Polyethylene

HH - Household

lps - Litres per second

MoWS - Ministry of Water Supply

NAST - National Academy of Science and Technology

OHT - Overhead Tank

PPA - Public Procurement Act

PU - Pokhara University

RCC - Reinforced Cement Concrete

RVT - Reinforced Vertical Tank

UPVC - Unplasticized Polyvinyl Chloride

WSSDO - Water Supply and Sanitation Division Office

WUSC - Water Users and Sanitation Committ.

CHAPTER 1. INTRODUCTION

1.1 Background

Access to safe and reliable drinking water is a fundamental human right and a key determinant of public health and socioeconomic development. In Nepal, the government has been actively investing in rural and urban water supply infrastructure through various institutions operating at the federal, provincial, and local levels. The Ministry of Water Supply (MoWS) is responsible for national policies and sectoral guidance, while implementation is carried out through the concerned government agencies. The Water Supply and Sanitation Division Office (WSSDO), Kailali, is one such government field office responsible for planning, designing, implementing, and supervising water supply and sanitation projects across Kailali and Kanchanpur districts in Sudurpashchim Province.

This internship was carried out at the Water Supply and Sanitation Division Office (WSSDO), Kailali, from 2083-02-05 BS to 2083-05-05 BS (May 19, 2026 AD to August 21, 2026 AD) as part of the 3-months internship requirement of the Bachelor of Civil Engineering (BCE) program at the National Academy of Science and Technology (NAST), affiliated with Pokhara University. The internship provided valuable practical exposure to government water supply engineering by bridging the gap between academic knowledge and professional engineering practice through participation in project planning, design review, construction observation, quantity estimation, engineering documentation, and project management activities.

WSSDO Kailali oversees multiple water supply schemes across Kailali and Kanchanpur district at different stages of implementation — ranging from DPR preparation and design to active construction and post-construction monitoring. During the internship period, multiple projects were studied and observed:

- **Dokebazar Khanepani Aayojana, Laljhadi Municipality (Wards 8 & 9), Kanchanpur:** A lift/gravity water supply scheme based on deep tubewells, designed to serve 759 households through a 225 m³ RCC overhead water tank (OHT) and a 16.3 km HDPE distribution network. During the internship, the project provided practical exposure to the study of the Detailed Project Report (DPR), engineering drawings, hydraulic design, quantity estimation, and field observation.

- **WSSDO Office Building Project (Godam and Residential Quarter), Dhangadhi, Kailali:** A government building construction project comprising a storage warehouse (Godam) on the ground floor and a residential quarter on the upper floor. During the internship, the project provided practical exposure to structural drawings, reinforcement detailing, formwork arrangements, field measurements, quantity verification, construction quality monitoring, and engineering documentation.
- **Basauti Water Supply Project, Kailali Rural Municipality–08, Kailali:** A community-based water supply scheme implemented under the supervision of the Water Supply and Sanitation Division Office (WSSDO), Kailali. During the internship, the project provided practical exposure to water supply construction activities, field measurements, quantity verification, BOQ-based billing procedures, Measurement Book (MB) preparation, and project documentation.
- **Hasuliya Water Supply Project, Kailali Rural Municipality–03, Kailali:** A community-based water supply scheme implemented under the supervision of the Water Supply and Sanitation Division Office (WSSDO), Kailali. During the internship, the project provided practical exposure to pipeline construction activities, valve chamber works, field measurements, quantity verification, Measurement Book (MB) preparation, BOQ-based billing procedures, and project documentation.
- **Bajinatha Sidanatha Water Supply Project, Suklaphanta Municipality–12, Kanchanpur:** A pumping water supply scheme implemented under the supervision of the Water Supply and Sanitation Division Office (WSSDO). During the internship, the project provided practical exposure to the study of engineering drawings, AutoCAD pipeline layouts, water supply system planning, quantity estimation, Bill of Quantities (BOQ), rate analysis, cost estimation, and other technical documents related to government water supply project planning and design.

1.2 Objectives

The primary objectives of this internship were:

- To understand the organizational structure, functions, and working procedures of the Water Supply and Sanitation Division Office (WSSDO), Kailali.
- To gain practical knowledge of water supply engineering projects through the study of DPR documents, project profiles, drawings, and technical specifications of different schemes.
- To understand the planning and estimation processes involved in water supply projects, including quantity estimation, Bill of Quantities (BOQ), rate analysis, and cost estimation procedures.
- To study engineering drawings and develop an understanding of water supply system components such as source structures, pumping systems, storage facilities, pipeline networks, valve chambers, and RCC overhead tanks.
- To observe ongoing construction activities and gain practical exposure to construction methods, quality control practices, and site supervision procedures.
- To enhance professional skills related to technical documentation, engineering report preparation, and practical application of civil engineering concepts in government infrastructure projects.

1.3 Scope

The scope of this internship was focused on gaining practical exposure to government water supply and infrastructure development activities carried out by the Water Supply and Sanitation Division Office (WSSDO), Kailali. The major areas covered during the internship were:

- Understanding the organizational functions, working procedures, and technical practices followed by WSSDO, Kailali.
- Studying Detailed Project Reports (DPRs), project profiles, engineering drawings, technical specifications, and other relevant documents of water supply schemes.

- Developing practical understanding of water supply system components, including source structures, pumping systems, transmission and distribution pipelines, storage facilities, valve chambers, and RCC overhead tanks.
- Studying quantity estimation, Bill of Quantities (BOQ), rate analysis, and cost estimation procedures used in water supply projects.
- Observing ongoing construction activities and gaining exposure to construction methods, site supervision practices, quality control procedures, and measurement documentation.
- Learning technical documentation practices such as Measurement Book (MB), billing procedures, completion documentation, and preparation of engineering reports.

1.4 Limitation.

The internship was carried out within a limited period and was mainly focused on observation and learning under the guidance of engineers. The major limitations of the internship were:

- The internship period was limited, which restricted detailed observation of all project stages from planning to completion.
- The intern was involved mainly in observation, study, and learning activities; therefore, independent design decisions, approvals, contract administration, and project management responsibilities were beyond the scope of the internship.
- Some project activities were at different stages of implementation, limiting direct field exposure to certain construction components.
- Access to confidential administrative, procurement, and contractual information of projects was limited.
- Some of the projects studied during the internship were at different stages of implementation, while others were completed, ongoing maintenance, or previously designed schemes. As a result, direct observation of all project phases of construction was not possible within the internship period.

- Some project activities remained pending or progressed slowly due to budget constraints, labour shortages, limited resources, and the unavailability of advanced equipment during the implementation period.

1.5 Report Organization

This report has been organized into four chapters in accordance with the Pokhara University School of Engineering Internship Guidelines, 2082.

- Chapter 1: Introduction presents the background of the internship, objectives, scope and limitations, and the overall organization of the report.
- Chapter 2: Organization Details and Literature Review provides an overview of the Water Supply and Sanitation Division Office (WSSDO), Kailali, including its organizational structure, working domains, and relevant literature related to water supply engineering.
- Chapter 3: Internship Activities describes the roles and responsibilities undertaken during the internship, weekly activities performed, projects studied, and technical knowledge gained through office and field exposure.
- Chapter 4: Conclusion and Learning Outcomes summarizes the internship experience, key observations, and the knowledge and skills acquired during the internship period.

CHAPTER 2: ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1 Introduction to Organization.

The Water Supply and Sanitation Division Office (WSSDO), Kailali is a provincial government office established in 2075 B.S. and located in Dhangadhi, Kailali. Operating under the Ministry of Physical Infrastructure Development, Sudurpashchim Province Government, the office is responsible for the planning, implementation, supervision, monitoring, and coordination of water supply and sanitation projects within Kailali and Kanchanpur districts.

The office plays an important role in improving access to safe drinking water and sanitation services through the development and management of water supply infrastructure. Its major activities include project identification, engineering surveys, preparation and review of project reports, cost estimation, construction supervision, quality control, technical coordination, and monitoring of ongoing and completed schemes.

WSSDO Kailali works in close coordination with local governments, Water Supply and Sanitation Users Committees (WUSCs), contractors, consultants, and other relevant stakeholders to ensure the effective implementation of water supply and sanitation projects. Through these activities, the office contributes to the sustainable development of water supply services and the improvement of public health and living standards within its service area.

During the internship period, the office provided an opportunity to gain practical exposure to various technical and administrative aspects of water supply project planning, design documentation, estimation, construction supervision, and project management practices.

2.2 Organizational Hierarchy.

The Water Supply and Sanitation Division Office (WSSDO), Kailali is headed by a Senior Divisional Engineer and supported by technical, administrative, and information technology personnel. The organizational hierarchy of the office is shown below.

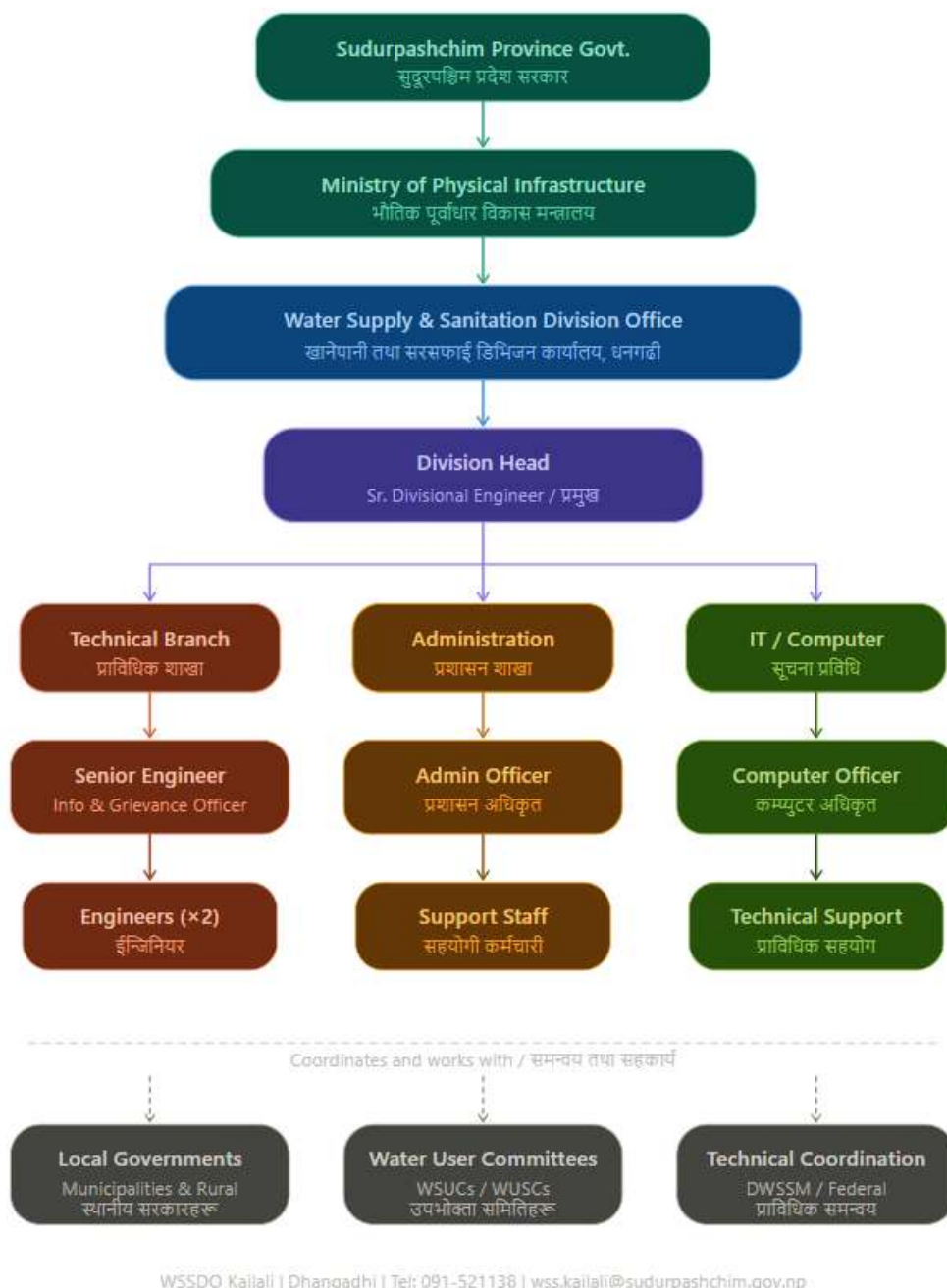


Figure 2.1 — Organizational Structure of WSSDO Kailali.

2.3 Working Domains of Organization.

The Water Supply and Sanitation Division Office (WSSDO), Kailali is responsible for the planning, implementation, supervision, and monitoring of water supply and sanitation projects within Kailali and Kanchanpur districts. The major working domains of the office include:

- Identification and prioritization of water supply and sanitation projects.

- Conducting engineering surveys, investigations, and feasibility studies.
- Preparation and review of Detailed Project Reports (DPRs), drawings, estimates, and Bills of Quantities (BOQs).
- Implementation and supervision of water supply and sanitation construction works.
- Monitoring project progress and ensuring quality control during construction.
- Coordination with local governments, Water Users and Sanitation Committees (WUSCs), contractors, and other stakeholders.
- Management of procurement and contract-related activities in accordance with prevailing regulations.
- Technical support, inspection, and evaluation of completed projects.
- Promotion of sustainable operation and maintenance of water supply and sanitation systems.

2.4 Description of Intern Department/Unit

The internship was conducted in the Technical Branch of WSSDO Kailali under the guidance of Er. Hansha Bahadur Thapa, Senior Engineer. The branch is responsible for technical activities related to water supply and sanitation projects, including planning, design, estimation, construction supervision, and monitoring. During the internship, the intern participated in document review, technical discussions, and field visits, gaining practical exposure to project implementation and management.

2.5 Literature Review

2.5.1 Water Supply Systems in Nepal

Water supply systems in Nepal are designed to provide safe and reliable drinking water to communities through the collection, treatment, storage, and distribution of water. Depending on geographical conditions and water availability, different types of water supply systems are adopted, including gravity-flow systems, pumping systems, and combined gravity-pumping systems. In the Terai region, where groundwater is abundant, deep tube wells and pumping systems are commonly used as water sources.

The planning and design of water supply projects in Nepal are guided by national policies, technical standards, and design guidelines issued by the Government of Nepal through the Ministry of Water Supply (MoWS) and the Department of Water Supply and Sewerage Management (DWSSM). These guidelines cover aspects such as water demand estimation, source selection, transmission and distribution systems, storage facilities, and operation and maintenance requirements. Proper planning and management of water supply systems are essential for ensuring sustainable access to safe drinking water and improving public health.

2.5.2 Components of a Typical Water Supply System

A typical water supply system consists of several components that work together to ensure the collection, conveyance, storage, and distribution of safe drinking water to consumers. In many water supply schemes of the Terai region, groundwater extracted from deep tube wells is used as the primary water source. The major components of such systems are described below.

I. Water Source (Deep Tube Well)

A deep tube well serves as the primary source of water in many Terai-based water supply schemes. Groundwater is extracted from suitable aquifers through drilled wells, providing a reliable source of water throughout the year.

II. Pumping Unit

The pumping unit is used to lift water from the deep tube well and convey it to storage facilities. It generally consists of pumps, motors, control panels, and associated electromechanical equipment required for efficient operation.

III. Transmission Pipeline

Transmission pipelines carry water from the source or pumping station to the overhead tank. These pipelines are designed to convey the required discharge while minimizing head loss and ensuring reliable water transfer.

IV. Overhead Tank (OHT)

The overhead tank is a storage structure used to maintain an adequate water reserve and provide the necessary pressure for distribution. Water pumped into the tank is later supplied to consumers through gravity flow.

V. Distribution Network

The distribution network consists of interconnected pipelines that convey water from the overhead tank to different service areas. Proper network design helps ensure equitable distribution and adequate pressure throughout the system.

The common types of water distribution networks are:

- **Dead-End System:** A simple network in which pipelines terminate at dead ends. It is economical but may cause water stagnation at the ends.
- **Grid Iron System:** Pipelines are interconnected in a grid pattern, allowing water to reach consumers through multiple paths and ensuring better pressure distribution.
- **Ring System:** Pipelines form a closed loop around the service area, providing reliable water supply and minimizing pressure loss.
- **Radial System:** Water is distributed from a central reservoir or tank through pipelines extending in different directions. It offers efficient pressure control and distribution.

During the internship, the Dead-End distribution system was observed to be commonly used in the water supply projects.

VI. Household Connections and Public Taps

Water is delivered to consumers through household connections and, in some cases, public tap stands. These facilities provide access to safe drinking water and represent the final point of service within the water supply system.

2.5.3 Hydraulic Design Basis

Hydraulic design is an important aspect of water supply engineering that ensures adequate flow, pressure, and efficient distribution of water throughout the system. The design involves determining key parameters such as water demand, design period, projected population, peak demand, storage capacity, and pressure requirements. These parameters are considered for the proper sizing of pipelines, reservoirs, and distribution systems.

Design Parameter	Standard Value	Remarks
Per capita water demand	45 L/capita/day	Water demand estimation
Design period	20 years	Future planning period
Population growth rate	2% per year	Population projection
Peak factor	1.5	Distribution design
Service reservoir capacity	1/3 of daily demand	Storage requirement
Minimum residual pressure	5–7 m head at tap	Adequate service pressure

Table 2.1: Common Design Parameters for Water Supply Systems:

2.5.4 Detailed Project Report (DPR)

A Detailed Project Report (DPR) is a technical document that presents the planning, design, and implementation details of a water supply project. It includes project background, survey information, design criteria, hydraulic analysis, drawings, BOQ, cost estimation, and implementation plans. DPRs provide a systematic basis for project execution, monitoring, and management.

2.5.5 Bill of Quantities (BOQ) and Cost Estimation

A Bill of Quantities (BOQ) is a detailed document that lists the quantities and descriptions of various items of work required for project implementation. It provides the basis for cost estimation, tender preparation, and project cost control.

Cost estimation involves calculating the total project cost based on estimated quantities, approved rates, material costs, labor costs, and other applicable expenses. Accurate BOQ preparation and cost estimation are essential for effective planning, budgeting, and successful implementation of construction projects.

2.5.6 Public Procurement Act (PPA) 2063 and Public Procurement Regulations (PPR) 2064

Public construction works in Nepal are governed by the Public Procurement Act (PPA) 2063 and its associated Public Procurement Regulations (PPR) 2064. These provide the legal and procedural framework for transparent, competitive, and

accountable procurement of goods, services, and construction works by public organizations. The procurement process generally includes preparation of bidding documents, invitation and evaluation of bids, contract agreement, and implementation of works according to approved procedures.

2.5.7 Construction Supervision and Quality Control

Construction supervision ensures that project activities are executed according to approved drawings, specifications, and contractual requirements. It involves monitoring progress, verifying measurements, checking workmanship, and maintaining quality standards to achieve safe and durable infrastructure.

CHAPTER 3: INTERNSHIP ACTIVITIES

3.1 Roles and Responsibilities

During the internship at WSSDO Kailali, the following activities were performed under the guidance and supervision of technical personnel:

- Studied DPR documents, technical drawings, BOQ, and cost estimation documents of various water supply schemes.
- Observed hydraulic design concepts, population projection, water demand estimation, RVT sizing, and pipe sizing procedures from project documents.
- Learned about technical documentation, project files, and progress record management practices followed in the office.
- Maintained weekly activity logs and documented technical observations during the internship period.
- Participated in site visits to water supply and building infrastructure projects to observe construction activities.
- Observed construction supervision practices, field measurements, quality control procedures, and progress monitoring activities.
- Interacted with engineers, sub-engineers, and field staff to understand practical aspects of project implementation and civil engineering practices.
- Site supervision of ongoing construction projects.

3.2 Weekly Log

The weekly activity log was maintained throughout the internship period to record the technical activities, observations, and learning experiences gained during the internship. The detailed week-wise activity log, prepared in the prescribed format, is presented in **Appendix A**.

3.3 Description of Projects Involved During Internship

3.3.1 Overview of Water Supply Schemes Reviewed During the Internship

During the internship period, I had the opportunity to study and observe several water supply schemes under the Water Supply and Sanitation Division Office (WSSDO),

Kailali. These schemes were at different stages of planning, design, and construction, providing practical exposure to various aspects of water supply engineering. The summary of the schemes studied during the internship is presented below:

S. N.	Project / Scheme Name	Location	Project Type	Nature of Exposure
1	Dokebazar Khanepani Aayojana	Laljhadi Municipality, Wards 8 & 9, Kanchanpur	Lift/Gravity Water Supply Scheme	Studied the already prepared DPR, engineering drawings, hydraulic design, BOQ, quantity estimation, and cost estimation.
2	WSSDO Office Building project (Godam & Residential Quarter)	WSSDO Compound, Dhangadhi, Kailali	Government Building Construction	Observed construction activities, structural drawings, reinforcement detailing, quantity estimation, Quality control and site supervision.
3	Basauti Water Supply Project	Kailali Rural Municipality–08, Kailali	Community Water Supply Scheme	Observed pipeline installation, valve chamber construction, field measurements, MB preparation, BOQ-based billing, and completion documentation.
4	Hasuliya Water Supply Project	Kailali Rural Municipality–03, Kailali	Community Water Supply Scheme	Observed pipeline works, valve chamber construction, site supervision, quantity verification, MB preparation, and project documentation.
5	Bajinatha Sidanatha Water Supply Project	Suklaphanta Municipality–12, Kanchanpur	Pumping Water Supply Scheme	Studied engineering drawings, water supply components, BOQ, quantity estimation, rate analysis, and cost estimation.

Table 3.1: Summary of Water Supply Schemes Studied During the Internship

3.3.2 Dokebazar Khanepani Aayojana — Detailed Description

The Dokebazar Khanepani Aayojana was one of the principal water supply schemes studied during the internship at the Water Supply and Sanitation Division Office (WSSDO), Kailali. The project is a lift water supply system that utilizes groundwater from deep tube wells and includes major components such as pumping units, a 225 m³ RCC Overhead Tank (OHT), transmission mains, HDPE distribution pipelines, valve chambers, and household service connections. The scheme was designed to provide a reliable and sustainable drinking water supply over a 20-year design period.

The project served as a primary reference for understanding the planning, design, estimation, and implementation processes of a government water supply scheme. As part of the internship, the Detailed Project Report (DPR) and its supporting documents were studied to understand engineering drawings, pipeline layout plans, hydraulic design calculations, Bill of Quantities (BOQ), quantity estimation, rate analysis, cost estimation, and technical specifications. Selected DPR drawings and technical documents are included in **Appendix B**.

In addition to document study, site visits were conducted to observe the overhead tank, pipeline alignment, and ongoing pipeline laying and jointing works. These observations provided practical insight into the implementation of a water supply project and the challenges encountered during construction. During the internship period, parts of the distribution network remained incomplete due to budget limitations and labour shortages.

The detailed salient features of the project are provided in **Appendix B**.

3.3.3 WSSDO Office Building Project (Godam and Residential Quarter) — Description.

The WSSDO Office Building Project (Godam and Residential Quarter) is a government building construction project located within the premises of the Water Supply and Sanitation Division Office (WSSDO), Dhangadhi, Kailali. The project comprises the construction of a Godam (storage warehouse) on the ground floor and a Residential Quarter on the upper floor to improve the office's storage capacity and residential facilities for technical personnel.

The project is being implemented in accordance with the approved engineering drawings, technical specifications, Bill of Quantities (BOQ), and the Government of Nepal's public procurement procedures. The major construction activities include reinforced cement concrete (RCC) structural works, brick masonry, reinforcement fixing, formwork, concrete casting, plastering, painting, and other associated building works.

During the internship, this project provided practical exposure to the construction and documentation processes of a government building project. Under the guidance of the supervising engineers, field visits and office-based discussions enhanced the understanding of structural drawings, reinforcement detailing, formwork arrangements, field measurements, quantity verification, construction quality monitoring, and the relationship between engineering drawings and actual site activities. The project also provided practical insight into the sequence of building construction activities and the engineering practices followed during project implementation. The detailed technical activities performed during the internship are presented in **Section 3.4.2**. Representative engineering documents and site photographs related to the project are provided in the respective appendices.

3.3.4 Basauti Water Supply Project— Description

The Basauti Water Supply Project is a community-based water supply scheme located in Kailali Rural Municipality–08, Kailali, and implemented under the supervision of the Water Supply and Sanitation Division Office (WSSDO), Kailali. The project aims to provide a reliable and safe drinking water supply to the local community through the construction of essential water supply infrastructure.

The major components of the project include pipeline excavation, backfilling, HDPE pipeline laying and jointing, valve chamber construction, transportation of construction materials, and installation of associated fittings. The project also involves field measurement, quantity verification, Measurement Book (MB) preparation, contractor billing, and completion documentation in accordance with the approved design, specifications, and Bill of Quantities (BOQ).

During the internship, the project provided practical exposure to the implementation and documentation processes of a government water supply scheme. Site visits were conducted to observe pipeline installation, valve chamber construction, and completed

works, while office-based activities included the study of field measurements, MB preparation, BOQ-based billing procedures, and completion documentation under the guidance of the supervising engineers.

The detailed technical documents and selected project records related to the Basauti Water Supply Project are provided in **Appendix D**.

3.3.5 Hasuliya Water Supply Project — Description.

The Hasuliya Water Supply Project is a community water supply scheme located in Kailali Rural Municipality–03, Kailali. The project aims to provide a reliable and sustainable drinking water supply to the local community through the construction of a distribution pipeline network and associated water supply infrastructure. Major project components include transmission and distribution pipelines, valve chambers, household service connections, and other appurtenant structures required for the operation of the distribution system.

During the internship, the project provided practical exposure to the construction and implementation stage of a government water supply scheme. Site visits were conducted to observe pipeline excavation, HDPE pipe laying and jointing, backfilling, valve chamber construction, and completed painting works. In addition, office-based activities related to field measurements, Measurement Book (MB) preparation, BOQ-based billing, work completion documentation, and project records were studied under the guidance of the departmental engineers.

The project served as a practical example for understanding the relationship between site execution and engineering documentation, particularly in the areas of quantity verification, contractor billing, and project documentation. Selected project documents are included in **Appendix E**.

3.3.6 Bajhnath Sidanatha Water Supply Project— Description.

The Bajhnath Sidanatha Water Supply Project is a pumping-based water supply scheme located in Suklaphanta Municipality–12, Kanchanpur, and implemented under the supervision of the Water Supply and Sanitation Division Office (WSSDO), Kailali. The project aims to provide a reliable and sustainable drinking water supply to the service area through the development of a pumping-based water supply system.

The project comprises major components including source works, pumping facilities, transmission pipelines, a 225 m³ reinforced concrete overhead water tank (OHT), distribution pipelines, valve chambers, household service connections, and other associated civil, mechanical, and electrical facilities required for the operation of the water supply system. The project was supported by a Detailed Project Report (DPR), engineering drawings, hydraulic design, quantity estimation, Bill of Quantities (BOQ), rate analysis, cost estimation, and other technical documents prepared for planning and implementation.

During the internship, the Bajhnath Sidanatha Water Supply Project served as one of the principal reference projects for understanding the planning and design of a government pumping water supply scheme. Project drawings, estimation documents, and other technical records were reviewed under the guidance of the supervising engineers to develop an understanding of engineering drawings, water supply system layout, quantity estimation, BOQ, rate analysis, cost estimation, and project documentation. The detailed technical activities performed during the internship are presented in **Section 3.4.5**, while selected project drawings and supporting technical documents are included in **Appendix F**.

3.4 Tasks and Activities Performed

3.4.1 Dokebazar Khanepani Aayojana – Technical Activities Performed

A. Initial Site Visit and Project Familiarization

The first technical activity carried out during the internship was a site visit to the Dokebazar Khanepani Aayojana. The purpose of the visit was to gain an overall understanding of the project and observe its construction status before studying the technical documents. During the visit, the major components of the scheme, including the source area, overhead tank (OHT), transmission and distribution pipeline alignments, and ongoing pipeline laying and HDPE pipe jointing works, were observed.

Discussions were held with the supervising engineer, the contractor, and members of the Water Users and Sanitation Committee (WUSC) regarding the progress and implementation of the project. It was observed that while major components such as the source works and overhead tank had progressed, portions of the distribution pipeline network remained incomplete. According to the information provided during

the site visit, the primary reasons for the delay were labour shortages and budget limitations.

The site visit provided practical exposure to the implementation of a government water supply project and helped establish a clear understanding of the existing field conditions before the detailed study of the DPR. It also provided an opportunity to relate the physical components of the scheme to the engineering drawings and technical documents studied later during the internship. Selected site photographs from the field visit are presented in **Appendix G**.

B. Study of the Detailed Project Report (DPR)

After gaining an overall understanding of the project through the site visit, the already prepared Detailed Project Report (DPR) of the Dokebazar Khanepani Aayojana was studied under the guidance of the supervising engineer. The primary objective of this activity was to understand the complete documentation process involved in the planning, design, estimation, and implementation of a government water supply project. The DPR served as a practical reference for learning how technical, financial, and administrative information is systematically compiled into a single project document prior to construction.

The study covered the overall structure and contents of the DPR, including the project background, salient features, socio-economic profile, source investigation, design criteria, population projection, water demand forecasting, hydraulic design, engineering drawings, pipeline layout, Bill of Quantities (BOQ), quantity estimation, rate analysis, cost estimation, implementation arrangements, environmental considerations, and the supporting annexes. Particular attention was given to understanding how various engineering calculations, design assumptions, technical drawings, and cost estimates are integrated to prepare a comprehensive DPR for approval and execution.

The supporting annexes were also studied to understand the preparation of cost summaries, detailed estimates, hydraulic calculations, reservoir sizing, pipeline data, rate analysis, and the application of the District Rate Schedule in project cost estimation. Discussions with the supervising engineer further enhanced the understanding of the purpose and practical application of these documents in government water supply projects.

This activity provided valuable practical knowledge of the complete DPR documentation process and improved the ability to interpret technical reports, engineering drawings, and estimation documents used during project planning and implementation. Selected DPR drawings and supporting technical documents referred to during the internship are presented in **Appendix B**.

C. Study of Hydraulic Design and Engineering Drawings

The hydraulic design and engineering drawings included in the already prepared DPR were studied under the guidance of the supervising engineer to understand the design and layout of a government water supply system. The study covered the engineering drawings, pipeline layout plans, KML-based pipeline alignment, longitudinal profiles, and distribution network drawings to understand the arrangement of the various components and the overall configuration of the water supply scheme.

Particular attention was given to understanding the hydraulic design calculations used in the project, including population projection, water demand forecasting, pipeline sizing, hydraulic grade line (HGL), head loss calculations, and the selection of appropriate pipe diameters. The study also included the design and location of major water supply components such as the overhead tank (OHT), transmission pipelines, distribution pipelines, valve chambers, washout chambers, air valves, and household service connections. Discussions with the supervising engineer further improved the understanding of the design assumptions, engineering considerations, and practical approaches adopted during the preparation of the DPR.

The engineering drawings and hydraulic design were also correlated with observations made during the site visit, which helped in understanding how the proposed design was implemented in the field and how the different components of the scheme were interconnected. This activity enhanced the ability to read and interpret engineering drawings, understand hydraulic design principles, and relate technical calculations to the actual implementation of a water supply project. Selected engineering drawings, pipeline layouts, and hydraulic design documents studied during the internship are included in **Appendix B**.

D. Study of BOQ, Quantity Estimation, Rate Analysis, and Cost Estimation

The Bill of Quantities (BOQ), quantity estimation sheets, rate analysis, and cost estimates included in the already prepared DPR were studied under the guidance of

the supervising engineer to understand the estimation procedures followed in government water supply projects.

The BOQ was reviewed to understand how quantities of excavation, pipeline installation, valve chambers, RCC works, fittings, and other construction activities were derived from the engineering drawings. The detailed estimate prepared using Microsoft Excel was also studied to understand quantity calculation, tabulation, and estimate preparation. In addition, the rate analysis sheets and District Rate Schedule were examined to understand the basis of unit rate preparation for materials, labour, equipment, transportation, overhead, and contractor's profit.

This activity improved the understanding of the relationship between engineering drawings, quantity estimation, BOQ preparation, rate analysis, and overall project cost estimation. Selected BOQ extracts, estimation sheets, and rate analysis documents studied during the internship are included in **Appendix B**.

E. Correlation Between DPR and Field Observation

An important technical activity during the internship was relating the information presented in the DPR with the actual field conditions observed during the site visit. The engineering drawings, pipeline layout, and major project components were compared with the existing site conditions to understand how the approved design was being implemented.

The overhead tank location, pipeline alignment, and other major components generally corresponded with the DPR drawings. However, portions of the distribution pipeline network remained incomplete during the internship period due to labour shortages and budget limitations. Comparing the DPR with the field conditions provided a practical understanding of how implementation progress may differ from the original project schedule because of site constraints.

This activity enhanced the ability to relate engineering drawings and technical documents to actual construction practices and strengthened the understanding of project implementation, site supervision, and engineering documentation. Selected site photographs and supporting technical documents are presented in **Appendices G and B**, respectively.

3.4.2 WSSDO Office Building Project (Godam and Residential Quarter) – Technical Activities Performed

The WSSDO Office Building Project (Godam and Residential Quarter) is a government building construction project located within the premises of the Water Supply and Sanitation Division Office (WSSDO), Dhangadhi, Kailali. The project comprises the construction of a Godam (storage warehouse) on the ground floor and a Residential Quarter on the upper floor to improve the office's storage capacity and residential facilities for technical personnel.

The project is being implemented in accordance with the approved engineering drawings, technical specifications, Bill of Quantities (BOQ), and the Government of Nepal's public procurement procedures. The major construction activities include reinforced cement concrete (RCC) structural works, brick masonry, reinforcement fixing, formwork, concrete casting, plastering, painting, and other associated building works.

During the internship, this project provided practical exposure to the construction and documentation processes of a government building project. Under the guidance of the supervising engineers, field visits and office-based discussions enhanced the understanding of structural drawings, reinforcement detailing, formwork arrangements, field measurements, quantity verification, construction quality monitoring, and the relationship between engineering drawings and actual site activities. The project also provided practical insight into the sequence of building construction activities and the engineering practices followed during project implementation. Representative technical documents are provided in **Appendix C**, and site photographs are presented in **Appendix G**.

3.4.3 Basauti Water Supply Project – Technical Activities Performed

A. Site Supervision and Field Observation

One of the major technical activities carried out during the internship was the site supervision of the Basauti Water Supply Project. Site visits were conducted under the guidance of the supervising engineers to observe the progress of construction works and understand the practical implementation of a community water supply scheme.

During the visits, pipeline excavation, HDPE pipeline laying and jointing, backfilling, valve chamber construction, and completed painting works were observed. The construction sequence, workmanship, and overall progress of the project were discussed with the engineering team to understand how the approved design was being implemented in the field. These observations provided practical insight into construction practices, quality control, and the day-to-day supervision responsibilities of engineers involved in water supply projects. Selected site photographs related to the Basauti Water Supply Project are provided in **Appendix D**, while additional site photographs are included in **Appendix G**.

B. Field Measurement and Quantity Verification

Field measurements of completed construction works were carried out under the guidance of the departmental engineers to understand the procedures followed for quantity verification in government water supply projects. The measurements covered pipeline excavation, backfilling, HDPE pipe laying and jointing, valve chamber construction, and completed painting works.

The measured quantities were verified with the approved engineering drawings and the Bill of Quantities (BOQ) before the preparation of project documentation and contractor payments. This activity provided practical exposure to field measurement techniques, quantity verification procedures, and their importance in ensuring the accuracy of construction records and payment certification. Representative field measurement records and related project documents are included in **Appendix D**.

C. Measurement Book (MB) Preparation

The preparation of the Measurement Book (MB) was carried out under the guidance of the departmental engineers using the verified field measurements obtained during site inspections. The activity involved recording measurement details in the prescribed MB format, calculating quantities, identifying the corresponding Bill of Quantities (BOQ) items, and documenting completed construction works for official records.

Participation in this process provided practical understanding of the standard procedures followed for MB preparation and demonstrated the importance of the Measurement Book as the primary document for quantity verification, contractor billing, and project documentation in government water supply projects. Sample

Measurement Book (MB) records and related project documents are included in **Appendix D**.

D. BOQ-Based Billing and Work Completion Documentation

BOQ-based contractor billing and work completion documentation were carried out under the guidance of the departmental engineers using the verified quantities recorded in the Measurement Book (MB). The activity included relating the measured quantities to the corresponding Bill of Quantities (BOQ) items and contract rates to understand the preparation of contractor payment bills.

In addition, work completion and acceptance documents were reviewed and prepared after verifying the completed construction activities and compiling the supporting records required for bill processing. This activity provided practical understanding of the relationship between field measurements, the Measurement Book (MB), the Bill of Quantities (BOQ), contractor billing, and project completion documentation followed in government water supply projects.

Representative BOQ sheets, billing documents, work completion records, and related project documents are included in **Appendix D**.

3.4.4 Hasuliya Water Supply Project – Technical Activities Performed

A. Site Supervision and Field Observation

Site supervision activities were carried out under the guidance of the Senior Engineer and departmental engineers to observe the progress and quality of construction works at the Hasuliya Water Supply Project. During the site visits, pipeline excavation, HDPE pipe laying and jointing, backfilling, valve chamber construction, and completed painting works were observed and monitored. The construction sequence, workmanship, and implementation of the approved drawings and technical specifications were also reviewed during field inspections. These activities provided practical exposure to the execution stage of a government water supply project and improved understanding of site supervision and quality control practices. Selected site photographs are included in **Appendix G**.

B. Field Measurement and Quantity Verification

Field measurements of completed construction works were carried out under the guidance of the Senior Engineer and departmental engineers to verify the quantities

executed by the contractor. The activities included the measurement and verification of completed pipeline excavation, backfilling, HDPE pipe laying and jointing, valve chamber construction, and painting works. The verified quantities were compared with the approved engineering drawings and the Bill of Quantities (BOQ) to ensure consistency before the preparation of project documentation and contractor payment. This activity provided practical experience in field measurement techniques, quantity verification procedures, and quality assurance practices followed in government water supply projects. Representative measurement records are included in **Appendix E**.

C. Measurement Book (MB) Preparation

The preparation of the Measurement Book (MB) was carried out under the guidance of the Senior Engineer using the verified field measurements. The activity involved recording measurement details in the prescribed format, calculating quantities, identifying the corresponding BOQ items, and documenting completed works for official records. Participation in this process improved understanding of the standard procedures followed for MB preparation and demonstrated the importance of the Measurement Book as the primary document for quantity verification and contractor payment. Sample MB records are included in **Appendix E**.

D. BOQ-Based Billing and Work Completion Documentation

BOQ-based contractor billing and work completion documentation were carried out under the guidance of the Senior Engineer and departmental engineers using the verified quantities recorded in the Measurement Book. The measured quantities were related to the corresponding BOQ items and contract rates to understand the preparation of contractor payment bills. In addition, work completion and acceptance documents were reviewed and prepared after verifying the completed construction activities and compiling the required supporting documents. This activity provided practical knowledge of the billing, documentation, and financial procedures adopted in government water supply projects. Representative billing and completion documents are included in **Appendix E**.

E. Transportation Cost Estimation

Transportation cost estimates for major construction materials used in the project were prepared under the guidance of the Senior Engineer by considering haulage distances, transportation routes, vehicle operating costs, and the applicable district

rate schedule. The activity improved understanding of transportation rate analysis and its contribution to the overall project cost estimation process. Representative transportation cost estimation sheets are included in **Appendix E**.

3.4.5 Bajhnatha Sidanatha Water Supply Project – Technical Activities Performed

A. Project Familiarization

The internship activities began with an orientation to the Bajhnath Sidanatha Water Supply Project under the guidance of the Senior Engineer. During this stage, the project profile and available technical documents were reviewed to understand the project objectives, scope, major components, and overall water supply system. This activity provided a general understanding of the planning and implementation of a government pumping water supply scheme.

B. Water Supply System Layout and Engineering Drawings

The project's engineering drawings, AutoCAD layouts, and system plans were reviewed under the guidance of the Senior Engineer to understand the arrangement and functions of the major water supply components. Learning activities included the transmission and distribution pipeline network, source works, pumping system, 225 m³ overhead water tank (OHT), valve chambers, deep tube well, pump house, and household service connections. Practical knowledge was also gained on how GPS survey data is incorporated into AutoCAD drawings for preparing pipeline layouts. These activities improved the ability to interpret engineering drawings and understand their application in water supply project planning. Representative engineering drawings are included in **Appendix F**.

C. Quantity Estimation and Bill of Quantities (BOQ)

Project estimation documents were reviewed to learn the preparation and organization of the Bill of Quantities (BOQ) and quantity estimation procedures. By examining engineering drawings and quantity calculation sheets, practical knowledge was gained on how quantities of major water supply structures, including the 225 m³ RCC overhead water tank, valve chambers, pump house, deep tube well platform, and other civil works, are derived and incorporated into project estimates. This activity strengthened the understanding of quantity take-off methods used in water supply projects.

D. Rate Analysis and Cost Estimation

The project's rate analysis sheets and detailed cost estimation documents were reviewed under the guidance of the Senior Engineer to understand the estimation procedures adopted in government water supply projects. Learning activities included item-wise rate analysis based on approved district rates, transportation cost calculations, market rates, pipeline procurement and laying estimation, household connection estimation, electrification cost estimation, and operation and maintenance cost estimation. Practical exposure to Excel-based estimation sheets improved understanding of how project costs are prepared and organized for implementation.

E. Technical Discussions and Learning Outcomes

Throughout the internship, regular discussions were held with the Senior Engineer to clarify concepts related to engineering drawings, quantity estimation, BOQ preparation, rate analysis, and project cost estimation. Project-related technical documents were reviewed to reinforce the knowledge gained during the internship and to develop a better understanding of the relationship between engineering design documents, estimation procedures, and practical water supply project implementation. These activities significantly enhanced the ability to interpret technical documents and understand the engineering practices followed during the planning and implementation of government water supply projects.

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES

4.1 Conclusion

The internship at the Water Supply and Sanitation Division Office (WSSDO), Kailali, provided valuable practical exposure to the planning, design, construction, quantity estimation, supervision, and technical documentation processes involved in government water supply projects. The opportunity to observe and study projects at different stages of implementation helped bridge the gap between theoretical knowledge acquired during the Bachelor of Civil Engineering program and its practical application in the field.

Throughout the internship, exposure was gained to a variety of technical activities through the review of engineering drawings, Detailed Project Reports (DPRs), quantity estimation sheets, Bills of Materials (BOMs), rate analysis, cost estimation documents, Measurement Books (MBs), and other project records. Site visits to ongoing construction projects further enhanced the understanding of pipeline installation, valve chamber construction, reinforced concrete works, quality control practices, and site supervision. Discussions with supervising engineers also provided valuable insight into engineering decision-making, project planning, and the procedures followed during the implementation of government-funded water supply schemes.

The internship also strengthened professional skills such as technical communication, engineering documentation, teamwork, time management, and the preparation of technical reports. Overall, the internship was a meaningful learning experience that enhanced both technical knowledge and professional competence while providing a clear understanding of the responsibilities of civil engineers working in the water supply sector. The knowledge and experience gained during this internship will serve as a strong foundation for future academic work and professional engineering practice.

4.2 Learning Outcomes

At the completion of the internship, the following learning outcomes were achieved:

- Developed an understanding of the planning, design, implementation, and documentation procedures of government water supply projects.
- Improved the ability to read and interpret engineering drawings, AutoCAD layouts, and project design documents.
- Gained practical knowledge of quantity estimation and the preparation and organization of Bills of Quantities (BOQs).
- Developed an understanding of rate analysis, detailed cost estimation, transportation cost calculations, and project budgeting procedures.
- Learned the purpose and application of Measurement Books (MBs), quantity verification, contractor billing, and project documentation.
- Gained practical exposure to pipeline laying, valve chamber construction, reinforced concrete structures, and quality control practices through site observations.
- Improved understanding of project planning, technical documentation, and engineering procedures followed by the Water Supply and Sanitation Division Office (WSSDO).
- Enhanced communication and teamwork skills through regular interaction with supervising engineers and technical staff.
- Strengthened the ability to relate theoretical civil engineering concepts to practical applications in water supply and infrastructure projects.
- Developed greater confidence in understanding engineering documents and professional practices required in the civil engineering profession.

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APPENDIX A
Weekly Activity Logs

APPENDIX B

Supporting Technical Documents of Dokebazar Khanepani

Aayojana

Source: Water Supply and Sanitation Division Office (WSSDO), Kailali. Internal project documents reviewed during the internship (used for academic purposes only).

APPENDIX C
Supporting Technical Documents of WSSDO Office
Building Project

Source: Water Supply and Sanitation Division Office (WSSDO), Kailali. Internal project documents reviewed during the internship (used for academic purposes only).

APPENDIX D

Supporting Technical Documents of Basauti Water Supply Project

Source: Water Supply and Sanitation Division Office (WSSDO), Kailali. Internal project documents reviewed during the internship (used for academic purposes only).

APPENDIX E

Supporting Technical Documents of Hasuliya Water Supply Project

Source: Water Supply and Sanitation Division Office (WSSDO), Kailali. Internal project documents reviewed during the internship (used for academic purposes only).

APPENDIX F

Supporting Technical Documents of Bajinatha Sidanatha

Water Supply Project

Source: Water Supply and Sanitation Division Office (WSSDO), Kailali. Internal project documents reviewed during the internship (used for academic purposes only).

APPENDIX G
Site Photographs

APPENDIX H
Internship Supporting Documents

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 1					
Name:	TEK RAJ JOSHI					
Supervisor:	Senior Er. Hansha Bahadur Thapa					
S.N.	Date (BS)	Activities Performed			Learning Outcomes	Remarks
1	Tuesday, 2083/02/05	Reported to WSSDO, Kailali for first day of internship. Attended orientation with supervisor; interacted with office staff and civil/technical personnel. Reviewed internship objectives, office protocols, and organizational hierarchy. Took overview of ongoing schemes, office operations, and probable scope of work during internship period.			Understood the hierarchy and functioning of WSSDO, Kailali; gained initial overview of departmental schemes and intern responsibilities.	Office
2	Wednesday, 2083/02/06	Visited the site and inspected the overhead tank . Observed ongoing pipe layout marking, excavation, and distribution line works. Interacted with the Head Contractor and Upabhokta Samiti members; reviewed the stage-wise construction brief. Works are progressing, however, pipeline laying remains pending due to labour shortage and budget constraints..			Gained field exposure to implementation challenges in community water supply schemes; understood the role of Upabhokta Samiti in community coordination and contractor management; observed how project delays arise due to labour shortage and budget constraints; noted the correlation between project specifications and actual site conditions.	Site Visited
3	Thursday, 2083/02/07	Studied project drawings and technical documents of Dokebazar scheme. Studied pipe sizing, hydraulic grade line, and design notes from office files.			Gained understanding of hydraulic design principles including pipe sizing and hydraulic grade line (HGL) analysis; developed ability to read and interpret technical drawings and design notes of a community water supply scheme.	office
4	Friday, 2083/02/08	Continued review of Dokebazar Khanepani Ayojana technical drawings; studied hydraulic grade line calculations, pipe sizing (dia. selection), and head loss computations. Discussed design queries with supervisor.			Deepened understanding of hydraulic grade line calculations, pipe diameter selection criteria, and head loss computations; developed confidence in interpreting design drawings and clarifying technical queries with supervising engineer.	Office (Fri)
5	Saturday, 2083/02/09	Office closed — no internship activity.			—	Off (Sat)
6	Sunday, 2083/02/10	Office closed — no internship activity.			—	Off (Sun)
7	Monday, 2083/02/11	Reviewed technical project documents, engineering drawings, and relevant water supply design guidelines under the guidance of the Engineers.			Familiarized with government documentation procedures and WSSDO technical guidelines.	Office

WEEKLY LEARNING SUMMARY	
1. Gained first-hand field exposure to Dokebazar Khanepani Ayojana urban water supply scheme. 2. Understood the organizational structure and workflow of WSSDO, Kailali. 3. Familiarized with project drawings and technical documentation of water supply schemes.	

VERIFICATION & SIGNATURES	
Student Signature	Supervisor / Mentor Signature
	Date of Verification
	2083-02-12

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 2					
Name:	TEK RAJ JOSHI					
Supervisor:	Senior Er. Hansha Bahadur Thapa					
S.N.	Date (BS)	Activities Performed	Learning Outcomes	Remarks		
1	Tuesday, 2083/02/12	Reviewed (Cost Summary & Financial Analysis) and (Detailed BOQ & Cost Estimate) of Dokebazar DPR; studied how quantities are measured and priced for each work item	Understood how BOQ items are structured, quantified, and priced in a water supply project; learned how cost summary is prepared from detailed estimates	Office		
2	Wednesday, 2083/02/13	Reviewed previously prepared DPR documents, including rate analysis and district rate schedule documents, to understand quantity estimation, cost estimation procedures, and standard government documentation. Also developed practical knowledge of Excel-based engineering calculations and technical tabulation.	Studied rate analysis methodology, including material, labour, equipment, transportation, and district rate components, and developed an understanding of their application in quantity estimation and project cost estimation.	office		
3	Thursday, 2083/02/14	Office closed — Bakri Eid (Public Holiday).		off day		
4	Friday, 2083/02/15	Office closed — Republic Day / International Everest Day (Public Holiday).		off day		
5	Saturday, 2083/02/16	Office closed — no internship activity.		Off (Sat)		
6	Sunday, 2083/02/17	Office closed — no internship activity.		Off (Sun)		
7	Monday, 2083/02/18	Studied the detailed estimate, structural design, structural drawings, reinforcement detailing, and quantity take-off sheets of the godown building and staff quarters at the WSSDO Kailali under the guidance of the Engineers.	Gained understanding of detailed estimation procedures, interpretation of structural drawings, reinforcement detailing, and quantity calculation methods used in building construction projects.	Office		

WEEKLY LEARNING SUMMARY	
During this week, I studied DPR documents, including BOQ, cost estimates, rate analysis, and district rate schedules, to understand quantity estimation, cost calculation, and standard documentation procedures. I also studied structural design, structural drawings, final estimate preparation, reinforcement detailing, and quantity take-off methods, which enhanced my understanding of construction planning and estimation practices.	

VERIFICATION & SIGNATURES	
Student Signature _____	Supervisor / Mentor Signature _____ Date of Verification _____ 2083-02-20 _____

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 3				Date Range:	2083/02/19 – 2083/02/25 (Jestha)
Name:	TEK RAJ JOSHI				Roll No.:	22040384
Supervisor:	Senior Er. Hansha Bahadur Thapa				College:	NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGAD
S.N.	Date (BS)	Activities Performed	Learning Outcomes	Remarks		
1	Tuesday, 2083/02/19	Studied the detailed estimate, structural design, structural drawings, reinforcement details, and quantity calculations of the godown and staff quarters building to understand construction planning and estimation procedures.	Gained understanding of detailed estimation procedures, interpretation of structural drawings, reinforcement detailing, and excell based quantity calculation methods used in building	Office,site		
2	Wednesday, 2083/02/20	Visited the site and observed trench excavation and layout marking for pipeline construction; checked excavation dimensions, alignment, and depth as per the design drawings.	Understood trench excavation procedures, layout setting-out, alignment control, and excavation requirements etc.	Site Visited,office		
3	Thursday, 2083/02/21	Observed footing reinforcement placement, rebar spacing, clear cover, and formwork preparation.	Understood footing reinforcement detailing — rebar spacing, lap length, and cover requirements in practice.	Site Visited,office		
4	Friday, 2083/02/22	Understood footing reinforcement detailing — rebar spacing, lap length, and cover requirements in practice.	Developed basic quantity take-off skills for isolated footing works.	Office (Fri)		
5	Saturday, 2083/02/23	Office closed — internship activity ongoing -construction	—	Off (Sat)		
6	Sunday, 2083/02/24	Office closed — internship activity-construction	—	Off (Sun)		
7	Monday, 2083/02/25	Visited site; observed column reinforcement cage placement and tie beam reinforcement; interacted with site supervisor on construction performance.	Visited site; observed column reinforcement cage placement and tie beam reinforcement; interacted with site supervisor on construction sequence.	Site Visited,office		
WEEKLY LEARNING SUMMARY						
Studied structural drawings, detailed estimates, and quantity calculations. Gained practical understanding of trench excavation, footing reinforcement, and column and tie beam construction through site observations.						
VERIFICATION & SIGNATURES						
Student Signature		Supervisor / Mentor Signature		Date of Verification		
				2083-02-27		

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 4		Date Range:	2083/02/26 – 2083/03/01		
Name:	TEK RAJ JOSHI		Roll No.:	22040384		
Supervisor:	Senior Er. Hansha Bahadur Thapa		College:	NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGADHI		
S.N.	Date (BS)	Activities Performed	Learning Outcomes		Remarks	
1	Tuesday, 2083/02/26	site supervision column concreting and brick masonry works and quality control work.	Gained practical understanding of column concreting, brick masonry construction, and quality control measures at the construction site.		Office	
2	Wednesday, 2083/02/27	Conducted site supervision and office work related to the godown and staff quarters construction project.	Gained practical understanding of construction supervision, progress monitoring, and project documentation procedures.		Site Visited	
3	Thursday, 2083/02/28	Conducted site supervision and office work related to the godown and staff quarters construction project.	Gained practical understanding of construction supervision, progress monitoring, and project documentation procedures.		Site Visited	
4	Friday, 2083/02/29	off day due to health condition	progress		off day	
5	Saturday, 2083/02/30	Office closed — no internship activity.	—		Off (Sat)	
6	Sunday, 2083/02/31	Office closed — no internship activity.	—		Off (Sun)	
7	Monday, 2083/03/01	Studied rate analysis for M20 concrete and reinforcement steel using District Rate 2083/83; prepared quantity take-off for tie beams and columns as per estimate at office.	Learned how unit rates for RCC works are built from material, labor, and equipment costs using district rates.		Office	

WEEKLY LEARNING SUMMARY
Gained practical experience in site supervision of column concreting and brick masonry works, including quality control and progress monitoring. Also improved understanding of quantity take-off, rate analysis, and cost estimation procedures for RCC construction works.

VERIFICATION & SIGNATURES	
Student Signature	Supervisor / Mentor Signature
	Date of Verification
	_____ 2083-03-02 _____

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 5					
Name:	TEK RAJ JOSHI	Date Range:	2083/03/02 – 2083/03/08	Roll No.:	22040384	
Supervisor:	Senior Er. Hansha Bahadur Thapa	College:	NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY, DHANGADHI			
S.N.	Date (BS)	Activities Performed	Learning Outcomes	Remarks		
1	Tuesday, 2083/03/02	Participated in the site supervision of godown and staff quarters construction works and monitored ongoing construction activities.	Gained practical understanding of site supervision and ongoing construction activities.	Office and site		
2	Wednesday, 2083/03/03	Participated in the field measurement and verification of completed painting, pipeline excavation, backfilling, pipe laying, and jointing works under the Basauti Water Supply Project.	Gained practical experience in field measurement and verification of completed water supply construction works.	Site visited		
3	Thursday, 2083/03/04	Participated in the measurement verification and bill review of completed pipeline and valve chamber construction works under the Basauti Water Supply Project under the guidance of the department engineer	Gained practical knowledge of field measurement verification and the bill review process for completed construction works	Site Visited		
4	Friday, 2083/03/05	Participated in office-based project discussions, reviewed project drawings and documents, collected site-related data using Google Earth, and observed CAD drawings under the guidance of the department engineer	Gained practical understanding of project planning, data collection, drawing review, and the use of digital tools in engineering projects.	Office (Fri)		
5	Saturday, 2083/03/06	Office closed — no internship activity.	—	Off (Sat)		
6	Sunday, 2083/03/07	Office closed — no internship activity.	—	Off (Sun)		
7	Monday, 2083/03/08	Observed and learned the preparation of transportation cost estimates for major construction materials used in the Hasuliya Water Supply Project under the guidance of the department engineer.	Gained practical knowledge of transportation cost estimation, haulage distance calculation, and rate analysis of construction materials	Office		

WEEKLY LEARNING SUMMARY

1. Gained practical knowledge of site supervision, field measurement, bill verification, project documentation, and transportation cost estimation for water supply projects.

VERIFICATION & SIGNATURES

Student Signature	Supervisor / Mentor Signature	Date of Verification
		_____ 2083-03-09 _____

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 6	Date Range: 2083/03/09 – 2083/03/15				
Name:	TEK RAJ JOSHI	Roll No.: 22040384				
Supervisor:	Senior Er. Hansha Bahadur Thapa	College: NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGADHI				
S.N.	Date (BS)	Activities Performed			Learning Outcomes	Remarks
1	Tuesday, 2083/03/09	Participated in the site supervision and quantity measurement of painting works and pipeline construction under the Hasuliya Water Supply Project with the engineer and senior engineer team.			Gained practical knowledge of measuring and verifying painting works, pipeline excavation, backfilling, and pipe laying quantities during site supervision, while enhancing understanding of valve chamber components and urban water supply system infrastructure.	Site Visited
2	Wednesday, 2083/03/10	Learned to prepare Measurement Books (MB) from field records and calculate quantities in Excel for painting and pipeline works under the Hasuliya Water Supply Project.			Enhanced understanding of MB preparation, quantity calculation, and documentation of painting and pipeline works under the Hasuliya Water Supply Project	office
3	Thursday, 2083/03/11	Learned the process of preparing contractor bills (वेक्का सम्बन्धि बिल) based on measured quantities, contract rates, and project documentation			Gained practical knowledge of contractor billing procedures, quantity-based payment calculations, and contract documentation	office
4	Friday, 2083/03/12	Participated in the preparation of the Work Acceptance/Completion Report (कार्य स्वीकार / सम्पन्न प्रतिवेदन) and bill documentation for painting works, pipeline excavation, backfilling, and pipe laying & jointing works under the Hasuliya Water Supply Project			Learned the procedures for preparing work completion reports, processing bills, and documenting completed painting and pipeline works	Office
5	Saturday, 2083/03/13	Office closed — no internship activity.			—	Off (Sat)
6	Sunday, 2083/03/14	Office closed — no internship activity.			—	Off (Sun)
7	Monday, 2083/03/15	learned the preparation of the Measurement Book (MB) for pipeline and valve chamber works of the Basauti Water Supply Project under the guidance of the department engineer.			Learned how to prepare a Measurement Book (MB), calculate quantities from field measurements, and apply BOQ rates to determine work values for billing purposes	Office

WEEKLY LEARNING SUMMARY	
1 Gained practical knowledge of site supervision, Measurement Book (MB) preparation, quantity calculation, contractor billing, work completion documentation, and bill processing for water supply projects.	

VERIFICATION & SIGNATURES		
Student Signature	Supervisor / Mentor Signature	Date of Verification
		2083-03-16

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 7	Date Range: 2083/03/16 – 2083/03/22				
Name:	TEK RAJ JOSHI	Roll No.: 22040384				
Supervisor:	Senior Er. Hansha Bahadur Thapa	College: NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY, DHANGADHI				
S.N.	Date (BS)	Activities Performed			Learning Outcomes	Remarks
1	Tuesday, 2083/03/16	Participated in the process of preparation of the final bill for the Basauti Water Supply Project using measured site data and carried out related office work in Microsoft Excel under the guidance of the engineer and senior engineer.			Gained practical experience in bill preparation, quantity tabulation, and Excel-based construction documentation	Office
2	Wednesday, 2083/03/17	Carried out site supervision of the godown building and staff quarters construction at the WSSDO Kailali premises, observing brick masonry work, sill level construction, and reinforcement placement as per approved .			Gained practical exposure to site sequencing of masonry work up to sill level, proper reinforcement placement and cover as per approved structural drawings, and quality control practices followed during construction supervision.	Site Visited
3	Thursday, 2083/03/18	Carried out site supervision of the ongoing godown building and staff quarters construction project at the WSSDO Kailali premises, observing brick masonry work up to sill level and reinforcement placement as per approved structural drawings.			Gained practical knowledge of brick masonry construction sequencing up to sill level and proper reinforcement placement as per approved structural drawings.	Site Visited
4	Friday, 2083/03/19	Carried out site supervision and quality control of ongoing godown building and staff quarters construction at WSSDO Kailali			Gained practical exposure to site supervision procedures and quality control practices followed during ongoing building construction.	Office (Fri)
5	Saturday, 2083/03/20	Office closed — no internship activity.			—	Off (Sat)
6	Sunday, 2083/03/21	Office closed — no internship activity.			—	Off (Sun)
7	Monday, 2083/03/22	Conducted site supervision and quality control of the ongoing godown building and staff quarters construction at WSSDO Kailali			Gained practical exposure to site supervision techniques and quality control practices followed during building construction.	Office

WEEKLY LEARNING SUMMARY	
1. Gained practical experience in final bill preparation, Excel-based quantity documentation, site supervision, quality control, and building construction up to sill level with reinforcement inspection.	

VERIFICATION & SIGNATURES	
Student Signature	Supervisor / Mentor Signature
	Date of Verification
	2083-03-22

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 8		Date Range: 2083/03/23 – 2083/03/29			
Name:	TEK RAJ JOSHI		Roll No.: 22040384			
Supervisor:	Senior Er. Hansha Bahadur Thapa		College: NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGADHI			
S.N.	Date (BS)	Activities Performed	Learning Outcomes		Remarks	
1	Tuesday, 2083/03/23	Supervised shuttering , bamboo staging , and reinforcement (ring) checks; verified rebar dimensions and column cage alignment; inspected excavated trench (waterlogged) and JCB earthwork. Formwork and reinforcement confirmed as per drawing; excavation of safety tank construction in adjacent building .	Gained practical knowledge of shuttering and bamboo staging arrangement, reinforcement ring/stirrup verification, and column reinforcement cage alignment as per approved structural drawings, along with field understanding of excavation and dewatering requirements for safety tank construction.		Office and site visited	
2	Wednesday, 2083/03/24	Observed site supervision, reinforcement checking, and formwork inspection.	Learned reinforcement and formwork quality checking before concreting.		Site Visited	
3	Thursday, 2083/03/25	Monitored reinforcement, shuttering, and slab casting preparations.	Gained practical knowledge of pre-concreting inspection procedures.		Site Visited	
4	Friday, 2083/03/26	Participated in quality inspection of slab casting preparations, including reinforcement, formwork, and Measurement Book (MB)	Developed practical understanding of quality control, MB preparation, and quantity verification procedures.		Office (Fri)	
5	Saturday, 2083/03/27	Office closed — no internship activity.	—		Off (Sat)	
6	Sunday, 2083/03/28	Office closed — no internship activity.	—		Off (Sun)	
7	Monday, 2083/03/29	Learned the process of Measurement Book (MB) preparation and completion work documentation.	Developed practical understanding of Measurement Book (MB) preparation and project completion documentation procedures.		Office	
WEEKLY LEARNING SUMMARY						
1. Developed practical understanding of formwork inspection, reinforcement verification, column alignment, excavation supervision, dewatering requirements, and construction quality control.						
VERIFICATION & SIGNATURES						
Student Signature		Supervisor / Mentor Signature		Date of Verification		
				2083-03-30		

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 9	Date Range: 2083/03/30 to 2083/04/04				
Name:	TEK RAJ JOSHI	Roll No.: 22040384				
Supervisor:	Senior Er. Hansha Bahadur Thapa	College: NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGADHI				
S.N.	Date (BS)	Activities Performed	Learning Outcomes		Remarks	
1	Tuesday, 2083/03/30	Studied the project profile of the Bajinatha Siddhanath Water Supply Project under the guidance of the supervising engineer. The study, together with a site visit, provided a practical understanding of the project's objectives, scope of work, major components, and	Gained an understanding of the project objectives, scope, and major components.		site visited,Office	
2	Wednesday, 2083/03/31	Learned the overall layout of the water supply system, including the source, pumping system, transmission main, overhead reservoir, and distribution network.	Gained an understanding of the overall configuration and operation of the water supply system.		office	
3	Thursday, 2083/04/01	Learned how GPS survey data is converted into AutoCAD drawings for preparing transmission and distribution pipeline layouts.	Developed an understanding of pipeline layout preparation using AutoCAD.		Office	
4	Friday, 2083/04/02	Learned the functions and engineering details of the overhead water tank, valve chambers, deep tube well, and pump house by reviewing their design drawings.	Improved understanding of the purpose and layout of major water supply structures.		office	
5	Saturday, 2083/04/03	Office closed — no internship activity.	—		Off (Sat)	
6	Sunday, 2083/04/04	Office closed — no internship activity.	—		Off (Sun)	
7	Monday, 2083/04/05	Learned to read and interpret detailed engineering drawings of water supply structures and discussed key observations with the supervising engineer.	Enhanced the ability to read engineering drawings and relate them to practical construction.		Office	
WEEKLY LEARNING SUMMARY						
1. During this week, I reviewed the DPR and project profile of the Bajhnath Siddhartha Water Supply Project, studied the overall water supply system layout, examined AutoCAD drawings of the pipeline network and associated structures, and gained an understanding of the design and function of the 225 m³ overhead water tank and its role within the water distribution system.						
VERIFICATION & SIGNATURES						
Student Signature		Supervisor / Mentor Signature			Date of Verification	
					2083-04-05	

INTERNSHIP WEEKLY ACTIVITY LOG									
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management									
Week No.:	Week 10	Date Range: 2083/04/06 – 2083/04/12							
Name:	TEK RAJ JOSHI	Roll No.: 22040384							
Supervisor:	Senior Er. Hansha Bahadur Thapa	College: NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGADHI							
S.N.	Date (BS)	Activities Performed	Learning Outcomes					Remarks	
1	Tuesday, 2083/04/06	Learned the preparation and organization of the Bill of Quantities (BOQ) for the Bajhnath Siddhartha Water Supply Project by reviewing project estimation documents.	Developed an understanding of the purpose and preparation of the BOQ.					Office	
2	Wednesday, 2083/04/07	Learned how quantities of construction items are calculated from engineering drawings using quantity estimation sheets.	Improved understanding of quantity take-off methods used in water supply projects.					office ,site	
3	Thursday, 2083/04/08	Learned the detailed quantity estimation of the 225 m³ RCC overhead water tank by reviewing quantity calculation sheets and design drawings.	Gained knowledge of quantity estimation for reinforced concrete water supply structures.					Office	
4	Friday, 2083/04/09	Learned the quantity estimation of valve chambers, pump house, and deep tube well platform from the project estimation documents.	Developed an understanding of quantity estimation for different water supply structures.					office	
5	Saturday, 2083/04/010	Office closed — no internship activity.						Off (Sat)	
6	Sunday, 2083/04/11	Office closed — no internship activity.						Off (Sun)	
7	Monday, 2083/04/12	Learned how detailed cost estimates are prepared using quantity calculations, approved district rates, market rates, and transportation costs.	Improved understanding of the preparation of detailed cost estimates for water supply projects.					Office	
WEEKLY LEARNING SUMMARY									
1. During this week, I reviewed valve chamber design drawings, studied quantity estimation procedures, examined the BOQ and rate analysis sheets, and reviewed the detailed cost estimate of the Bajhnath Siddhartha Water Supply Project, gaining practical knowledge of estimation, costing, and project planning processes used in water supply engineering.									
VERIFICATION & SIGNATURES									
Student Signature		Supervisor / Mentor Signature						Date of Verification	
—								2083-04-12	

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 11	Date Range:	2083/04/13 – 2083/04/19			
Name:	TEK RAJ JOSHI		Roll No.:	22040384		
Supervisor:	Senior Er. Hansha Bahadur Thapa		College:	NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGADHI		
S.N.	Date (BS)	Activities Performed	Learning Outcomes		Remarks	
1	Tuesday, 2083/04/13	Learned the preparation of rate analysis for various construction items by reviewing project estimation sheets and approved district rates.	Developed an understanding of item-wise rate analysis used in water supply projects.		Office	
2	Wednesday, 2083/04/14	Learned how transportation costs for construction materials are estimated using haulage charts, transportation distances, and approved rates.	Gained knowledge of transportation cost estimation for project materials.		Site Visit ✓	
3	Thursday, 2083/04/15	Learned the estimation of pipeline procurement, laying works, and pipe fittings by reviewing detailed estimation sheets.	Improved understanding of pipeline material estimation and installation costs.		Office	
4	Friday, 2083/04/16	Learned the quantity and cost estimation of household connections by reviewing estimation sheets for pipes, fittings, and accessories.	Developed an understanding of estimating materials required for household water connections.		office	
5	Saturday, 2083/04/17	Office closed — no internship activity.			Off (Sat)	
6	Sunday, 2083/04/18	Office closed — no internship activity.			Off (Sun)	
7	Monday, 2083/04/19	Learned the estimation of operation and maintenance costs, including pump operation and electrification costs, by reviewing project cost estimation documents.	Gained an understanding of operation and maintenance cost estimation for water supply systems.		Office	
WEEKLY LEARNING SUMMARY						
1. Developed a comprehensive understanding of rate analysis, material transportation costing, pipeline and household connection estimation, and operation and maintenance cost estimation through the review of engineering estimation documents and approved district rate schedules.						
VERIFICATION & SIGNATURES						
Student Signature		Supervisor / Mentor Signature			Date of Verification	
					2083-04-19	

INTERNSHIP WEEKLY ACTIVITY LOG						
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management						
Week No.:	Week 12		Date Range:	2083/04/20 – 2083/04/26		
Name:	TEK RAJ JOSHI		Roll No.:	22040384		
Supervisor:	Senior Er. Hansha Bahadur Thapa		College:	NATIONAL ACADEMY OF SCIENCE AND TECHNOLOGY,DHANGADHI		
S.N.	Date (BS)	Activities Performed	Learning Outcomes		Remarks	
1	Tuesday, 2083/04/20	Reviewed the project drawings, BOQ, quantity estimates, and cost estimation documents with the supervising engineer to clarify technical concepts.	Strengthened understanding of engineering drawings, estimation, and project documentation		Office	
2	Wednesday, 2083/04/21	Learned how different technical documents, estimation sheets, and design drawings are integrated during project planning and implementation.	Developed a better understanding of the relationship between design, estimation, and project execution.		Office	
3	Thursday, 2083/04/22	Reviewed project-related estimation sheets, rate analysis, and design documents to reinforce previously learned concepts.	Improved understanding of project estimation and engineering documentation.		Site Visit ✓	
4	Friday, 2083/04/23	Started compiling internship records, organizing project documents, drawings, photographs, and field notes for the preparation of the final internship report.	Organized the information and supporting documents required for report writing and final documentation.		office	
5	Saturday, 2083/04/24	Office closed — no internship activity.	—		Off (Sat)	
6	Sunday, 2083/04/25	Office closed — no internship activity.	—		Off (Sun)	
7	Monday, 2083/04/26	Continued final report writing and documentation.	Drafted major sections of the internship report.		Office	
WEEKLY LEARNING SUMMARY						
1. Integrated the knowledge gained throughout the internship by reviewing engineering drawings, estimation documents, BOQ, and technical records, while systematically compiling the final internship report and supporting documentation.						
VERIFICATION & SIGNATURES						
Student Signature		Supervisor / Mentor Signature			Date of Verification	
					2083-04-26	

INTERNSHIP WEEKLY ACTIVITY LOG				
Water Supply & Sanitation Division Office (WSSDO), Kailali Dept. of Water Supply & Sewerage Management				
Week No.:	Week 14	Date Range:2083/05/04 to 2083/05/05		
Name:	TEK RAJ JOSHI			
Supervisor:	Senior Er. Hansha Bahadur Thapa			
S.N.	Date (BS)	Activities Performed	Learning Outcomes	Remarks
1	Tuesday, 2083/05/04	Documentaion work	Progressed.	Office
2	Wednesday, 2083/05/05	Submitted the final internship report, received feedback from the Senior Engineer and college coordinator, completed the required documentation, and formally concluded the internship at WSSDO, Kailali.	Successfully completed all internship requirements and gained practical knowledge of water supply project planning, design review, cost estimation, construction practices, and operation and maintenance activities.	End of internship
3				
4			—	
5			—	Off (Sat)
6			—	Off (Sun)
7				
WEEKLY LEARNING SUMMARY				
1. During this final week, I completed the remaining documentation, submitted my internship report, received feedback from the Senior Engineer, and formally concluded my internship at WSSDO, Kailali. The internship helped me gain practical knowledge and experience in various aspects of water supply engineering.				
VERIFICATION & SIGNATURES				
Student Signature		Supervisor / Mentor Signature		Date of Verification
				2083-05-05

SALIENT FEATURES

Doke Bazzar Water Supply and Sanitation Project — Detailed Engineering Design Estimate

S.N.	Items	Description
1	Name of the Project	Doke Bazzar Water Supply and Sanitation Project
2	Type	Lift/Gravity Flow System
3	Study Level	Detailed Engineering Design
4	Location Area	
	Region	Mid-Western
	Zone	Mahakali
	District	Kanchanpur
	VDC/Municipality	Baise Bichwa VDC (Now Laljhadi Municipality)
	Ward	Ward No. 8, 9(8)
5	Available Facilities	
	Road	Connected with district feeder road
	Supply Water	STW/ Dug Well-Point source, seasonal source (poor quality)
	System Electricity	Yes
	Communication	Yes
	Health Services	Yes
	Banking Facilities	No
6	Source Characteristics	
	Source Name	Ground Water
	Source Type	Deep Tube-Well
	Source Location	Within Community
	Safe Yield (lps)	Expected to be 8-10 lps
7	Type of Structures	
	Storage Tank	RCC 225 OHT-1 No.
	Intake/ Boring with Plate form	2 No. Deep Tube-well- 250mm x 200mm x 150 M deep
	Pump House and Office	3 Nos.
	Valve Chambers (nos.)	14 nos.
	Washout/Air valve Chamber (no.)	2 Nos.
	Compound Wall	210 M
	HH Connections	759 Nos.
	Crossing GI	12 Nos.

S.N.	Items	Description
	Total Length of Pipe:	
	Transmission Main	0.2 Km
	Distribution Network	16.3 Km
	Pipe Used	
	HDPE (size) 16.1 Km	200-140 4 Kg/cm ² & 125-50mm mm dia. (6 kg/cm ²)
	GI (size) (0.3 Km)	100, 65 mm dia./32 mm (medium class)
	DI (Size 0.1 Km)	100 mm dia
8	Demography	
	Present Population (2017)	4512
	Base Year Population (2019)	4694
	Design Year Population (2038)	6839
	Growth Rate %	2.0%
	Present Household Numbers (2017)	759
9	Total Cost of the Project (including contingencies)	NRs. 9,44,58,755.00
10	Cost Sharing arrangement	
	Grant Component (GoN/DWSS, Foreign Aid)	NRs. 7,54,33,378.00 (79.86%)
	WUSC's Contribution both cash & kind	NRs. 1,90,25,377.00 (20.14%)
11	Tariff	Rs. 160.00 for base year 2019
14	Cost per unit	
	Per Capita Cost (Base Year - 2019)	NRs. 19,520.00
	Per Capita Cost (Design year - 2038)	NRs. 13,399.00

Note: The above project summary is reproduced from the original DPR for academic reference. While some project data correspond to the period of DPR preparation, the document serves as a valuable reference for understanding the engineering design process, project estimation, and technical documentation practices followed in government water supply projects.

Doke Bazzar Water Supply and Sanitation Project



Figure __. Pipeline Layout of Dokebazar Khanepani Aayojana(KML based formate)

Summary of Pipeline and Hydraulic Design

Doke Bazar Water Supply and Sanitation Project

A. Pipeline Network Overview

Parameter	Value
Total number of pipe segments	58
Total actual (surveyed) length	16,115.0 m
Total design length	17,726.5 m
Pipe material and class	HDPE, Class 4 & 6 kg/cm ² (distribution network); GI, MD/HD (house connections)
Outer diameter (OD) range	32 mm – 315 mm
Hazen-Williams roughness coefficient (C)	140
Velocity range (representative segments)	0.04 m/s – 0.60 m/s
Friction (head loss) range per km (representative segments)	0.09 m – 5.72 m per km

B. Cost Estimate of Pipe

Pipe Category	Quantity (m)	Weight (kg)	Amount (NPR)
HDPE pipe (sub-total)	17,726.5	45,954.11	98,34,201
GI pipe (sub-total)	228	1,629	2,63,651
DI pipe (sub-total)	0	0	0
Total (HDPE + GI + DI)	17,955	47,583	1,00,97,852
Local transportation for pipes & fittings @ 3%	-	-	3,33,229
Total cost of pipe including fittings	-	-	1,04,31,081

C. Network Hydraulic Performance (Demand, Elevation, Head & Pressure)

Parameter	Value
Total number of junctions (demand nodes)	46
Source / reservoir	1 (Overhead Tank, OHT-200)
Total system demand	25.77 LPS
Ground elevation range	348.151 m (J-2, lowest) – 353.100 m (J-6, highest)
Hydraulic grade line / head range at nodes	359.44 m (J-4) – 365.62 m (J-27); Reservoir HGL = 366.00 m
Residual pressure range at nodes	7.79 m (J-38, minimum) – 14.68 m (J-27, maximum); Reservoir = 16.00 m
Average residual pressure (all nodes)	11.38 m

Note: The values presented in this table are summarized from the Hydraulic Design of Pipeline Network section of the DPR. Velocity and head loss values are based on the available hydraulic design data for pipe segments P-38 to P-58, while the remaining parameters represent the complete pipeline network. For detailed hydraulic calculations and pipe-wise design data, refer to the original DPR.

Doke Bazaar Water Supply & Sanitation Project
SUMMARY OF RATE ANALYSIS

Sr. No	Norms No.	Description of works	Analyzed for	UNIT RATE
1 Surface clearance				
1	(1-7)	Trimming projections, filling	1 Sq.m.	4.43
2	(1-8)	Top Soil Excavation 15-20 cm and disposal	1 Sq.m.	70.84
		Site clearance	1 Sqm	75.27
2 Excavation Work				
4	2-(9)	Earthwork in Excavation in Soft Soil	1 Cum	456.03
5	2-(12)	Earthwork in Excavation in Hard Soil	1 Cum	570.04
6	2-(14)	Earthwork in Excavation in BMS	1 Cum	725.09
7	(2-15Ka)	Soft rock cutting in foundation, Drain	1 Cum	1,368.10
8	(2-15Kha)	Hard rock cutting	1 Cum	11,492.02
3 Back Filling				
9	(2-39Ka)	with soft Soil	1 Cum	228.02
10	2(39-Ga)	Backfilling with Hard Soil	1 Cum	287.30
11	2(39-Gha)	Earthwork in Backfilling with BMS	1 Cum	273.62
12	2(39-Na)	Backfilling with Soft Rock	1 Cum	501.64
12-1	2 (42)	Sand filling wor with watering & compation all complete	1 Cum	6,570.36
4 Brick Works				
13	5-1Ka1	First class Brick masonry in 1:3 cement mortar	1 Cum	14,502.70
14	5-1Ka2	First class Brick masonry in 1:4 cement mortar	1 Cum	13,887.54
15	5-1Ka3	Brick masonry in 1:6 cement mortar	1 Cum	13,329.29
16	5-1Ka3	Dry Brick masonry	1 Cum	9,832.35
16a	5-1Ka4	Mud Brick masonry	1 Cum	10,085.93
16b	11-15Ka	Flat brick solling with sand	1 m2	1,086.93
16c	11-15Kha	Brick sollig on edge with sand	1 m2	1,695.55
4 Stone Works				
17	6-2 (2)	R.R. Masonry in mud mortar	1 Cum	9,773.93
18	6-2 (2)	Dry stone masonry work	1 Cum	9,409.66
19	6 (5)	Dry boulder packing work	1 Cum	9,223.03
20	11-16	Boulder soling with sand	1 m2	14,114.61
21	6-1 (2)	R.R. masonry in 1:4 cement mortar	1 Cum	17,243.94
22	6-1 (3)	R.R. masonry in 1:6 cement mortar	1 Cum	16,883.68
23	6-1 (1)	R.R. masonry in 1:3 cement mortar	1 Cum	18,570.53
24	15-9 (gha)	Base course packing with sand mixing below 40 mm down	1 Cum	7,131.45
25		Gravel packing with sand mixing below 63 mm down	1 m2	1,402.76
5 Cement Concret Works				
26-1	7-4 Kha	Plain cement concrete (1:1:5:3)	1 Cum	18,833.65
26-2	7-4-adj	Plain cement concrete (1:1:5:3) for upto 4m height	1 Cum	19,276.40
26-3	7-4-adj	Plain cement concrete (1:1:5:3) for 4m to 8m height	1 Cum	19,719.15
26-4	7-4-adj	Plain cement concrete (1:1:5:3) for 8m to 12m height	1 Cum	21,490.15
26-5	7-4-adj	Plain cement concrete (1:1:5:3) for 12m to 16m height	1 Cum	22,375.65
26-6	7-4-adj	Plain cement concrete (1:1:5:3) for 16m to 20m height	1 Cum	23,261.15
26-7	7-4-adj	Plain cement concrete (1:1:5:3) for 20m to 22m height	1 Cum	24,146.65
26-8	7-4-adj	Plain cement concrete (1:1:5:3) for 22m to 26m height	1 Cum	25,032.15
27	7-2 Gha	Plain cement concrete (1:2:4)	1 Cum	15,932.86
28	7-2 ga	Plain cement concrete (1:3:6)	1 Cum	14,098.85
29	7-5	TMT Fe-500 steel reinforcement bar	1 kg	125.33
30		Reinforcement including for Ferro-cement works	1 Sqm	1,170.35
6 Form works				
31	8-2-ka	Form work for concrete work for upto 4 m height	1 Sqm	615.10
31-1	8-2-gha-adj	Form work for concrete work for 4m to 8 m slab wall height	1 Sqm	1,149.46
31-2	8-2-gha-adj	Form work for concrete work for 8m to 12 m slab wall height	1 Sqm	2,221.05

Sr. No	Norms No.	Description of works	Analyzed for	UNIT RATE
31-3	8-2-gha-adj	Form work for concrete work for 12m to 16 m slab wall height	1 Sqm	2,511.46
31-4	8-2-gha-adj	Form work for concrete work for 16m to 20 m slab wall height	1 Sqm	3,536.06
31-5	8-3-ka-adj	Form work for concrete work for column errction all complete work	1 Sqm	887.83
31-6	8-1-cha-adj	Form work for concrete work for dome portion all complete	1 Sqm	2,852.27
31-7	8-1-ja-adj	Form work for concrete work for stair upto 4m height all complete	1 Sqm	948.52
31-8	8-1-ja-adj	Form work for concrete work for stair 4m to 8 m height all complete	1 Sqm	1,200.63
31-9	8-1-ja-adj	Form work for concrete work for stair 8m to 12m height all complete	1 Sqm	2,254.39
31-10	8-1-ja-adj	Form work for concrete work for stair 12m to 16 m height all complete	1 Sqm	2,536.52
31-11	8-1-ja-adj	Form work for concrete work for stair 16m to 20 m height all complete	1 Sqm	2,812.14
31-12	8-1-ja-adj	Form work for concrete work for stair 20m to 22 m height all complete	1 Sqm	3,043.48
31-13	8-4-kha-adj	Form work for concrete work for beams upto 4 m height	1 Sqm	742.87
31-14	8-4-kha-adj	Form work for concrete work for beams 4m to 8 m height	1 Sqm	990.86
31-15	8-4-kha-adj	Form work for concrete work for beams 8m to 12m height	1 Sqm	2,040.48
31-16	8-4-kha-adj	Form work for concrete work for beams 12m to 16m height	1 Sqm	2,318.48
31-17	8-4-kha-adj	Form work for concrete work for beams 16m to 20m height	1 Sqm	2,589.96
31-18	8-4-kha-adj	Form work for concrete work for beams 20m to 22 m height all complete	1 Sqm	2,865.58
32		Form work for concrete work for ferro cement tank	1 Sqm	264.98
7 Roofing				
33-1	9-1	CGI sheet roofing 26 gauge	1 Sqm	642.22
34	9-2	GI plain sheet ridge work, 26 gauge	1 R.m.	734.63
8 Wood works				
35	(10-1)	Sallwood work for frames all complete		
35-1		door size 0.9*2.1m (Size of wood : 100 x 75 mm)	1 No.	8,370.12
35-2		for door & window	1 Cum	206,497.51
36	(10-2)	38 mm thick Sallwood frame for pannel Shutter size : 1.07 x 1.982 = 2.12074 m2	1 m2	10,657.51
36-1	(10-3)	Glazed shutter with 38 mm thick Sallwood frame & glass	1 m2	7,415.41
36-2	(10-7)	Making & fixing commerical plywood flush shutter 38 mm thick sallwood frame	1 m2	6,668.86
37	(10-7)	38 mm thick Sallwood frame using 24 SWG G.I. wire mesh for mosquito size 1.092*2.05m	1 m2	3,942.02
38	(10-10)	38 mm thick wooden shutter with 24 SWG plain sheet on one side	1 Sqm	7,051.79
39	10-17	Wood works for beam, rafters etc. local wood	1 Cum	67,748.08
10 Flooring				
40	11-1Ka	Floring with 25 mm thick PCC (1:2:4)	1 Sqm	496.95
41	11-7	Paving of porcelite tile in (1:4)	1 Sqm	2,137.60
42	11-9	Paving of 37.5 mm thick tile in 1:4 cement mortar	1 Sqm	1,536.45
43	11-13	Paving of brick on edge in 1:6 cement mortar and pointing in 1:2 cement	1 Sqm	1,750.42
44	11-15Kha	Paving of brick on edge with sand	1 Sqm	1,695.55
45	11-19 Ka	Filling work including watering and compaction with sand	1 Cum	7,144.21
46	2-26	Filling work including watering and compaction with outer soil	1 Cum	1,152.88
11 Plastering Works				
47	12-1 ka	12.5mm thick cement plaster (1:2)	1 Sqm	423.32
48	12-1-kha	12.5mm thick cement plaster (1:3)	1 Sqm	365.12
49	12-1-ga	12.25mm thick cement plaster (1:4)	1 Sqm	355.88

Sr. No	Norms No.	Description of works	Analyzed for	UNIT RATE
50	12-4-ka	20mm thick cement plaster (1:3) with 2% Water Proffing compound	1 Sqm	597.18
51	12-4-ka	20mm thick cement plaster (1:4)	1 Sqm	505.16
52	12-4-kha	20mm thick cement plaster (1:6)	1 Sqm	485.78
53	12-4-ga	20mm thick cement plaster (1:6)	1 Sqm	440.55
54	12-4-gha	Flush ruled pointing work in 1:3 C/M	1 Sqm	234.24
55	11-20	3 mm thick net cement punning	1 Sqm	231.31
12 Painting				
55-1	13-(4kha-1ka)	Water proof cement painting over plastered surface double coat on top of single coat white cement point as primer	1 Sqm	301.12
56	13-12 kha	Two coats cement painting	1 Sqm	276.29
56-1	13-10 Kha	Two coats of anti corrosive paint inside the over head tank on top of with one coat of primer	1 Sqm	54.21
57	13-5 ka	Two coats of enamel painting with one coat primer	1 Sqm	219.66
58	13-1Ka	Two coats of bitumen painting	1 Sqm	52.56
59	13-1Ka	Two coats of aluminium painting	1 Sqm	226.89
60		Laying of one layer of 500 gauge of Polythene sheet	1 Sqm	322.65
13 Protection Work				
61	16-2 ka	28.1 Gabion preparation, all complete (size 2*1*1)	1 Cum	2,035.47
62	16-11	Filling of Gabion box with stone	1 Cum	8,067.04
14 Pipe laying and Jointing				
63	17-2	HDPE pipe jointing works 20,25 mm dia	1 m	2.16
64	17-2	32 mm dia	1 m	2.64
65	17-2	40,50 mm dia	1 m	3.24
66	17-2	63,90 mm dia	1 m	42.82
67	17-2	110, 125 mm dia	1 m	64.05
68	17-2	140, 160, 180 mm dia & above	1 m	85.39
68	17-1	200,225 mm dia & above	1 m	106.73
69	17-4-ka	15,20 mm dia GI	1 m	46.58
70	17-4-ka	25,32 mm dia	1 m	63.50
71	17-4-ka	GI pipe jointing works 40,50 mm dia	1 m	86.79
72	17-4-ka	65,80 mm dia	1 m	116.11
73	17-4-ka	100 mm dia	1 m	145.44
74	17-3-Na	DI pipe jointing works 100 mm dia	1 m	102.12
75	17-3-ka	125 mm dia	1 m	113.44
76	17-3-ka	150 mm dia	1 m	141.80
77	17-3-ka	200 mm dia	1 m	170.16
78	17-3-ka	250 mm dia	1 m	226.88
79	17-3-kha	300 mm dia	1 m	283.60
80	17-3-ga	350 mm dia	1 m	340.32
15 Fencing Works				
78-1	19-2	Dismantling work of brick masonry in c/m	1 m3	975.20
79	24-7	Barbed wire fencing on wooden posts (2 diagonal & 5 horizontals) 3m	1 Rm	1,514.21
80	24-7	Hexagonal GI Wire fencing on matalic angle pole	1 Rm	1,003.36
81	24-6	Barbed wire fencing	1 Rm	179.47
82	24-7	Barbed wire fencing on RCC Posts (100*100*22mm) @ 3 m c/c 2 dia	1 Rm	2,349.04
16 Preparation and Packing of Filter Media				
83	7-4	Packing of Gravel in the Sand Filter	1 Cum	9,083.36
84	24-7	Packing of Filter Sand in the Sand filter	1 Cum	4,081.89
17 Iron Works				
85		Installation of GI Pipe Railing 40 mm dia.	1 m	2,181.91
86		Installation of Grill in the Window	1 m2	2,690.86

Sr. No	Norms No.	Description of works	Analyzed for	UNIT RATE
18 Road Graveling Works				
87	15-4 (ka)	Sub grade preparation work upto 10 cm thick earth cutting all complete	1 m ²	53.13
87-1	15-9 (Ka)	Laying of 100 mm thick (compacted) sand mixed rived bed gravel as	1 m ²	565.65
88	15-12 ang	Laying of 120 mm thick (compacted) WBM base course using crushed stone and stone dust	1 m ²	702.34
19 Gabion Works				
89	16-1Ka	Gabion work all complete work (box 2*1*1)	1 box	20,927.90
90	16-1gha	Gabion work all complete work (box 3*1*.5)	1 box	17,135.70

Note: The detailed labour, material, transportation, and item-wise cost estimates are available in the original Detailed Project Report (DPR)

ANALYSIS OF RATES

Collection and Transportation of Construction Material

Availability of Construction Materials

Cement and other construction materials :	Mahendranagar- 77 Km
Sand :	Mahakali- 91 Km
Boulders and Aggregates :	Mahakali/ Bhansi-91/75Km
Brick :	Belaury/Punarbhas -25Km
Cost of per hour working of truck :	NRs./hr 1265.33

Transportation by Truck

Truck Capacity	=	6.00 MT
Speed at Black Topped Road	=	30.00 Km/hr
Speed at Gravel Road	=	15.00 Km/hr
Speed at Earthen Road	=	10.00 Km/hr
Loading and Unloading Time	=	0.75 hr
Time for single trip by truck	=	(2*(distance/speed)+loading/unloading time)) in hours

Sand transported from -91Km

Haulage Distance		
Black Topped Road	=	32.00 km
Gravel Road	=	53.00 km
Earthen Road	=	6.00 km
Time for single trip	=	11.9 hr

Capacity of Truck at single trip & time/unit content

<u>Materials</u>	Capacity Unit	Time/M ³	Unit	Cost/ M ³
Sand	= 4.00 M ³	2.98 hr		3764.34

Hence cost of M³ transportation of sand = hours X cost per hour **3764.34 Rs.**

Stone transported from -91 Km

Haulage Distance		
Black Topped Road	=	32.00 km
Gravel Road	=	53.00 km
Earthen Road	=	6.00 km
Time for single trip	=	11.9 hr

Capacity of Truck at single trip & time/unit content

<u>Materials</u>	Capacity Unit	Time/M ³	Unit	Cost/ M ³
Sand	= 3.50 M ³	3.40 hr		4302.11

Hence cost of M³ transportation of sand = hours X cost per hour **4302.11 Rs.**

ANALYSIS OF RATES

Collection and Transportation of Construction Material

Aggregate transported from -75Km

Haulage Distance		
Black Topped Road	=	20.00 km
Gravel Road	=	53.00 km
Earthen Road	=	2.00 km
Time for single trip	=	10.3 hr

Capacity of Truck at single trip & time/unit content

<u>Materials</u>		Capacity Unit	Time/M ³	Unit	Cost/ M ³
Aggregate 40 mm	=	4.87 M ³	2.11 hr		2676.15
Aggregate 20 mm	=	4.87 M ³	2.11 hr		2676.15
Aggregate 10 mm	=	5.79 M ³	1.78 hr		2250.92
Aggregate 5 mm	=	5.79 M ³	1.78 hr		2250.92
Gravel	=	4.69 M ³	2.20 hr		2778.86

Cement and other construction materials from Mahendranagar-77 Km

<u>Haulage Distance</u>		
Black Topped Road	=	22.00 km
Gravel Road	=	53.00 km
Earthen Road	=	2.00 km
Time for single trip	=	10.4333 hr

Capacity of Truck at single trip & time/unit content for different material

<u>Transportation of easy carrying materials - For 1 kg</u>	Capacity Unit	Time/unit	Unit	Cost/ Kg
Cement	= 4.00 MT	2.608333 hr		3.300
Loading and unloading charge as per district rate= Rs.9 per bag				0.180
Hence cost per Kg				3.480
Rod, other materials	4.00 MT	2.608333 hr		3.300
Loading and unloading charge as per district rate= Rs.23 per quintal				0.460
Hence cost per Kg				3.760
<u>Transportation of non-easy carrying materials</u>	= 1.5 times of easily carrying materials			5.411
<u>HDPE pipe, MS rod</u>				
24 gauge C.G.I. Sheet	= 1127.00 Sq.m.	0.0093 hr		11.71

Brick collected from -25 Km

<u>Haulage Distance</u>		
Black Topped Road	=	km
Gravel Road	=	23.00 km
Earthen Road	=	2.00 km
Time for single trip	=	4.96667 hr

ANALYSIS OF RATES

Collection and Transportation of Construction Material

Capacity of Truck at single trip & time/unit content for different material

<u>Materials</u>		Capacity Unit	Time/brick Unit	Cost/ brick
Brick	=	2400.00 No.	0.0021 hr	2.62

Wood from - 77 Km

Haulage Distance

Black Topped Road	=	22.00 km
Gravel Road	=	53.00 km
Earthen Road	=	2.00 km

Time for single trip	=	10.43 hr
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Capacity of Truck at single trip & time/unit content for different material

<u>Materials</u>		Capacity Unit	Time/unit Unit	Transportati Cost/ M ³
Salwood (1 ton = 1.25 m ³)	=	7.50 M ³	1.391111 hr	1760.21
Other woods (1 tone = 1 m ³)	=	6.00 M ³	1.7389 hr	2200.26

Doke Bazzar Water Supply & Sanitation Project

Rate Analysis for External Materials

Rate Analysis for External Materials									
			Unit Wt.	Trans. Cost	Material District Rate				
Sn	Item	Unit	in	Trans. Rate	Trans- Amount	Without	With	Material District Rate	Remarks
			kg	Rs.	Rs.	VAT	13% VAT		
			2	3	(2 x 3) = 4			Without Vat (Nrs)	
1	Cement	Bag	50	3.48	174.02	800.00	104.00	974.02	OPC
2	White Cement	kg	1	3.76	3.76	25.00	3.25	28.76	
3	TMT Fe 500. Bar	kg	1	5.41	5.41	86.00	11.18	91.41	
4	Good Quality Plywood (12.0 mm)	m2	1.5	5.41	8.12	570.20	74.13	578.31	
5	Good Quality Plywood (4 mm)	m2	0.25	5.41	1.35	215.17	27.97	216.52	
6	GI Wire 9	kg	1	5.41	5.41	95.00	12.35	100.41	medium
7	Salvage Wire 7	kg	1	5.41	5.41	95.00	12.35	100.41	medium
8	Bind Wire 12	kg	1	5.41	5.41	125.00	16.25	130.41	
9	Binding Wire (white)	kg	1	5.41	5.41	125.00	16.25	130.41	
10	CK Wire Mesh	Sqm.	0.40	5.41	2.16	173.00	22.49	175.16	
11	Barbed Wire- 12 gauze	m.	0.10	5.41	0.54	114.00	14.82	114.54	
12	Plain Wire	kg	1	5.41	5.41	95.00	12.35	100.41	medium
13	Nails/Screws	kg	1	3.76	3.76	140.00	18.20	143.76	
14	Polythene Sheet	Sqm.	0.5	3.76	1.88	200.00	26.00	201.88	500 gauze
15	Bitumen Paint	Lit.	1	3.76	3.76	190.00	24.70	193.76	Black Japan
16	Water Proof Compound	Lit.	1	3.76	3.76	413.00	53.69	416.76	Silica
17	WPC Powder	kg	1	3.76	3.76	86.00	11.18	89.76	
18	Fuel	Lit.	1	3.76	3.76	92.00	11.96	95.76	
19	Cement Paint (Snowcem)	kg	1	3.76	3.76	350.00	45.50	353.76	Varnes synthetic
20	Enamel Paint	Lit.	1	3.76	3.76	550.00	71.50	553.76	
21	Primer- Apex	Lit.	1	3.76	3.76	320.00	41.60	323.76	
22	Aluminium Paint	Lit.	1	3.76	3.76	520.00	67.60	523.76	
23	Sand Paper	No.	1	3.76	3.76	21.00	2.73	24.76	
24	GI Wire 8 - 10	kg	1	5.41	5.41	91.00	11.83	96.41	commercial
25	GI Wire 12	kg	1	5.41	5.41	91.00	11.83	96.41	commercial
26	M.S. Angle	kg	1	5.41	5.41	120.00	15.60	125.41	
27	Grill for Windows	kg	1	5.41	5.41	137.00	17.81	142.41	
28	C. GI Sheet (26 Gauge)	Sqm.	3.65	5.41	19.75	318.89	41.46	338.64	
29	GI Sheet (26 Gauge) for ridge	rm.	1	5.41	5.41	318.89	41.46	324.30	
30	GI Plain Sheet (26G)	Sqm.	3.65	5.41	19.75	346.66	45.07	366.41	
31	Holdfast	No	0.2	3.76	0.75	15.71	2.04	16.47	7 / kg
32	Screw Nails	No	0.01	3.76	0.04	4.00	0.52	4.04	
33	Hinge 150 mm	No	0.1	3.76	0.38	60.00	7.80	60.38	

Doke Bazzar Water Supply & Sanitation Project

Rate Analysis for External Materials

Rate Analysis for External Materials									
			Unit Wt.	Tran. Cost		Material District Rate			
Sn	Item	Unit	in	Trans. Rate	Trans- Amount	Without	With	Material District Rate Without Vat (Nrs)	Remarks
			kg	Rs.	Rs.	VAT	13% VAT		
			2	3	(2 x 3) = 4				
34	Tower bolt 150mm	No	0.15	3.76	0.56	20.00	2.60	20.56	
35	Tower bolt 300mm	No	0.25	3.76	0.94	40.00	5.20	40.94	
36	Nut & Bolts	kg	0.01	3.76	0.04	200.00	26.00	200.04	
37	J-Hook	No	0.01	3.76	0.04	10.00	1.30	10.04	
38	U-Hook	No	0.01	3.76	0.04	11.00	1.43	11.04	
39	Bitumen Washer	No.	0.01	3.76	0.04	0.30	0.04	0.34	
40	GI Wire Mesh	Sqm.	0.40	3.76	1.50	210.00	27.30	211.50	
41	Lockset 250 mm	No	0.1	3.76	0.38	800.00	104.00	800.38	Mortise
42	Handle-6"	No	0.1	3.76	0.38	68.00	8.84	68.38	Almunium
43	Spring Set	No	0.1	3.76	0.38	312.00	40.56	312.38	
44	GI Mesh Wire Screen	m2	1.5	3.76	5.64	300.00	39.00	305.64	
45	10 mm Suspender Cable	m	0.36	5.41	1.95	437.50	56.88	439.45	
46	14 mm Suspender Cable	m	0.70	5.41	3.79	562.50	73.13	566.29	
47	Flat Stone for Small Chamber	Set	1	5.41	5.41	27.50	3.58	32.91	
48	Glass 4 mm thick	m2	1	5.41	5.41	667.02	86.71	672.43	
49	Wooden Listic	m	0.05	5.41	0.27	24.00	3.12	24.27	
50	Mosquito Net	m2	0.5	5.41	2.71	260.00	33.80	262.71	Kailali rate/ good quality
51	Porcelene Tile	m2	2.5	5.41	13.53	645.50	83.92	659.03	
52	Cement Tile Decorative	m2	3	5.41	16.23	753.09	97.90	769.32	
53	1 1/2" dia. GI pipe	rm.	3.73	5.41	20.18	567.00	73.71	587.18	

Doke Bazaar Water Supply & Sanitation Project

Rate Analysis for External Materials

Sn	Item	Unit	Unit Wt.		Trans. Cost		Material District Rate			Remarks
			in	kg	Trans. Rate	Trans- Amount	Without	With		
			2	3	Rs.	Rs.	VAT	13% VAT	Material District Rate Without Vat (Nrs)	
						(2 x 3) = 4				

Rate for Local Materials

Sn	Item	Unit	Qty.	Trans. Rate	Trans- Amount	Without VAT	With VAT	Rate	Remarks
1	First Class Chimney Brick	1000/Pc	1000		2,618.52	9,900.00		12,518.52	
2	Block Stone	cum	1		4,302.11	1,900.00		6,202.11	
3	Bond Stone	cum	1		4,302.11	2,280.00		6,582.11	
4	40 mm Aggregate(from crusher)	cum	1		2,676.15	1,200.00		3,876.15	
5	20 mm Aggregate (from crusher)	cum	1		2,676.15	1,485.00		4,161.15	
6	10 mm Aggregate "	cum	1		2,250.92	1,000.00		3,250.92	
7	Sand	cum	1		3,764.34	1,184.62		4,948.96	
8	Mud	cum	1		150.00	375.00		525.00	
9	Sal Wood	cum	1		1,760.21	140,000.00		141,760.21	
10	Local Wood (salla)	cum	1		2,200.26	43,200.00		45,400.26	
11	Bamboo (30 ft)	No	1		20.00	210.00		230.00	
12	Skilled Labour (Mason)	Manday						590.00	
13	Unskilled Labour (Mason)	Manday						385.00	
14	Skilled Labour (Painter)	Manday						590.00	
15	Unskilled Labour (Painter)	Manday						590.00	
16	Skilled Labour (Plumber)	Manday						590.00	
17	Katigad (Juniour Plumber)	Manday						400.00	Semi Skilled
18	Skilled Labour (Carpenter)	Manday						590.00	
19	Unskilled Labour (Carpenter)	Manday						385.00	
20	Stone dust	m3						600.00	

Doke Bazzar Water Supply and Sanitation Project
Total Water Demand the Project Area

Survey year: 2017

Base Year: 2019

Design year: 2038

Annex -

SN	Year	HHs	Population	Consumers' Demand as per type										Total domestic demand	Commer/Inst demand LPD	System losses LPD	Total Revenuable demand	Total demand
				Yard Connection					Community taps									
				%	HHs	Pop'n	LPCD	demand	%	HHs	Pop'n	LPCD	demand					
1	2017	759	4512	60	455	2707	100	270720	40	304	1805	100	180480	451200	47460	87999	496650	586659
2	2018	775	4602	60	465	2761	100	276134	40	310	1841	100	184090	460224	48362	89750	508586	598336
3	2019	791	4694	60	475	2817	100	281657	40	316	1878	100	187771	469428	49283	91537	518711	610249
4	2020	807	4788	62	500	2969	100	296867	38	307	1820	100	181950	478817	50222	93360	529039	622399
5	2021	823	4884	63	518	3077	100	307688	37	305	1807	100	180706	488393	51179	95219	539572	634791
6	2022	839	4982	64	537	3188	100	318823	36	302	1793	100	179338	498161	52156	97115	550317	647432
7	2023	856	5081	65	556	3303	100	330281	35	300	1778	100	177844	508124	53152	99049	561276	660325
8	2024	873	5183	66	576	3421	100	342069	34	297	1762	100	176218	518287	54169	101022	572456	673478
9	2025	890	5287	67	596	3542	100	354197	33	294	1745	100	174455	528653	55205	103034	583858	686891
10	2026	908	5392	68	617	3667	100	366674	32	291	1726	100	172552	539226	56263	105086	595489	700575
11	2027	926	5500	69	639	3795	100	379507	31	287	1705	100	170503	550010	57341	107180	607351	714531
12	2028	945	5610	70	662	3927	100	392707	30	284	1683	100	168303	561010	58441	109315	619451	728766
13	2029	964	5722	71	684	4063	100	406284	29	280	1659	100	165947	572231	59563	111493	631794	743287
14	2030	983	5837	72	708	4202	100	420246	28	275	1634	100	163429	583675	60708	113715	644383	758098
15	2031	1003	5953	73	732	4346	100	434605	27	271	1607	100	160744	595349	61875	115981	657224	773204
16	2032	1023	6073	74	757	4494	100	449369	26	266	1579	100	157887	607256	63066	118292	670322	788614
17	2033	1043	6194	75	782	4646	100	464551	25	261	1549	100	154850	619401	64280	120650	683681	804330
18	2034	1064	6318	76	809	4802	100	480160	24	255	1516	100	151629	631789	65519	123054	697308	820362
19	2035	1085	6444	77	835	4962	100	496207	23	250	1482	100	148218	644425	66782	125507	711207	836714
20	2036	1107	6573	78	863	5127	100	512704	22	244	1446	100	144609	657313	68071	128009	725384	853393
21	2037	1129	6705	79	892	5297	100	529663	21	237	1408	100	140796	670459	69386	130561	739845	870406
22	2038	1152	6839	80	922	5471	100	547095	20	230	1368	100	136774	683869	70727	133164	754596	887760

Doke Bazzar Water Supply and Sanitation Project
Domestic Demand of the Project Area

Survey year: 2017

Base Year: 2019

Design year: 2038

SN	Year	HHs	Population	Water Demand Assessment										total Demand
				Fully Plumbed Connection					Yard Connection/Shared taps					
				%	HHs	Pop'n	LPCD	demand	%	HHs	Pop'n	LPCD	demand	
1	2017	759	4512	60	455	2707	100	270720	40	304	1805	100	180480	451200
2	2018	775	4602	60	465	2761	100	276134	40	310	1841	100	184090	460224
3	2019	791	4694	60	475	2817	100	281657	40	316	1878	100	187771	469428
4	2020	807	4788	62	500	2969	100	296867	38	307	1820	100	181950	478817
5	2021	823	4884	63	518	3077	100	307688	37	305	1807	100	180706	488393
6	2022	839	4982	64	537	3188	100	318823	36	302	1793	100	179338	498161
7	2023	856	5081	65	556	3303	100	330281	35	300	1778	100	177844	508124
8	2024	873	5183	66	576	3421	100	342069	34	297	1762	100	176218	518287
9	2025	890	5287	67	596	3542	100	354197	33	294	1745	100	174455	528653
10	2026	908	5392	68	617	3667	100	366674	32	291	1726	100	172552	539226
11	2027	926	5500	69	639	3795	100	379507	31	287	1705	100	170503	550010
12	2028	945	5610	70	662	3927	100	392707	30	284	1683	100	168303	561010
13	2029	964	5722	71	684	4063	100	406284	29	280	1659	100	165947	572231
14	2030	983	5837	72	708	4202	100	420246	28	275	1634	100	163429	583675
15	2031	1003	5953	73	732	4346	100	434605	27	271	1607	100	160744	595349
16	2032	1023	6073	74	757	4494	100	449369	26	266	1579	100	157887	607256
17	2033	1043	6194	75	782	4646	100	464551	25	261	1549	100	154850	619401
18	2034	1064	6318	76	809	4802	100	480160	24	255	1516	100	151629	631789
19	2035	1085	6444	77	835	4962	100	496207	23	250	1482	100	148218	644425
20	2036	1107	6573	78	863	5127	100	512704	22	244	1446	100	144609	657313
21	2037	1129	6705	79	892	5297	100	529663	21	237	1408	100	140796	670459
22	2038	1152	6839	80	922	5471	100	547095	20	230	1368	100	136774	683869

Source: Water Supply and Sanitation Division Office (WSSDO), Kailali. Extracted from the original Detailed Project Report (DPR) reviewed during the internship.

 Sudurpashchim Province Government Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal					
<i>Project Name: Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)</i>					
Abstract of Cost (Godam)					
S.N	Description	Unit	Quantity	Rate	Amount
A.	Provision Sum (insurance for work, plant and materials damage, owner and consultant staff, construction equipments and machines, third party, As Built Drawing, lab test, Dismantling Works and others) As p	Job	1	327096.25	327096.25
B	Construction of Godown				
1	Site clearance work including cutting & uprooting shrubs, thick herbs & vegetables, top soil cutting with maintaining of soil grade for drain work & removal of waste as per instruction of site engineers.	sqm	225	19.96	4491
2	Excavation (using hydraulic excavator) including disposal up to 10m up to 1.5 m etc., trimming bottom and side slopes in accordance with requirements of lines, grades and cross sections all complete as per specification.	cum	105.14	50.54	5313.77
3	Excavation for foundations, pipe/conduit trenches, wall foundation trenches and all underground structures from ground level to bottom of building under footings and where required as per drawings, in any kind of soil including haulage with the use of labors to the dimensions and levels shown in the drawings.	cum	26.28	626.00	16451.28
4	One Layer of brick Soling of 150 mm in foundation and floor all complete as per drawing, specification & instructions of engineer all complete.	sqm	305.52	820.87	250792.2
5	Back filling with excavated dumped soil and transported soil (if required) in to the foundation and flooring of the building (for good materials if needed, removal of extra soil, debris, and organic materials shall be removed by bidder). Back filling materials shall be applied on layers by layers of 150 mm thick, watering and compaction, including loading, unloading and transportation cost and others all complete as per specification, drawings and instruction of engineer.	cum	115.44	435.62	50287.97

Estimated By:

Checked By:

Approved By:

 Sudurpashchim Province Government Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal						
Project Name: <i>Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)</i>						
Abstract of Cost (Godam)						
S.N	Description	Unit	Quantity	Rate	Amount	Remarks
6	P.C.C. M10 (1:3:6) for R.C.C. with cement sand and stone ballast including mixing with concrete mixture, compacting the concrete with vibrator machine, lying, finishing to approved level lines and dimensions and curing as per instruction & specification complete work	cum	34.08	11010.19	375227.27	
7	Providing and placing machine mixed at site cement concrete M20 grade (1:1.5:3) grade work in Footings, Columns, Beams, Slab with transportation up to 30m	cum	115.64	14320.12	1655978.67	
8	Providing and placing machine mixed at site cement concrete M15 grade (1:2:4) work in Sill/Lintel Band with transportation up to 30m	cum	3.1	12756.35	39544.68	
9	Reinforcement Steel Works: Providing, straightening, cleaning, cutting, fabricating, bending, stirrups & tie TMT (Fe 500) for main reinforcement bars and binding the bars with 18 gauge annealed wire for tying them at each junction as per design and details for all types of RCC elements including all waste and cut pieces, provision for spacers, chairs, providing and placing cover blocks cement mortar (1:1) to keep the bars in the intended position at all levels etc. all complete as per specification, drawings and instruction of engineer.	M.T	15.43	149350.50	2304478.21	
10	Form Work: Form Work Column post fitting and dismantling, removing with supplying wood and transporting up to 30m	sqm	136.8	1310.40	179262.72	
11	Formwork: Form work for beams, slab, footings, sill, lintels including, screening, fitting, oiling, dismantling, removing and transportation up to 30m	sqm	623.61	1375.97	858068.65	

Estimated By:

Checked By:

Approved By:

 Sudurpashchim Province Government Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal						
<i>Project Name: Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)</i>						
Abstract of Cost (Godam)						
S.N	Description	Unit	Quantity	Rate	Amount	Remarks
12	Brickworks: Brickwork(SN 01-2035) CLASS "A" in cement sand mortar in 1:4 super - structure with 1 st class bricks in perfect line and level including wetting the bricks, packing the joints and curing the work in all types and thickness of walls, column etc. as per instruction & specification complete work.	cum	32.67	14556.87	475572.94	
13	Brickworks: Chimney Brickwork(SN 01-2035) CLASS "A" in cement sand mortar in 1:6 sub - structure with 1 st class bricks in perfect line and level including wetting the bricks, packing the joints and curing the work in all types and thickness of walls, column etc. as per instruction & specification complete work.	cum	12.61	13601.65	171516.8	
			Sub-Total		6386986.16	
			VAT		830308.2008	
			Grand T		7544390.611	

Estimated By:

Checked By:

Approved By:



Sudurpashchim Province Government
Ministry of Physical Infrastructure Development
Water Supply and Sanitation Division Office
Dhangadhi, Kailali, Nepal

Project Name: Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (**WSSDO Kailali Office Building Project**)

Summary of Cost

S.No.	Description of work	Amount (Rs.)	Remarks
A	Works		
	Godam	6386986.16	
	Sub Total	6386986.16	
B	PROVISIONAL SUM		
1	Providing Insurance for Proposed Building, Materials, Third Party (People and Property), Employees and workers etc for complete construction and liability period.	165,499.55	
2	As Built Drawing	23,876.70	
3	Lab Test	102,720.00	
4	Project Information Board	5,000.00	
5	Health and Safety Equipment's	10,000.00	
	Total (B)	307,096.25	
C	Contingencies & work charge establishment 4% to be funded by NG	255,479.45	
D	Physical Contingencies (10% of A)	638,698.62	
E	Price Escalation (10% of A)	638,698.62	
	Total (B+C+D+E)	1,839,972.93	
F	VAT @ 13% (A+B)	870,230.71	
G	Grand Total (A+B+C+D+E+F)	9,404,286.05	ok

Estimated By:

Checked By:

Approved By:

 Sudurpashchim Province Government Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal											
Project Name:Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)											
S.N.	Description	No. of section	Dia (mm)	No. of Rebar	Length of single rebar	Develop ment	Lap length	Total Length (m)	Unit weight	Total quantity	Remark
	Isolated Footing(1F1)										
	Bottom Reinforcement										
	Horizontal	2	12	19	2.25	1.17	0.00	129.78	0.89	115.22	Kg
	Vertical	2	12	19	2.25	1.17	0.00	129.78	0.89	115.22	Kg
	Isolated Footing(1F2)										
	Bottom Reinforcement										
	Horizontal	6	12	17	2	1.17	0.00	322.85	0.89	286.63	Kg
	Vertical	6	12	17	2	1.17	0.00	322.85	0.89	286.63	Kg
	Isolated Footing(1F3)										
	Bottom Reinforcement										
	Horizontal	4	12	15	1.7	1.17	0.00	171.91	0.89	152.62	Kg
	Vertical	4	12	15	1.7	1.17	0.00	171.91	0.89	152.62	Kg
1	Lower Tie Beam										
	Horizontal										
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	Stirrups	3	8	108	0.77			248.18	0.39	97.93	Kg
	Vertical										
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	Stirrups	4	8	88	0.77			269.63	0.39	106.39	Kg

 Sudurpashchim Province Government Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal											
Project Name:Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)											
S.N.	Description	No. of section	Dia (mm)	No. of Rebar	Length of single rebar	Develop ment	Lap length	Total Length (m)	Unit weight	Total quantity	Remark
2	Upper Tie Beam										
	Horizontal										
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	Stirrups	3	8	108	1.63			527.47	0.39	208.13	Kg
	Vertical										
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	Stirrups	4	8	88	1.63			573.06	0.39	226.12	Kg
1	Ground Floor Beam										
	Horizontal										
	TH TOP & BOTTOM	5	16	6	19	0.78	1.60	641.30	1.58	1012.19	Kg
	EX	5	16	6	19	0.78	1.60	641.30	1.58	1012.19	Kg
	Stirrups	3	8	128	1.63			625.15	0.39	246.68	Kg
	Vertical										
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	EX	3	16	2	16	0.78	1.60	110.26	1.58	174.03	Kg
	Stirrups	3	8	108	1.63			527.47	0.39	208.13	Kg
7	Column										
a)	Ground Floor Column										
	C1	10	16	12	6.95	0.78	0.80	1023.21	1.58	1614.98	Kg
		2	16	12	6.05	0.78	0.80	183.04	1.58	288.90	Kg

 Sudurpashchim Province Government Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal											
Project Name:Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)											
S.N.	Description	No. of section	Dia (mm)	No. of Rebar	Length of single rebar	Develop ment	Lap length	Total Length (m)	Unit weight	Total quantity	Remark
	ties (outer)	12	2	48	2.08			1198.08	0.02	29.55	Kg
	ties (outer)	12	2	22	2.08			549.12	0.02	13.54	Kg
3	Slab										
i)	First Floor Slab										
	FOR VERTICAL BAR CALCULATION										
	Bottom Bar										
	Horizontal	1	10	129	19	0.49	1.00	2642.63	0.62	1629.28	Kg
	Vertical	1	8	153	16	0.39	0.80	2629.82	0.39	1037.69	Kg
	Top rebar										
	Horizontal	1	8	65	9.5	0.39	0.40	668.75	0.39	263.88	Kg
	Vertical	1	10	77	8	0.49	0.50	691.88	0.62	426.57	Kg
5	Sill/Lintel Band										
	Ground Floor										
	Total Length of wall	2	8	2	49.00		1.80	203.20	0.39	80.18	Kg
	Parapet wall	2	8	2	49.00		1.80	203.20	0.39	80.18	Kg
						Assuming 10% wastage:				1402.92	Kg
							Total			15432.08	Kg

 Sudurpashchim Province Government Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal											
Project Name:Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)											
S.N.	Description	No. of section	Dia (mm)	No. of Rebar	Length of single rebar	Develop ment	Lap length	Total Length (m)	Unit weight	Total quantity	Remark
	Isolated Footing(IF1)										
	Bottom Reinforcement										
	Horizontal	2	12	19	2.25	1.17	0.00	129.78	0.89	115.22	Kg
	Vertical	2	12	19	2.25	1.17	0.00	129.78	0.89	115.22	Kg
	Isolated Footing(IF2)										
	Bottom Reinforcement										
	Horizontal	6	12	17	2	1.17	0.00	322.85	0.89	286.63	Kg
	Vertical	6	12	17	2	1.17	0.00	322.85	0.89	286.63	Kg
	Isolated Footing(IF3)										
	Bottom Reinforcement										
	Horizontal	4	12	15	1.7	1.17	0.00	171.91	0.89	152.62	Kg
	Vertical	4	12	15	1.7	1.17	0.00	171.91	0.89	152.62	Kg
1	Lower Tie Beam										
	Horizontal										
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	Stirrups	3	8	108	0.77			248.18	0.39	97.93	Kg
	Vertical										
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	Stirrups	4	8	88	0.77			269.63	0.39	106.39	Kg

S.N.	Description	No. of section	Dia (mm)	No. of Rebar	Length of single rebar	Development	Lap length	Total Length (m)	Unit weight	Total quantity	Remark
2	Upper Tie Beam										
	Horizontal										
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	Stirrups	3	8	108	1.63			527.47	0.39	208.13	Kg
	Vertical										
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	TH TOP & BOTTOM	4	16	4	13	0.78	1.60	246.03	1.58	388.32	Kg
	Stirrups	4	8	88	1.63			573.06	0.39	226.12	Kg
3	Ground Floor Beam										
	Horizontal										
	TH TOP & BOTTOM	5	16	6	19	0.78	1.60	641.30	1.58	1012.19	Kg
	EX	5	16	6	19	0.78	1.60	641.30	1.58	1012.19	Kg
	Stirrups	3	8	128	1.63			625.15	0.39	246.68	Kg
	Vertical										
	TH TOP & BOTTOM	3	16	6	16	0.78	1.60	330.78	1.58	522.09	Kg
	EX	3	16	2	16	0.78	1.60	110.26	1.58	174.03	Kg
	Stirrups	3	8	108	1.63			527.47	0.39	208.13	Kg

S.N.	Description	No. of section	Dia (mm)	No. of Rebar	Length of single rebar	Development	Lap length	Total Length (m)	Unit weight	Total quantity	Remark
4	Column										
a)	Ground Floor Column										
	C1	10	16	12	6.95	0.78	0.80	1023.21	1.58	1614.98	Kg
		2	16	12	6.05	0.78	0.80	183.04	1.58	288.90	Kg
	ties (outer)	12	2	48	2.08			1198.08	0.02	29.55	Kg
	ties (outer)	12	2	22	2.08			549.12	0.02	13.54	Kg
5	Slab										
i)	First Floor Slab										
	FOR VERTICAL BAR CALCULATION										
	Bottom Bar										
	Horizontal	1	10	129	19	0.49	1.00	2642.63	0.62	1629.28	Kg
	Vertical	1	8	153	16	0.39	0.80	2629.82	0.39	1037.69	Kg
	Top rebar										
	Horizontal	1	8	65	9.5	0.39	0.40	668.75	0.39	263.88	Kg
	Vertical	1	10	77	8	0.49	0.50	691.88	0.62	426.57	Kg
6	Sill/Lintel Band										
	Ground Floor										
	Total Length of wall	2	8	2	49.00		1.80	203.20	0.39	80.18	Kg
	Parapet wall	2	8	2	49.00		1.80	203.20	0.39	80.18	Kg
						Assuming 10% wastage:				1402.92	Kg
						Total				15432.08	Kg

 Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali, Nepal						
Project Name: Procurement of Consulting Service for Preparation of Master Plan, Detailed Architectural and Engineering Design of Office Building of WSSDO, Kailali (WSSDO Kailali Office Building Project)						
LAB TEST						
S.N.	Description of Work	Quantity	Unit	Rate	Amount	Remark
1	Concrete Test					
a	Cubes or cylinder crushing					
i	Design Mix sampling and test	6.00				
ii	Foundation					
	2 days @ 6 cubes per day	6.00				
iii	Tie and Footing Beam					
	1 days @ 3 cubes per day	6.00				
iv	Column					
	2 days @ 3 cubes per day per floor					
	For all Five floor	6.00				
v	Slab and beam					
	2 days @ 3 cubes per day per floor					
	For all Five floor	6.00				
		6.00	Nos	150.00	900.00	
2	Brick Test					
a	Brick Crushing Test	6.00	Nos	1100.00	6600.00	
b	Water Absortion Test	6.00	Nos	700.00	4200.00	
3	Aggregate Test					
a	Aggregate Crushing Value Test	6.00	Nos	420.00	2520.00	
b	Aggregate Impact Value	6.00	Nos	320.00	1920.00	
4	Sieve Analysis					
a	Sieve Analysis (Wet)	6.00	Nos	840.00	5040.00	
b	Sieve Analysis (Dry)	6.00	Nos	840.00	5040.00	
5	U.T.M. Tensile Strength Test , Bend Test, Rebend Test and Bond Test		Nos			
a	8mm Ø	6.00	Nos	2800.00	16800.00	
b	10mm Ø	6.00	Nos	2800.00	16800.00	
c	12mm Ø	6.00	Nos	2800.00	16800.00	
d	16mm Ø	6.00	Nos	2800.00	16800.00	
6	Cement Test					
a	Cement mortar cube Crushing 3 Nos.	6.00	Nos	650.00	3900.00	
b	Intial setting & Final Setting	6.00	Nos	900.00	5400.00	
					102720.00	



भौतिक पूर्वाधार विकास मन्त्रालय

म.ले.प.फारम न. ५०३

खानेपानी तथा सरसफाई डिभिजन कार्यालय, कैलाली

कार्यालय कोड नं. : ३३७०५७७०१७

नापी किताब (Measurement Book)

नापी नं:

समझौता विवरणः

आयोजनाको नाम:- बसौटि खानेपानी योजना, कैलारी गा -८, कैलाली ।

सामितिकोको नामः

समितिकोको नामः श्री बसौटि खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,

सम्प्रति नं: 2082/083-01, WSSDO, kailali

कार्यादिश मिति:-

पाइपलाइन , भल्म च्याम्बर

कार्य सरु गर्नपने मिति:-

आयोजना कार्य स्थल: कैलारी गा .पा. -८, कैलाली

कार्य सम्पन्न गर्नुपर्ने मिति:-

कार्य सम्पन्न भएको मिति:-

नापी लिङ्गएको मिति :-

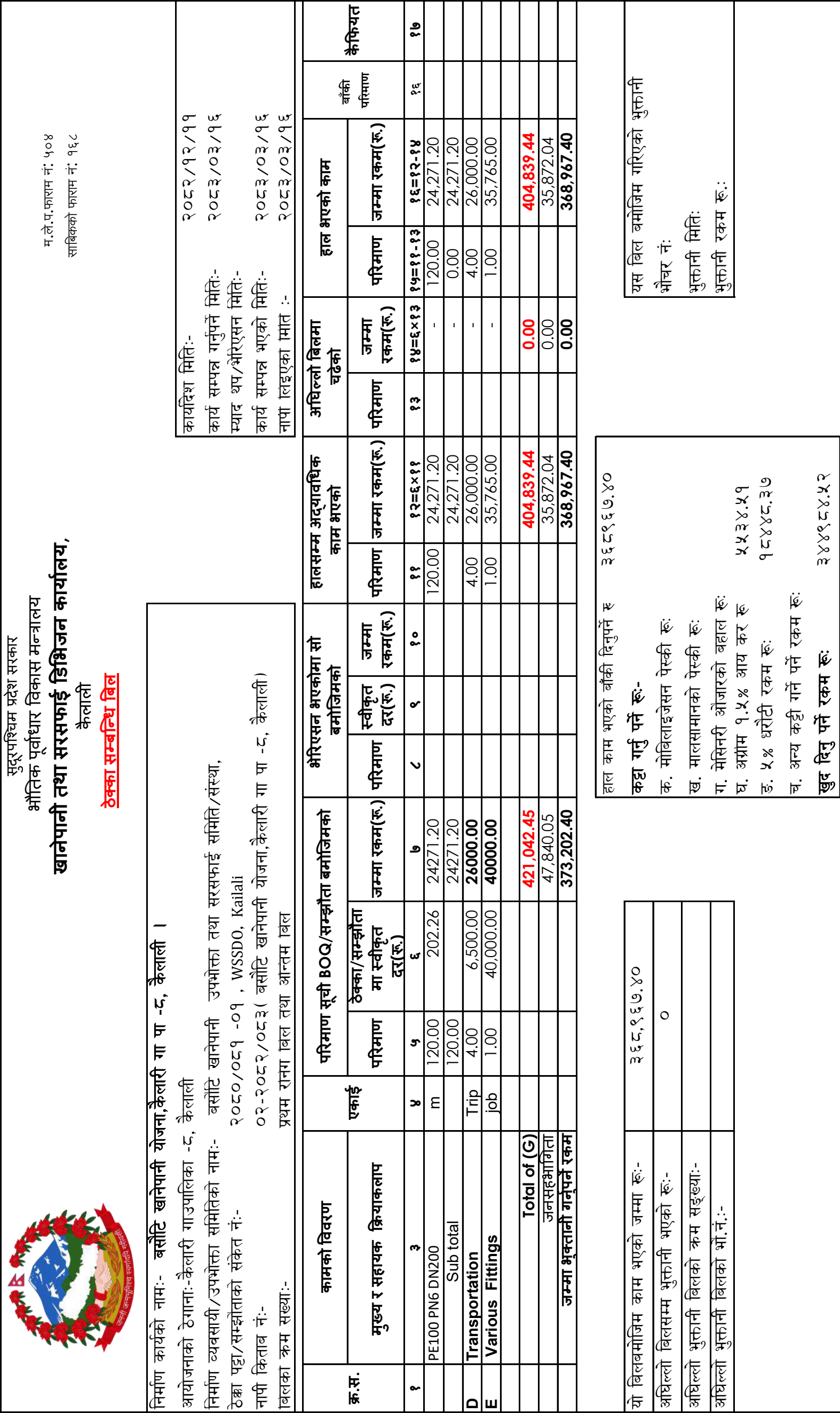
36/30/3202

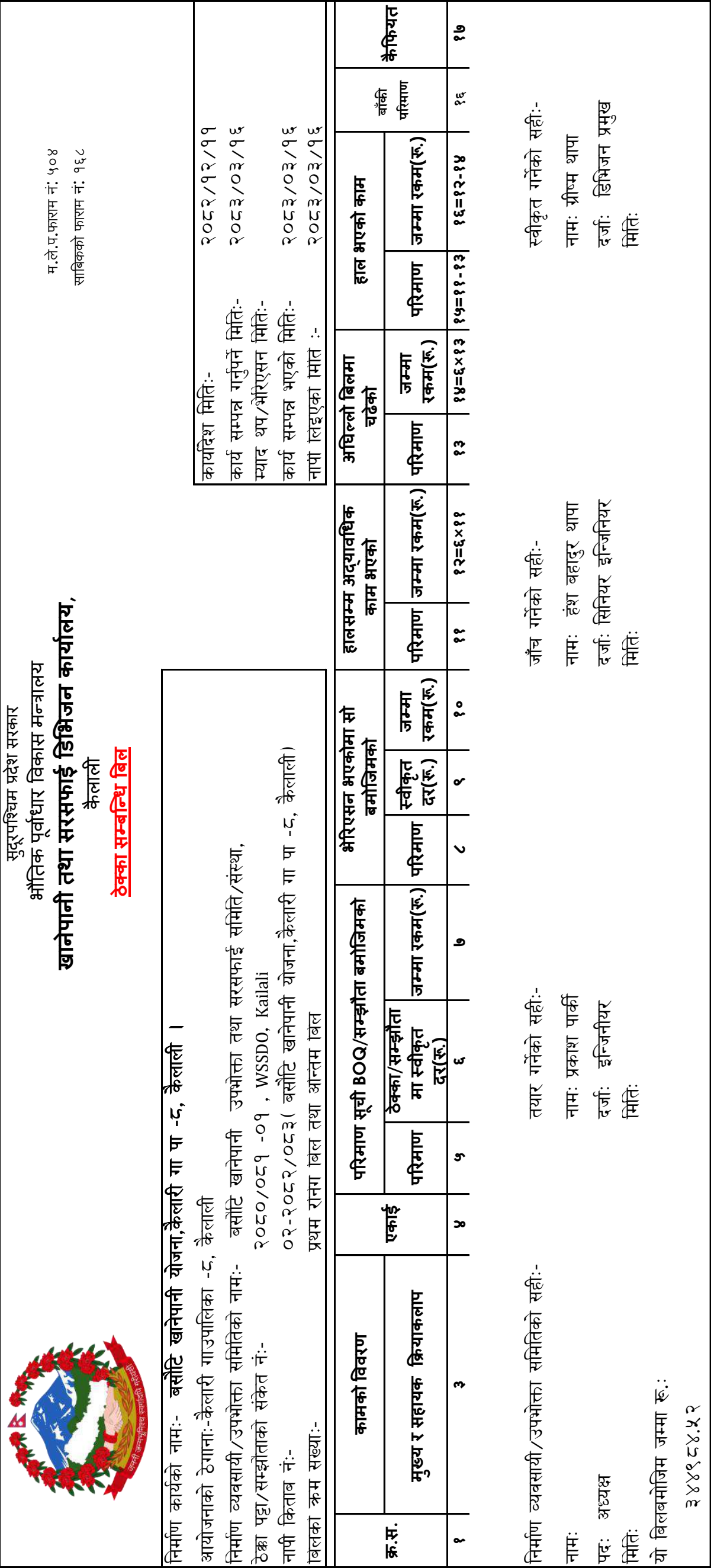
Details of measurements, calculations of quantities and Payment details are as follows:

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क्र.सं.	काम/सेवाको विवरण/क्रियाकलाप	अघिल्लो नापी किताबमा चढेको परिमाण	यस नापीको मापनको आयाम					हालसम्म अध्ययवधिक काम भएको परिमाण	ईकाइ	परिमाण सूची (BoQ) मा उल्लिखित दर	जम्मा रकम
			संख्या	लम्बाइ	चौडाइ	उचाइ	यस नापीबाट कायम भएको परिमाण				
१	२	३	४	५	६	७	८	९	१०	११	१२=९*११
	Inner Side		4	1.9		0.9	6.84				
	top		1	8		0.15	1.20				
	Total						8.04	8.04	m ²	362.28	2912.76
8	Supply and installation Fittings of VC		1				1.00	1	job	5000.00	5000.00
	Sub Total For No.s									Rs.	45112.96
	Sub Total for 6 No.s									Rs.	270677.77
B	Earthworks										
1	E/W in Excavation for pipe trench in Ordinary soil			120.00	0.60	0.72	51.84				
							51.84	51.84	m ³	777.65	40313.38
2	E/W in backfilling for pipe trench in ordinary soil (75 % of E W)			120.00			38.88	38.88	m ³	194.41	7558.66
	Sub total									Rs.	47872.04
C	Laying & Jointing of pipe										
	HDPE Pipe										
	PE100 PN6 DN200	120						120.00	m	202.26	24271.20
	Sub total	120								Rs.	24271.20
D	Transportation Of pipe and Fittings										
	Fittings and pipe From Office To site										
E	Fittings			4.00				4.00	Trip	6500.00	26000.00
	Die set 1/2* 1	1	set	3836.00	3836.00						35765.00
	Pipe wrench 12";	1	pc	690.00	690.00						404586.00
	Pipe wrench 14 ";	1	pc	690.00	690.00						35872.04
	Pipe wrench 18";	1	pc	849.00	849.00						368713.97
	HDPE HDPE flange Set 200mm	3	set	6000.00	18000.00						0.00
	Gasket Rubber	12	kg	350.00	4200.00						368713.97
	Nut Volts	30	kg	250.00	7500.00						
					35765.00						18435.70
	क. जम्मा रकम										350278.27
	ख. जनसहभागिता										
	ग. जम्मा भुक्तानी गर्नुपर्ने रकम (क-ख)										
	घ. अघिल्लो नापिबाट भुक्तानी भएको रकम										
	ङ. बाँकि भुक्तानी गर्नुपर्ने रकम (ग-घ)										
	घटाउनु पर्ने										
	च. ५% धरोटी रकम										
	छ. खुद भुक्तानी गर्नुपर्ने रकम (ङ-च)										

 सुदूरपश्चिम प्रदेश सरकार भौतिक पूर्वाधार विकास मन्त्रालय खानेपानी तथा सरसफाई डिभिजन कार्यालय, कैलाली ठेक्का सम्बन्धि बिल										म.ले.प.फासम नं: ५०४ साबिकको फासम नं: १६८
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सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय
कैलाली

म.ले.प.फा.सम नं: ५१०
साबिकको फा.सम नं: १७६

कार्य स्वीकार / सम्पन्न प्रतिवेदन

आयोजनाको नाम/कार्य स्थल:-बसौटी खानेपानी योजना,कैलारी गा पा -८, कैलाली ।	
कामको नाम :-	Paintning works ,Pipeline Excavation, Backfilling, Laying & Jointing Work.
उपभोक्ता समितिको नाम:-	बसौटी खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,
लागत अनुमानको कुल रकम:-	रु. ४ र१ ०४२.४५
खर्च भएको कुल रकम:-	रु. ४ ०४ ८३९.४४
सुरु लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-	१.०१ %
भरिएसपछिको कुल लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-	

कार्यदिश मिति:-	२०८२/१२/११
कार्य सम्पन्न गनुपने मिति:-	२०८३/०३/१६
म्याद थप/भरिएसप मिति:-	
कार्य सम्पन्न भएको मिति:-	२०८३/०३/१६
नापी लिइएको मिति :-	

क्र.सं	कामको विवरण	एकाई	सुरु लागत अनुमान वमोजिमको			भरिएसन भएकोमा सोवमोजिमको कुल लागत अनुमान			खुद भएको काम				फरक		कैफियत
			परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	
१	२	३	४	५	६	७	८	९	१०	११	१२	१६=७-१०	१७=८-११	१८=९-१२	१९
A Construction of Valve Chamber															
1	Earthwork in excavation in ordinary soil	m3	4.00	777.65	3110.60				4.00	777.65	3110.60	0.00	0.00	0.00	
2	Flat brick soling with sand packing	m2	3.61	823.78	2973.85				3.61	823.78	2973.85	0.00	0.00	0.00	
3	PCC in (1:2:4)	m3	0.65	11747.05	7635.58				0.65	11747.05	7635.58	0.00	0.00	0.00	
4	Brick masonry in (1:6) cement mortar	m3	1.39	12707.18	17662.98				1.39	12707.18	17662.98	0.00	0.00	0.00	
5	Reinforcement for slab	Kgs	23.12	127.41	2945.72				23.12	127.41	2945.72	0.00	0.00	0.00	
6	Shuttering work for RCC	m2	4.37	666.76	2913.74				4.37	666.76	2913.74	0.00	0.00	0.00	
7	12 mm thick cement Plaster (1:4)	m2	8.04	362.28	2912.73				8.04	362.28	2912.73	0.00	0.00	0.00	
8	Supply and installation Fittings of VC	job	1.00	5000.00	5000.00				1.00	5000.00	5000.00	0.00	0.00	0.00	
	Sub Total for 1 No.s				45155.20						45155.20	0.00	0.00	0.00	
	Sub Total for 9 No.s				270931.20						270931.20			0.00	
B Earthworks															
1	E/W in Excavation for pipe trench in Ordinary soil	m3	64.80	777.65	50391.72				51.84	777.65	40313.38	12.96	0.00	10078.34	
2	E/W in backfilling for pipe trench in ordinary soil (75 % of E W)	m3	48.60	194.41	9448.33				38.88	194.41	7558.66	9.72	0.00	1889.67	
					59840.05						47872.04			11968.01	



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय
कैलाली

म.ले.प.फारम नं: ५१०
साबिकको फारम नं: १७६

कार्य स्वीकार / सम्पन्न प्रतिवेदन

आयोजनाको नाम/कार्य स्थल:-बसौटी खानेपानी योजना,कैलारी गा पा -८, कैलाली ।	
कामको नाम :-	Paintning works ,Pipeline Excavation, Backfilling, Laying & Jointing Work.
उपभोक्ता समितिको नाम:-	बसौटी खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,
लागत अनुमानको कुल रकम:-	रु. ४ २१ ०४२.४५
खर्च भएको कुल रकम:-	रु. ४ ०४ ८३९.४४
सुरु लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-	१.०१ %
भेरिएसनपछिको कुल लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-	

कार्यदिश मिति:-	२०८२/१२/११
कार्य सम्पन्न गनुपने मिति:-	२०८३/०३/१६
म्याद थप/भेरिएसन मिति:-	
कार्य सम्पन्न भएको मिति:-	२०८३/०३/१६
नापी लिइएको मिति :-	

क्र.सं	कामको विवरण	एकाई	सुरु लागत अनुमान वमोजिमको			भेरिएसन भएकोमा सोवमोजिमको कुल लागत अनुमान			खुद भएको काम			फरक		कैफियत
			परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	दर(रु.)	जम्मा रकम(रु.)	
१	२	३	४	५	६	७	८	९	१०	११	१२	१७=८-११	१८=९-१२	१९
C Laying & Jointing of pipe														
HDPE Pipe														
	PE100 PN6 DN200 /225	m	120	202.26	24271.2				120.00	202.26	24271.20	0.00	0.00	
	Sub total				24271.20						24271.20	0.00	0.00	
D	Transportation	trip	4.00	6500.00	26000.00				4.00	6500.00	26000.00	0.00	0.00	
E	Fittings	job	1.00	40000.00	40000.00				1.00	35765.00	35765.00	1.00	4235.00	
	जम्मा रकम (A+.....)				421,042.45						404,839.44			16,203.01
	जनसहभागिता				47 840.05						35,872.04			11,968.01
	नेपाल सरकार/प्रदेश सरकार				373,202.40						368,967.40			4,235.00

क) स्वीकृत लागत अनुमान (इष्टिमेट) भन्दा घटि(-) बढी(+) रु.	-16203.01	प्रतिशत (%)	1.01 %
घटी/बढीको पुस्त्याई:-	१.	उपर्युक्त बमोजिम भएको कार्य सुरु सम्झौताको स्वीकृत डिजाइन र स्पेसिफिकेसन बमोजिम भएको छ । भेरिएशन आदेश र सर्तहरूबमोजिम भएको हुँदा उक्त कार्य सम्पन्न भएको स्वीकार गरिएको छ ।	
	२.		
स्वीकृत लागत अनुमान, डिजाइन र स्पेसिफिकेसनअनुसार काम भएको छ भनी सिफारिस गर्ने:			
तयार गर्ने:-	जाँच गर्ने:-	स्वीकृत गर्ने:-	
सही:	सही:	सही:	
नाम:	नाम:	नाम:	



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय
कैलाली

म.ले.प.फारम नं: ५१०
साबिकको फारम नं: १७६

कार्य स्वीकार / सम्पन्न प्रतिवेदन

आयोजनाको नाम/कार्य स्थल:-बसौटी खानेपानी योजना,कैलारी गा पा -८, कैलाली ।	
कामको नाम :-	Painting works ,Pipeline Excavation, Backfilling, Laying & Jointing Work.
उपभोक्ता समितिको नाम:-	बसौटी खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,
लागत अनुमानको कुल रकम:-	रु. ४ २१ ०४२.४५
खर्च भएको कुल रकम:-	रु. ४ ०४ ८३९.४४
सुरु लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-	१.०१ %
भरिएसपछिको कुल लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-	

कार्यदिश मिति:-	२०८२/१२/११
कार्य सम्पन्न गर्नुपर्ने मिति:-	२०८३/०३/१६
म्याद थप/भरिएसप मिति:-	
कार्य सम्पन्न भएको मिति:-	२०८३/०३/१६
नापी लिइएको मिति :-	

क्र.सं	कामको विवरण	एकाई	सुरु लागत अनुमान वमोजिमको			भरिएसप भएकोमा सोवमोजिमको कुल लागत अनुमान			खुद भएको काम			फरक			कैफियत
			परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	
१	२	३	४	५	६	७	८	९	१०	११	१२	१६=७-१०	१७=८-११	१८=९-१२	१९
दर्जा:	दर्जा:					दर्जा:					दर्जा:				
मिति:	मिति:					मिति:					मिति:				



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय
कैलाली

Cost Estimate for Valve Chamber Construction, Pipeline Works, Transportation, and Associated Items – Basauti Water Supply Project

fy 082/083

क्र.सं.	काम/सेवाको विवरण/क्रियाकलाप	अघिल्लो नापी कित्ताबमा चढेको परिमाण	यस नापीको मापनको आयाम				हालसम्म अध्ययनवधिक काम भएको परिमाण	ईकाइ	परिमाण सूची (BoQ) मा उल्लिखित दर	जम्मा रकम
			सँख्या	लम्बाइ	चौडाइ	उचाइ				
१	२	३	४	५	६	७	८	९	११	१२=९*११
A	Construction of Valve Chamber									
1	Earthwork in excavation in ordinary soil		1	2	2	1	4.00	4	777.65	3110.60
2	Flat brick soling with sand packing		1	1.9	1.9		3.61	3.61	823.78	2973.85
3	PCC in (1:2:4)									
	On the Floor		1	1.9	1.9	0.08	0.29			
	For RCC slab		1	1.9	1.9	0.1	0.36			
	Total						0.65	0.6498	11747.05	7633.23
4	Brick masonry in (1:6) cement mortar									
	Long Wall		2	1.9	0.23	0.9	0.79			
	Short Wall		2	1.45	0.23	0.9	0.60			
	Total						1.39	1.387	12707.18	17623.59
5	Reinforcement for slab									
	Both span 10 mm dia bar 0.12 mm c/c		13x2x0.62				19.84			
	stirups		12x0.7x0.39				3.28			
	Total						23.12	23.116	127.41	2945.21
6	Shuttering work for RCC									
	Bottom		1	1.9	1.9		3.61			
	Sides		4	1.9		0.1	0.76			
	Total						4.37	4.37	666.76	2913.72
7	12 mm thick cement Plaster (1:4)									



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय
केलाली

Cost Estimate for Valve Chamber Construction, Pipeline Works, Transportation, and Associated Items – Basauti Water Supply Project

fy 082/083

क्र.सं.	काम/सेवाको विवरण/क्रियाकलाप	अचिल्लो नापी कित्ताबमा चढेको परिमाण	यस नापीको मापनको आयाम				हालसम्म अध्ययनवधिक काम भएको परिमाण	ईकाइ	परिमाण सूची (BoQ) मा उल्लिखित दर	जम्मा रकम
			सँख्या	लम्बाइ	चौडाइ	उचाइ	यस नापीबाट कायम भएको परिमाण			
१	२	३	४	५	६	७	८	१०	११	१२=९*११
	Inner Side		4	1.9		0.9	6.84			
	top		1	8		0.15	1.20			
	Total						8.04	m ²	362.28	2912.76
8	Supply and installation Fittings of VC	1					1.00	job	5000.00	5000.00
	Sub Total For No.s								Rs.	45112.96
	Sub Total for 6 No.s								Rs.	270677.77
B	Earthworks									
1	E/W in Excavation for pipe trench in Ordinary soil			120.00	0.60	0.90	64.80			
							64.80	m ³	777.65	50391.72
2	E/W in backfilling for pipe trench in ordinary soil (75 % of E W)			120.00			48.60	m ³	194.41	9448.33
	Sub total								Rs.	59840.05
C	Laying & Jointing of pipe									
	HDPE Pipe									
	PE100 PN6 DN200	120								
	Sub total	120						m	202.26	24271.20
									Rs.	24271.20
D	Transportation Of pipe and Fittings									
	Fittings and pipe From Office To site			4.00				Trip	6500.00	26000.00
E	Fittings									
					40000.00					40000.00
	क. जम्मा रकम									420789.01
	ख. जनसहभागिता									47840.05
	ग. जम्मा भुक्तान									372948.97

प्रमाणित गर्नेको सही:-
नाम:-बीष्म थापा
पद:- डिभिजन प्रमुख
मिति:-

जाँच गर्नेको सही:-
नाम:- हंश बहादुर थापा
पद:- सिनियर इन्जिनियर
मिति:-

नापी लिनेको सही:-
नाम:-प्रकाश पार्की
पद:- इन्जिनियर



Government of Sudurpashchim Province
Ministry of Physical Infrastructure Development
Water Supply and Sanitation Division Office
Kailali

cost Estimate

Name of Project : Basauti Loharpur Water Supply Project
Name of Work : Distribution Pipeline Work and transportation

S.N.	Description of Items	No.	Length (m)	Breadth (m)	Height (m)	Quantity	Unit	Rate	Amount
A	Earthwork								
1	Earthwork in excavation in ordinary soil for pipeline trench	1	6388	0.45	0.7	2012.22	m ³	777.65	1564802.88
2	Refilling with ordinary soil		75% of Excavation			1509.17	m ³	194.41	293396.77
	Sub Total								1858199.65
B	Laying and Jointing Work								
	HDPE Pipe								
	PE100 PN6 DN110		90			90.00	m	100.47	9042.30
	PE100 PN6 DN90		1198			1198.00	m	67.42	80769.16
	PE100 PN6 DN75		1700			1700.00	m	67.42	114614.00
	PE100 PN10 DN63		900			900.00	m	5.01	4509.00
	PE100 PN10 DN50		900			900.00	m	5.01	4509.00
	PE100 PN10 DN40		1000			1000.00	m	5.01	5010.00
	PE100 PN10 DN32		600			600.00	m	4.09	2454.00
			6388			6388.00			
	Sub Total								220907.46
	Total(A+B)							Rs.	2164329.50
	Government Contribution							Rs.	914316.87
	Community Contribution							Rs.	1250012.64

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prepared By

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Checked By

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Approved By



Government of Sudurpashchim Province
Ministry of Physical Infrastructure Development
Water Supply and Sanitation Division Office
Kailali

Name of project: Basauti watersupply project

Estimate For Transportation Of Pipe

1	HDPE Pipe	Length	Unit Weight	Weight		Rate	Amount
	PE100 PN10 DN50	500	@ 0.506 kg/m	253			
	PE100 PN10 DN40	1000	@ 0.328 kg/m	328			
	PE100 PN10 DN32	1000	@ 0.210 kg/m	210			
	Total			791.00	Kg		
	Total			1.00	Trip	5957.33	5957.33
	VAT 13%						774.45
	Total amount						6731.78

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prepared By

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Checked By

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Approved By



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय
कैलाली

म.ले.प.फारम नं: ५१०
साबिकको फारम नं: १७६

कार्य स्वीकार / सम्पन्न प्रतिवेदन

आयोजनाको नाम/कार्य स्थल:-हसुलिया खानेपानी योजना,कैलारी गा पा -३, कैलाली ।
कामको नाम :-
उपभोक्ता समितिको नाम:-
लागत अनुमानको कुल रकम:-
खर्च भएको कुल रकम:-
सुरु लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-
भरिएसनपछिको कुल लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-

Painting works ,Pipeline Excavation, Backfilling, Laying & Jointing Work.

हसुलिया खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,

रु. २५ द६ ९दद.३६

रु. २४ ९४ ५९९.५५

२.३३ %

-३.७० %

कार्यादेश मिति:-			२०८२/०४/२१
कार्य सम्पन्न गर्नुपर्ने मिति:-			२०८३/०१/१५
म्याद थप/भरिएसन मिति:-			२०८२/०३/२५
कार्य सम्पन्न भएको मिति:-			२०८३/०३/०५
नापी लिइएको मिति :-			२०८३/०३/०५

क्र.सं	कामको विवरण	एकाई	सुरु लागत अनुमान वर्मोजिमको				भरिएसन भएकोमा सोमोजिमको कुल लागत अनुमान				खुद भएको काम				फरक				कैफियत
			परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)		
१	२	३	४	५	६	७	८	९	१०	११	१२	१६=७-१०	१७=८-११	१८=९-१२	१९				
A Paiting Works																			
1	Compound wall	Sqm	529.50	526.87	278977.67				529.50	526.87	278,977.67	0.00	0.00	0.01					
2	Toilet and Bathroom	Sqm	52.13	526.87	27465.73				52.13	526.87	27,465.73	0.00	0.00	0.00					
3	Pump House	Sqm	125.06	526.87	65890.36				125.06	526.87	65,890.36	0.00	0.00	0.00					
	Enamel Paint Works in Doors & Windows	Sqm	5.94	308.84	1834.51				5.94	308.84	1,834.51	0.00	0.00	0.00					
4	Office Building	Sqm	115.47	526.87	60837.68				115.47	526.87	60,837.68	0.00	0.00	0.00					
	Enamel Paint Works in Doors & Windows	Sqm	13.89	308.84	4289.79				13.89	308.84	4,289.79	0.00	0.00	0.00					
5	OHT 225 m3	Sqm	673.20	526.87	354688.88				673.20	526.87	354,688.88	0.00	0.00	0.00					
6	Name writing of Project with materials	job	1.00	5000.00	5000.00				1.00	5000.00	5,000.00			0.00			0.00		
	Total				798984.62						798,984.62			0.00			0.00		
B Construction of Valve Chamber																			
1	Earthwork in excavation in ordinary soil	m3	2.10	777.65	1633.07				2.10	777.65	1,633.07	0.00	0.00	0.00					
2	Flat brick soling with sand packing	m2	2.10	823.78	1729.94				2.10	823.78	1,729.94	0.00	0.00	0.00					
3	PCC in (1:2:4)	m3	0.30	11747.05	3524.12				0.30	11747.05	3,524.12	0.00	0.00	0.01					
4	Brick masonry in (1:6) cement mortar	m3	1.12	12707.18	14232.04				1.12	12707.18	14,232.04	0.00	0.00	0.00					
5	Reinforcement for slab	Kgs	22.33	127.41	2845.07				22.33	127.41	2,845.07	0.00	0.00	0.00					
6	Shuttering work for RCC	m2	2.73	666.76	1820.25				2.73	666.76	1,820.25	0.00	0.00	0.00					
7	12 mm thick cement Plaster (1:4)	m2	7.2	362.28	2608.42				7.2	362.28	2,608.42	0.00	0.00	0.00					
8	Supply and installation Fittings of VC	job	1.00	5000.00	5000.00				1.00	5000.00	5,000.00	0.00	0.00	0.00					
	Sub Total for 1 No.s				33392.91						33,392.90	0.00	0.00	0.01					
	Sub Total for 9 No.s				300536.19						300,536.06			0.13					
C Earthworks																			
1	E/W in Excavation for pipe trench in O	m3	1260.00	777.65	979839.00				1160.17	777.65	902,206.20	99.83	0.00	77632.80					
2	E/W in backfilling for pipe trench in ord	m3	945.00	194.41	183717.45				870.13	194.41	169,161.97	74.87	0.00	14555.48					
					1163556.45						1,071,368.17			92188.28					
C Laying & Jointing of pipe																			
	HDPE Pipe																		



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय
कैलाली

म.ले.प.फारम नं: ५१०
साबिकको फारम नं: १७६

कार्य स्वीकार / सम्पन्न प्रतिवेदन

आयोजनाको नाम/कार्य स्थल:-हसुलिया खानेपानी योजना,कैलारी गा पा -३, कैलाली ।

कामको नाम :-

उपभोक्ता समितिको नाम:-

लागत अनुमानको कुल रकम:-

खर्च भएको कुल रकम:-

सुरु लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-

भरिएसनपछिको कुल लागत अनुमान भन्दा घटी(-) बढी(+)को प्रतिशत:-

Painting works ,Pipeline Excavation, Backfilling, Laying & Jointing Work.

हसुलिया खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,

रु. २५ द६ ९दद.३६

रु. २४ ९४ ५९९.५५

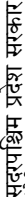
२.३३ %

-३.७० %

कार्यादेश मिति:-			२०८२/०४/२१
कार्य सम्पन्न गनुपर्ने मिति:-			२०८३/०१/१५
म्याद थप/भरिएसन मिति:-			२०८२/०३/२५
कार्य सम्पन्न भएको मिति:-			२०८३/०३/०५
नापी लिइएको मिति :-			२०८३/०३/०५

क्र.सं	कामको विवरण	एकाई	सुरु लागत अनुमान वर्माजिमको			भरिएसन भएकोमा सोवमोजिमको कुल लागत अनुमान			खुद भएको काम				फरक		कैफियत		
			परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)	परिमाण	दर(रु.)	जम्मा रकम(रु.)			
१	२	३	४	५	६	७	८	९	१०	११	१२	१३=७-१०	१४=८-११	जम्मा रकम(रु.)			
	PE100 PN6 DN200 /225	m	1440	202.26	291254.4				1440	202.26	291,254.40	0.00	0.00	१८=९-१२	१९		
	PE100 PN6 DN110	m	110	100.47	11051.7	110									0.00	0.00	
	PE100 PN6 DN50	m	500.00	5.01	2505.00				500.00	5.01	2,505.00	0.00	0.00	0.00			
	PE100 PN10 DN40	m	1000.00	5.01	5010.00				1000.00	5.01	5,010.00	0.00	0.00	0.00			
	PE100 PN10 DN32	m	1000.00	4.09	4090.00				951.00	4.09	3,889.59	49.00	0.00	200.41			
	Sub total				313911.10						313,710.69	0.00	0.00	200.41			
D	Transportation	trip	2.00	5000.00	10000.00				2.00	5000.00	10,000.00	0.00	0.00				
	जम्मा रकम (A+.....)				2,586,988.36						2,494,599.55			92,388.81			
	जनसहभागिता				758,556.45						726,443.18			32,113.27			
	नेपाल सरकार/प्रदेश सरकार				1,828,431.91						1,768,156.37			60,275.54			

क) स्वीकृत लागत अनुमान (इष्टिमेट) भन्दा घटि(-) बढी(+) खर्च रु.	-९२३८८.८१	प्रतिशत (%)	२.३३ %
घटी/बढीको पुस्त्याई:-	१.		
	२.		
स्वीकृत लागत अनुमान, डिजाइन र स्पेसिफिकेसनअन्सार काम भएको छ भनी सिफारिस गर्ने:			
तयार गर्ने:-	जाँच गर्ने:-		
सही:	सही:		
नाम:	नाम:		
दर्जा:	दर्जा:		
मिति:	मिति:		



भौतिक पूर्वाधार विकास मन्त्रालय

म. ले. प. फारम न. ५०३

भौतिक पूर्वाधार विकास मन्त्रालय
खाद्योपानी तथा सरसफाई डिभिजन कार्यालय, कैलाली
कार्यालय कोड नं. : ३३७०५७७०१७
नापी किताब (Measurement Book)

नापी नं:

आयोजनाको नाम:- हसलिया खानेपानी योजना, कैलारी गा. पा. -०३, कैलाली ।

सम्झौता नं: 2082/083-01, WSSDO.kailali

Painting works and pipeline

आयोजना कार्य स्थल: कैलारी गा पा. -०३ कैलाली

कार्यदिशा मिति:-	२०८२/०४/२१
कार्य सुरु गर्नुपर्ने मिति:-	२०८३/०१/१५
कार्य सम्पन्न गर्नुपर्ने मिति:-	१०-०१-२०८३
कार्य सम्पन्न भएको मिति:-	२०८३/०३/०५

Details of measurements, calculations of quantities and Payment details are as b

क्र.सं.	काम/सेवाको विवरण/क्रियाकलाप	अघिल्लो नापी किताबमा चढेको परिमाण	यस नापीको मापनको आयाम						हालसम्म अद्यावधिक काम भएको परिमाण	ईकाइ	परिमाण सूची (BoQ) मा उल्लिखित दर	जम्मा रकम
			सँख्या	लम्बाइ	चौडाइ	उचाइ	यस नापीबाट कायम भएको परिमाण					
१	२	३	४	५	६	७	८	९	१०	११	१२=९*११	
1	Compound wall											
	No.1											
	Two Coat Weather Coat	Painting over primer										
	Wall above GL		2	112		1.4	313.60					
	Top of wall		1	112	0.23		25.76					
	Gate pillar faces		3	1.6		1.4	6.72					
	Top of Gate pillars		3	0.4	0.4		0.48					
	No.2											
	Two Coat Weather Coat	Painting over primer										
	Wall above GL		2	58		1.4	162.40					
2	Top of wall		1	58	0.23		13.34					
	Gate pillar faces		3	1.6		1.4	6.72					
	Top of Gate pillars		3	0.4	0.4		0.48		Sqm	526.87	278977.67	
	Total amount					Total	529.50	529.50				
	Toilet and Bathroom											
	Outer											
	Inner		1	11	2.5		27.50					
	chajja		2	4.8	2.5		24.00					
	Deduction :	D	1	11	0.45		4.95					
		V	-2	2.1	0.9		-3.78					
3			-2	0.6		0.45	-0.54					
						Net quantity	52.13	52.13	Sqm	526.87	27465.73	
	Total amount											
	Pump House											
	Outer		1	23	2.9		66.70					
	Chajja		1	23	0.6		13.80					
	R1		1	13.6	2.9		39.44					
	Ceiling		1	2.9	3.9		11.31					
						Total	131.25	131.25				

क्र.सं.	काम/सेवाको विवरण/क्रियाकलाप	अघिल्लो नापी किताबमा चढेको परिमाण	यस नापीको मापनको आयाम					हालसम्म अध्यवधिक काम भएको परिमाण	ईकाइ	परिमाण सूची (BoQ) मा उल्लिखित दर	जम्मा रकम
१	२	३	सँख्या	लम्बाइ	चौडाइ	उचाइ	यस नापीबाट कायम भएको परिमाण	४	५	११	१२=१११
	Deduction :										
			2	0.9		2.1	3.78				
			1	1.5		0.8	1.20				
			1	1.1		1.1	1.21				
							6.19				
						Net Quantity	125.06	125.06	Sqm	526.87	65890.36
B.	Enamel Paint Works in Doors &										
	Windows										
	Window		2	1.2		0.9	2.16				
	Door		2	0.9		2.1	3.78				
							5.94	5.94	Sqm	308.84	1834.51
	Total amount										
4	Office Building										
	Front		1	8.8	2.9		25.52				
			1	1.6	2		3.20				
			1	4.1	2.9		11.89				
	side and Back		1	16.8	2.9		48.72				
			1	8.8	0.1		0.88				
	R1		1	1.7		2.3	3.91				
	R2 and R3		2	3.5		2.9	20.30				
	Chajja		1	25.6		0.6	15.36				
						Total	129.78				
	Deduction										
	Window										
	W1		8	1.2		0.9	8.64				
	Door										
	D1		2	0.9		2.1	3.78				
	D2		1	0.9		2.1	1.89				
						Total Deduction	14.31				
						Net quantity	115.47	115.47	Sqm	526.87	60837.68
B.	Enamel Paint Works in Doors &										
	Windows										
	Window										
	W1		8	1.2		0.9	8.64				
	Door										
	D1		2	0.9		2.1	3.78				
	D2		1	0.7		2.1	1.47				
							13.89	13.89	Sqm	308.84	4289.79
	Total amount										

क्र.सं.	काम/सेवाको विवरण/क्रियाकलाप	अघिल्लो नापी किताबमा चढेको परिमाण	यस नापीको मापनको आयाम					हालसम्म अद्यावधिक काम भएको परिमाण	ईकाइ	परिमाण सूची (BoQ) मा उल्लिखित दर	जम्मा रकम
१	२	३	४	५	६	७	८	९	१०	११	१२=९*११
5	OHT 225 m3										
	Two coat weather painting outside of Tank		8	19.6			250.88				
			32	1.8			103.68				
			1	27			105.30				
			1	(17.1+0)2			25.65				
			1	(25.8+4.2)2			58.50				
			1	(26.55+19.7)2	2		46.25				
			1	27			7.56				
			1	25	1.1		27.50				
			1	18.4	0.45		8.28				
			1	27	0.45		12.15				
	Landing below		11	0.75	0.75		6.19				
	Post		44	0.4			17.60				
	Stair big below		1	4.3			3.66				
	Total quantity						673.20	673.20	Sq m	526.87	354688.88
	Total amount										
6	Name writing of Project with materials						1.00	1	job	5000.00	5000.00
											798984.62
B	Earthworks										
1	E/W in Excavation for pipe trench in Ordinary soil			500.00	0.45		135.00				
				400.00	0.45		126.00				
				1551.00	0.45		453.67				
							714.67	714.67	m³	777.65	555761.18
2	E/W in backfilling for pipe trench in ordinary soil (75 % of E/W)			2451.00			536.00	536.00	m³	194.41	104203.88
	Sub total										659965.06
C	Laying & Jointing of pipe										
	HDPE Pipe										
	PE100 PN10 DN50		500					500.00	m	5.01	2505.00
	PE100 PN10 DN40		1000					1000.00	m	5.01	5010.00
	PE100 PN10 DN32		951					951.00	m	4.09	3889.59
	Sub total									Rs.	11404.59
	क. जम्मा रकम										1470354.27
	ख. जनसहभागिता										463885.06
	ग. जम्मा भुक्तानी गर्नपर्ने रकम (क-ख)										1006469.21
	घ. अघिल्लो नापिबाट भुक्तानी भएको रकम										0.00
	ङ. बाँकि भुक्तानी गर्नपर्ने रकम (ग-घ)										1006469.21
	घटाउनु पर्ने										
	च. १% धरौटी रकम										50323.46
	छ. खुद भुक्तानी गर्नपर्ने रकम (ङ-च)										956145.75

निर्माण व्यवसायी/उपभोक्ता समितिको सही:-
नाम:-
पद:-
अध्यक्ष
मिति:-

नापी लिनेको सही:-
नाम:-
पद:-
मिति:-

जाँच गर्नेको सही:-
नाम:-
हंश बहादुर थापा
पद:-
सिनियर इन्जिनियर
मिति:-

प्रमाणित गर्नेको सही:-
नाम:-
वीष्म थापा
पद:-
डिभिजन प्रमुख
मिति:-



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय,
कैलाली
ठेक्का सम्बन्धि बिल

म.ले.प.फारम नं: ५०४
साबिकको फारम नं: १६८

निर्माण कार्यको नाम:- हसुलिया खानेपानी योजना,कैलारी गा पा -३, कैलाली ।

आयोजनाको ठेगाना:-कैलारी गाउपालिका -३, कैलाली

निर्माण व्यवसायी/उपभोक्ता समितिको नाम:- हसुलिया खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,

२०८०/०८१ -०१ , WSSDO, Kailali

ठेक्का पट्टा/सम्झौताको संकेत नं:- ०२-२०८२/०८३(हसुलिया खानेपानी योजना,कैलारी गा पा -३, कैलाली)

नापी किताब नं:-

बिलको क्रम सख्या:- प्रथम रोलिंग बिल

कायदेशि मिति:-	२०८२/०४/२१	
कार्य सम्पन्न गर्नुपर्ने मिति:-	२०८३/०१/१५	
म्याद थप/भैरएसन मिति:-	२०८२/०३/२५	
कार्य सम्पन्न भएको मिति:-	२०८३/०३/०५	
नापी लिइएको मिति :-		

क्र.स.	कामको विवरण		एकाई	परिमाण सूची BOQ सम्झौता बमोजिमको		भरिएसन भएकोमा सो बमोजिमको		हालसम्म अद्यावधिक काम भएको		अचिल्लो बिलमा चढेको		हाल भएको काम		बाँकी पर्माण	कैफियत
	मुख्य र सहायक क्रियाकलाप			परिमाण	ठेक्का/सम्झौतामा स्वीकृत दर(रु.)	जम्मा रकम(रु.)	परिमाण	स्वीकृत दर(रु.)	जम्मा रकम(रु.)	परिमाण	जम्मा रकम(रु.)	परिमाण	जम्मा रकम(रु.)		
१	३		४	५	६	७	८	९	१०	१३	१४=६×१३	१५=११-१३	१६=१२-१४	१६	१७
A	Painting Works														
		1	Compound wall	529.50	526.87	278977.67			529.500	278,977.67		529.50	278,977.67		
		2	Toilet and Bathroom	52.13	526.87	27465.73			52.130	27,465.73		52.13	27,465.73		
		3	Pump House	125.06	526.87	65890.36			125.060	65,890.36		125.06	65,890.36		
			Enamel Paint Works in Doors & Windows	5.94	308.84	1834.51			5.940	1,834.51		5.94	1,834.51		
4	Office Building		Sqm	115.47	526.87	60837.68			115.470	60,837.68		115.47	60,837.68		
	Enamel Paint Works in Doors & Windows		Sqm	13.89	308.84	4289.79			13.890	4,289.79		13.89	4,289.79		
5	OHT 225 m3		Sqm	673.20	526.87	354688.88			673.200	354,688.88		673.20	354,688.88		
6	Name writing of Project with materials		job	1.00	5000.00	5000.00			1.000	5,000.00		1.00	5,000.00		
	Total					798,984.62					798,984.62				
B	Construction of Valve Chamber														
		1	Earthwork in excavation in ordinary soil	2.10	777.65	1633.07			2.10	1633.07		2.100	1633.070		
		2	Flat brick soling with sand packing	2.10	823.78	1729.94			2.10	1729.94		2.10	1,729.94		
		3	PCC in (1:2:4)	0.30	11,747.05	3524.12			0.30	3524.12		0.30	3,524.12		
		4	Brick masonry in (1:6) cement mortar	1.12	12,707.18	14232.04			1.12	14232.04		1.12	14,232.04		
		5	Reinforcement for slab	22.33	127.41	2845.07			22.33	2845.07		22.33	2,845.07		
		6	Shuttering work for RCC	2.73	666.76	1820.25			2.73	1820.25		2.73	1,820.25		
		7	12 mm thick cement Plaster (1:4)	7.20	362.28	2608.42			7.20	2608.42		7.20	2,608.42		
8	Supply and installation Fittings of VC		job	1.00	5,000.00	5000.00			1.00	5000.00		1.00	5,000.00		
	Sub Total for 1 No.s				-	33392.91				33392.91			33,392.91		
	Sub Total for 9 No.s				-	300536.19				300536.19			300,536.19		
C	Earthworks														
		1	E/W in Excavation for pipe trench in Ord	1260.00	777.65	979839.00			1160.17	902,206.20		445.50	346,443.08		
2	E/W in backfilling for pipe trench in ordin		m3	945.00	194.41	183717.45			870.13	169,161.97		334.13	64,958.21		
						1163556.45				1,071,368.17			411,401.29		
C	Laying & Jointing of pipe HDPE Pipe														
			m	1440.00	202.26	291254.40			1440.00	291,254.40		1440.00	291,254.40		
			m	110.00	100.47	11051.70			110.00	11,051.70		110.00	11,051.70		
			m	500.00	5.01	2505.00			500.00	2,505.00		500.00	2,505.00		

निर्माण व्यवसायी/उपभोक्ता समितिको सही:-

तयार गर्नेको सही:-

जाँच गर्नेको सही:-

प्रमाणित गर्नेको सही:-



सुदूरपश्चिम प्रदेश सरकार
भौतिक पूर्वाधार विकास मन्त्रालय
खानेपानी तथा सरसफाई डिभिजन कार्यालय,
कैलाली
ठेक्का सम्बन्धि बिल

म.ले.प.फारम नं: ५०४
साबिकको फारम नं: १६८

निर्माण कार्यको नाम:- हसुलिया खानेपानी योजना,कैलारी गा पा -३, कैलाली ।

आयोजनाको ठेगाना:-कैलारी गाउपालिका -३, कैलाली

निर्माण व्यवसायी/उपभोक्ता समितिको नाम:- हसुलिया खानेपानी उपभोक्ता तथा सरसफाई समिति/संस्था,

२०८०/०८१ -०१ , WSSDO, Kailali

ठेक्का पट्टा/सम्झौताको संकेत नं:- ०२-२०८२/०८३(हसुलिया खानेपानी योजना,कैलारी गा पा -३, कैलाली)

नापी किताब नं:- प्रथम रोतांग बिल

बिलको क्रम सख्या:-

क्र.स.	कामको विवरण		एकाई	परिमाण सूची BOQ सम्झौता बमोजिमको		भरिएसन भएकोमा सो बमोजिमको भएको		हालसम्म अद्यावधिक काम		अघिल्लो बिलमा चढेको		हाल भएको काम		बाँकी परिसमाण	कैफियत
	मुख्य र सहायक क्रियाकलाप			परिमाण	ठेक्का/सम्झौतामा स्वीकृत दर(रु.)	जम्मा रकम(रु.)	परिमाण	स्वीकृत दर(रु.)	जम्मा रकम(रु.)	परिमाण	जम्मा रकम(रु.)	परिमाण	जम्मा रकम(रु.)		
१	३	४	४	५	६	७	८	९	१०	११	१२=६×११	१३	१४=११-१३	१५=१२-१४	१६
	PE100 PN10 DN40		m	1000.00	5.01	5010.00			5,010.00	1000.00	5,010.00		1000.00	5,010.00	
	PE100 PN10 DN32		m	1000.00	4.09	4090.00			3,889.59	951.00	3,889.59		951.00	3,889.59	
	Sub total				-	313911.10			313,710.69		302,306.10			11,404.59	
D	Transportation		trip	2.00	5,000.00	10000.00			10,000.00	2.00	10,000.00				
	Total of (G)														
	जम्मा रकम (A+.....)					2,586,988.36			2,494,599.67		1,024,243.58			1,470,356.09	
	जनसहभागिता					2,586,988.36			2,494,599.67		1,024,243.58			1,470,356.09	
						758,556.45			726,443.18		262,556.29			463,886.89	
	जम्मा भुक्तानी गर्नुपर्ने रकम					1,828,431.91			1,768,156.49		761,687.29			1,006,469.20	

हाल काम भएको बाँकी दिनुपर्ने रु १००६४६९.२०

कट्टा गर्नु पर्ने रु:-

क. मोबिलाइजेसन पेस्की रु:

ख. मालसामानको पेस्की रु:

ग. मेसिनरी औजारको बहाल रु:

घ. अग्रीम १.५% आय कर रकम

ङ. ५% धरौटी रकम रु:

च. अन्य कट्टी गर्ने पर्ने रकम रु:

खुद दिनु पर्ने रकम रु:

यस बिल बमोजिम गरिएको भुक्तानी भौचर नं: भुक्तानी मिति: भुक्तानी रकम रु.:

हाल काम भएको बाँकी दिनुपर्ने रु १००६४६९.२०

कट्टा गर्नु पर्ने रु:-

क. मोबिलाइजेसन पेस्की रु:

ख. मालसामानको पेस्की रु:

ग. मेसिनरी औजारको बहाल रु:

घ. अग्रीम १.५% आय कर रकम

ङ. ५% धरौटी रकम रु:

च. अन्य कट्टी गर्ने पर्ने रकम रु:

खुद दिनु पर्ने रकम रु:

यो बिलबमोजिम काम भएको जम्मा रु:-	१,७६८,१५६.४९
अघिल्लो बिलसम्म भुक्तानी भएको रु:-	७६१,६८७.२९
अघिल्लो भुक्तानी बिलको क्रम सङ्ख्या:-	
अघिल्लो भुक्तानी बिलको भौ.नं.:-	

निर्माण व्यवसायी/उपभोक्ता समितिको सही:-

नाम: हनुमान चौधरी

पद: अध्यक्ष

मिति:

यो बिलबमोजिम जम्मा रु.: ९४१०४८.७१

तयार गर्नेको सही:-

नाम: प्रकाश पार्की

दर्जा: इन्जिनियर

मिति:

जाँच गर्नेको सही:-

नाम: हंश बहादुर थापा

दर्जा: सिनियर इन्जिनियर

मिति:

स्वीकृत गर्नेको सही:-

नाम: ग्रीष्म थापा

दर्जा: डिभिजन प्रमुख

मिति:

निर्माण व्यवसायी/उपभोक्ता समितिको सही:-

तयार गर्नेको सही:-

जाँच गर्नेको सही:-

प्रमाणित गर्नेको सही:-

TRANSPORTATION COST OF MAJOR CONSTRUCTION MATERIALS

FY 082/083

Name of Project : Hasuliya Water Supply Project

Metalled Road Head: Banbehada

Gravel Roadhead :

Earthen Road Head:

Distance in Km : 11

Distance in Km :

Distance in Km : 0

Sand quarry: Khutiya

Stone quarry : Khutiya

Aggregate quarry: Khutiya

Brick : kuchaine

	Metal	Gravel	Earthen
	4	2	
	4	2	
	4	2	
	4	2	

S.N.	Particulars	Unit	Unit Wt Kg.unit	Transportation Cost							Handelling		Total cost of Transportation & Handelling (NRs./Unit)	Material District Rate	Total Cost Per	
				Capacit y 6 Ton Truck	Avg. Haulage Dist. (Km.)			Total Distance (Km)	Time taken for Trans ⁿ (hour)	Cost of Trans ⁿ (NRs.)	Loading & Unloading Coefficient	Cost of Loading/Unloadi ng (NRs./Truck)				
					Metalled Road	Gravel Road	Earthen Road									
1	Reinforcement	M. Ton	1000.0	6	11	0	0	11	1.30	482.09	0.633	477.92	561.74	98000.00	98561.74	M.Ton
2	Cement	Bag	50.0	120	11	0	0	11	1.30	24.10	0.633	477.92	28.08	750.00	778.08	Bag
3	Bricks	Nos.	3.0	2000	4	2	0	6	1.15	1.28	0.418	315.59	1.44	12.50	13.94	Nos.
4	Aggregate (5-10) mm	Cum	1596.0	3.76	4	2	0	6	1.15	680.63	0.418	315.59	764.58	1400.00	2164.58	Cum
5	Aggregate (10-20) mm	Cum	1596.0	3.76	4	2	0	6	1.15	680.63	0.418	315.59	764.58	1890.00	2654.58	Cum
6	Aggregate (20-40) mm	Cum	1596.0	3.76	4	2	0	6	1.15	680.63	0.418	315.59	764.58	1350.00	2114.58	Cum
7	Sand	Cum	1500.0	4.00	4	2	0	6	1.15	639.69	0.418	315.59	718.59	800.00	1518.59	Cum
8	Local Wood	Cum	700.0	8.57	11	0	0	11	1.30	337.46	0.633	477.92	393.22	37000.00	37393.22	Cum
9	HDPE pipe	Kg	1.0	6000	11	0	0	11	1.30	0.48	0.633	477.92			3370.44	trip
10	Stone	Cum	1700.0	3.53	4	2	0	6	1.15	724.99	0.418	315.59	814.41	1057.00	1871.41	Cum

NOTES

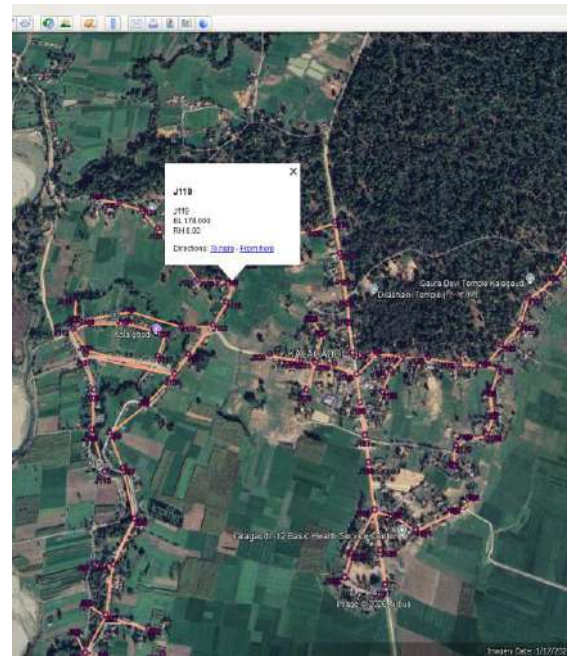
- * A Truck Capacity of 6 Ton has been taken for the calculation
- * Truck Speed of 40, 20 and 10 km per hour has been assumed for Metalled Road, Gravel Road and Earthen Road, respectively.
- * Loading and unloading time of 0.5 km has been assumed
- * Average time for one trip = 2 times traveling time + Loading and un loading time has been taken
- * 400 Nos. of Bricks = 1 cubic meter of Bricks
- * Transportation Rate
- * Unskilled Labour Wage
- * Constant Loading and unloading coefficient of 2.5 has been taken for Cement & reinforcement bar.
- 2225.02 (Nrs./Hour) has been taken for Truck
- 755.00 (NRs./Day) has been taken

Layout and distribution line – GPS Field Survey Mapping (KML)

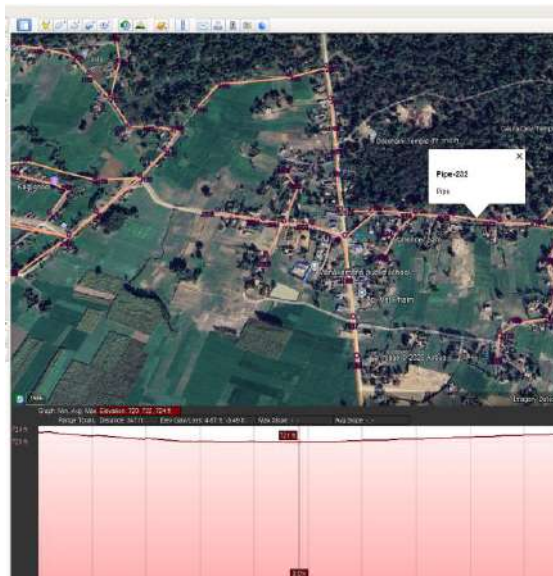
Legend		
J = Node	V = Valve	R = Reservoir



G.1 Overall Network – Kalagaudi (RVT2)



G.2 Kalagaudi Village Network (J119)



G.3 Pipeline Alignment – Elevation Profile



G.4 Valve/Reservoir Overview (V4)

1. Site Visit: Dokebazar Khanepani Aayojana



Photo 1 — Dokebazar OHT (225 m³ RCC), 2083-02-06 –field visit photograph
(:location of source of water supply project)



Fig: Inlet, Outlet and Pump Assembly of Overhead Tank – Field Visit



Fig: Underside View of OHT Structure with RCC Columns – Field Visit
(Recent condition)



Fig: Newly Constructed Hand Pump with Masonry Apron

2.Site supervision and office work-warehouse and staff quarters, **WSSDO,kailali**



Fig: — Overall View of Column and Tie Beam Reinforcement – Construction Site,
WSSDO Kailali(site visited)



Figure . Measurement of the timber ledger supporting the steel shuttering before slab casting.



Figure : Bamboo scaffolding (staging) erected beneath the beam/slab formwork to transfer construction loads to the ground during casting.



Figure : Site supervision at the construction area, with bamboo staging and column reinforcement cage visible in the background.



Figure : Verification of reinforcement bar diameter and hook length within the beam formwork using a steel measuring tape



Figure : Checking the height of a stirrup (ring) using a steel tape; reading of approximately 500 mm confirmed against design.



Figure : JCB backhoe loader (4x4) engaged in excavation and earthwork clearance at the site.



Figure : Excavated foundation trench adjacent to the existing RCC wall, showing accumulated turbid water requiring dewatering before further work.



Figure : Erection of vertical column reinforcement cage inside the formwork; worker aligning the main longitudinal bars before shuttering closure.



Figure : Top view of the tied reinforcement cage at a column-beam junction, showing main bars and stirrups held within steel shuttering plates.



Figure : Measurement of the steel shuttering plate width over the timber support beam using a steel measuring tape.



Figure : Workers assembling timber/plywood shuttering panels alongside the erected column reinforcement cage.



Figure : Timber formwork planks placed for beam/lintel casting, with column reinforcement bars crossing above, near the building corner



Figure __. Site supervision of Ground Floor Slab Casting of the WSSDO Kailali Office Building Project

3.Hasuliya Water Supply Project, Kailari Rural Municipality–03, Kailali, Nepal.



Figure: Overhead Tank (OHT) after Final Painting Work



Figure: Overhead Tank (OHT) after Final Painting Work



Figure . Valve chamber with installed fittings.



Figure . Measurement and verification of completed painting works of the compound wall, entrance gate, and office building.



Figure: Sluice valve fitted on GI pipeline connection at pump house outlet



Figure: pipeline flange joint, valve chamber submerged in water, and empty valve chamber pit during site inspection



Figure: pump/motor components, submersible pump column pipe, and cables stored at pump house/store during equipment inspection



Figure : Final painting of the compound wall, entrance gate, and office building



Figure: submersible pump column pipe at pump house during equipment inspection



Figure :. Valve chamber submerged in water during site inspection at the Hasuliya Water Supply Project.



Figure: HDPE pipe (DN 200, PN 16) stored at site with manufacturer markings visible



Figure: Minor crack observed at boundary wall corner junction due to tree roots, and measurement tape reading taken during site inspection

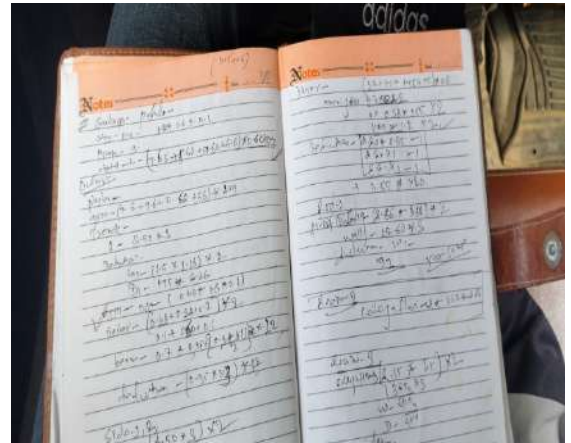
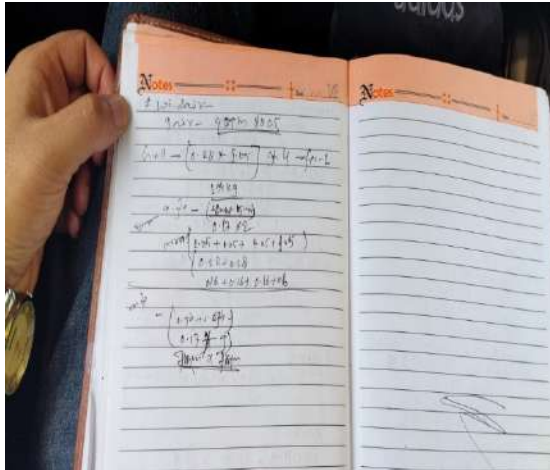


Figure: Handwritten site measurement book showing calculations recorded during field visit

4. Basauti Khanepani Yojana, Kailari Rural Municipality-8, Kailali



Figure: Overhead Tank (OHT) and office/pump house building of the water supply scheme (Basauti Khanepani Yojana) during site visit



Figure: Buried pipeline at intersection at site and precast RCC cover slabs for valve chamber



Figure :. Completed valve chamber at the Basauti Water Supply Project.



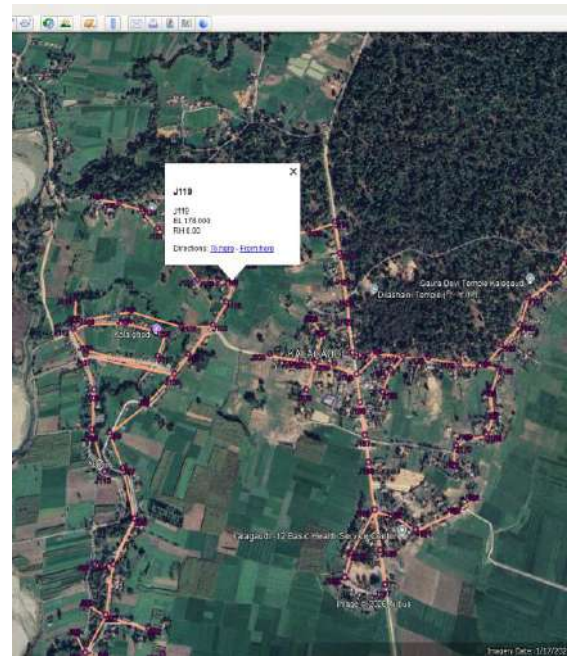
Figure: Newly constructed valve chamber with valve fitting, and site supervisor taking measurement during inspection

Layout and distribution line – GPS Field Survey Mapping (KML)

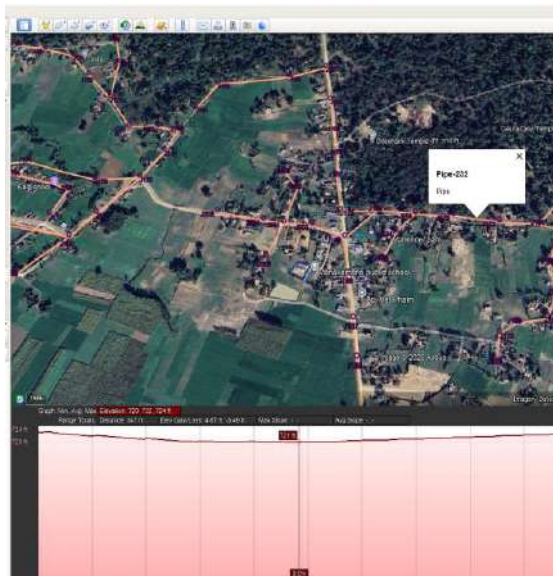
Legend		
J = Node	V = Valve	R = Reservoir



G.1 Overall Network – Kalagaudi (RVT2)



G.2 Kalagaudi Village Network (J119)



G.3 Pipeline Alignment – Elevation Profile



G.4 Valve/Reservoir Overview (V4)

Salient Features of Project

Name of Project: Bajinatha Siddhanatha Water Supply Project

1. District: Kanchanpur

2. Name of Municipality, Wards and Locality Covered: Suklaphata Municipality, Ward No. 12

3. Package: 1 - Haat

S.N.	Parameters	Unit	Total
A.	Household / Population		
1	Households (survey year: 2026)	No.	1,121
2	Population (survey year: 2025)	No.	6,720
3	Households (base year: 2028)	No.	1,222
4	Population (base year: 2028)	No.	6,933
5	Household (design year: 2044)	No.	1,922
6	Population (design year: 2044)	No.	8,891
7	No. of institution	No.	5
8	Design period	Yrs.	20
9	Annual growth rate	%	1.3
B.	Source Information		
1	Name of source	-	Ground Water
2	Type of source & number	Type/No.	DTW
3	Measured/Tapped discharge of source	Lps	10 LPS
4	Approximate distance from project site	M	1,000
5	Water quality of source	-	-
C.	Water Provision & Demand		
1	Total water demand (Base year)	L/day	1,001,992.00
2	Total water demand (design year)	L/day	1,284,954.54
D.	Project Components		
1	Actual length of transmission line	km	0.00
2	Actual length of distribution pipe line	km	23.910
3	Existing household connection	No.	0.00
4	Number of private connection provided (base year)	No.	1,222.00
5	Capacity and number of Over Head Tank	m ³ /no	1,450 / 1
6	Capacity and number of reservoirs	m ³ /no	-
7	Total number of DC, IC, Valve Box, VC & AVC	No.	13
8	Number of treatment plant operation building	No.	-
9	Number of WUA office	No.	1
E.	Project Cost Data		
1	Total estimated cost	NRs.	6,99,72,119.05
2	Average per capita cost at base year (E1/A4)	NRs.	10,092.62
3	Average per capita cost at design year (E1/A6)	NRs.	7,869.99



Sudurpaschim Province
Ministry of Physical Infrastructure
Water Supply and Sanitation Division Office
Kailali
BILL OF QUANTITY

Bajinatha siddhanatha Water Supply Project

Construction VCs, pipeline works, electromechanical works, supply of pipes and fittings, etc.

SN	Description of Items	Total Quantity	Unit	Rate (NRs.) In Figures	Amount (NRs.)	Remark
	Part A: Provisional Sum					
1	Insurances for the loss of damage to works, plant, material, equipment, property and personal injury or death and as stipulated in contract Data	1.00	P.S.	120000.00	120000.00	
2	Utility Relocation, Reinstallation, etc. , Third Party Inspection cost, claim, quality test, all items not covered by BOQ, etc.	1.00	P.S.	200000.00	200000.00	
3	Power Consumption cost for Pump or Utility Operation as per NEA bill for a period of 12 months (Operation and maintenance period) for scheme	1.00	PS	120000.00	120000.00	
	B. General Items					
4	Provide and maintain cube testing facilities including moulds for the duration of the contract and preparation and testing of cubes, marking, curing, storing and transport to lab for tests and maintain full records (Equipment and moulds to remain in the contractor's property), Paid as number of cubes for all complete work as per direction of Engineer	39.00	No.	350.00	13650.00	
5	Provide photographs (Digital CD) and colour prints, sized A4, 24 nos., each month for inclusion in periodical progress report to the satisfaction of the engineer. (6 sheet A4 size color Photo Paper per months)	10.00	Months	500.00	5000.00	
6	Provide safety equipment for the Engineer at each of the site offices of the types and number specified for the period of the Contract as per Specification.	24.00	Months	2500.00	60000.00	
	Part C: Days Work					
7	0.00	20.00	man-days	755.00	15100.00	
8	Hire of Skilled Labour (Plumber, Blacksmith, carpenter and other skilled Labour)	20.00	man-days	1130.00	22600.00	
	Distribution Pipe Line					
9	Excavation in foundation in ordinary soil for pipeline/foundation trench upto 2m from ground level and disposal of surplus materials to the specified site as directed by the Engineer and including dewatering all complete as per Clause 2.2 and other clause of the specification.	5336.04	m ³	626.010	3340412.52	
10	Earth filling work with all type of stacked soil in 20cm thick layer by layer with water sprinkling, compaction with ramming and site clearance for foundation and other trenches all complete as per Clause 2.2,3 and other clause of the specification and as directed by the Engineer.	4002.03	m ³	223.570	894733.34	

Note: Only the first page of the BOQ (valve chamber, pipeline works, and pipe supply) is included for reference. For complete details, refer to the original Excel-based BOQ file.



Sudurpashchim Province Government
Ministry of Physical Infrastructure Development
WSSDO
Dhangadhi, Kailali

HAULAGE CHART (BY TRUCK & MANUAL)

(F.Y. - 2082-83)

By Truck(6 MT capacity)			
Permissible speed taken in Km.			
Metalled Road (v)			40
Gravel Road (v)			30
Earthen Road (v)			20

Multiplying Factor For Loading & Unloading			
Metalled Road			1.00
Unmetalled Road(Gravel / Earthen)			1.00
Running cost/hr.			1904.03
Ideal charge/hr.			744.01

Running time/trip,hr. = (De/Ve + Dg/Vg + Dm/Vm)+0.75+(De/1.25Ve + Dg/1.25Vg + Dm/1.25Vm)

S.N.	Materials	Unit	Location		By 6 MT truck										Manual			Total Transportation cost per unit			
					From	To	Distance (Road Length), Km				Running hour/trip	Truck Ideal hour	Running hour cost.	Ideal hour cost	Labour for Load & unload	Total labour cost	Total cost per trip		Material/ trip(6MT)	Cost/ Unit	Distance (Km)
			Metalled	Unmetalled			Total														
				Gravel				Earthen													
	EXT. MATERIALS																				
1	Cement	Kg	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.545	411.48	812.57	6000.00	0.13	0.00	1.35	0.00	0.1300
2	Steel	Kg	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.545	411.48	812.57	6000.00	0.13	0.00	1.35	0.00	0.1300
3	Binding wire	Kg	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.545	411.48	812.57	6000.00	0.13	0.00	1.35	0.00	0.1300
4	G.I.Wire	Kg	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.545	411.48	812.57	6000.00	0.13	0.00	1.35	0.00	0.1300
5	Nails	Kg	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.545	411.48	812.57	6000.00	0.13	0.00	1.35	0.00	0.1300
7	Geo-Textile	Sq.m.	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.347	261.99	663.08	6000.00	0.11	0.00	1.35	0.00	0.1100
8	Brick	Nos	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.347	261.99	663.08	2000.00	0.33	0.00	1.35	0.00	0.3300
9	Rubber seal 6" wide	Rm	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.232	228.48	172.61	0.347	261.99	663.08	3000.00	0.22	0.00	1.35	0.00	0.2200
10	wood	Cum	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.176	228.48	130.95	0.168	126.84	486.27	10.90	44.61	0.00	1.35	0.00	44.6100
11	Plywood	Sq.m.	Jhalahari	Site	2.00	0.50	0.00	2.50	0.120	0.176	228.48	130.95	0.545	411.48	770.90	1500.00	0.51	0.00	1.35	0.00	0.5100

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Water Supply and Sanitation Division Office
Dhangadhi, Kailali
Detail Estimate & Cost -Bajinath sidnath wp

Name of the Work: Compound Wall

S.N.	Description	No	Length	Breadth	Height	Quantity	Unit	Rate	Amount
1	Site Clearance	1	80.00	0.80		64.00	Sqm	138.92	8890.88
2	Earthwork in Excavation in Soft soil for: foundation of walls 1st Step	1	80.00	0.60	0.65	31.20			
						31.20	Cum	626.01	19531.51
3	Flat Brick soling in foundation foundation of walls	1	80.00	0.60		48.00			
						48.00	Sqm	860.04	41281.87
4	Lean Concrete (1:3:6)								
	Base of wall	1	80.00	0.60	0.05	2.40			
						2.40	Cum	10563.56	25352.54
5	PCC(1:2:4) works								
	Top of Wall pillar	32	0.35	0.35	0.05	0.20			
	Top of wall (capping)	1	68.80	0.23	0.05	0.79			
	Deduction	0	0.35	0.12	0.05	0.00			
						0.99	Cum	12212.31	12055.99
6	1st class Brick work in 1:4 Cement motar Wall below PL								
	foundation of walls 1st Step	1	80.00	0.45	0.30	10.80			
	foundation of walls 2nd Step	1	80.00	0.35	0.30	8.40			
	Wall above PL								
	wall	1	68.80	0.23	1.50	23.74			
	Pillar	32	0.35	0.35	1.50	5.88			
						48.82	Cum	13981.59	682525.30
7	Backfilling of Earth in Foundation		25% of Total Excavation			7.80	Cum	245.94	1918.33
8	12.5 mm thick cement sand(1:4) plaster								
	Wall above GL	2	80.00		1.50	240.00			
	Top of wall	1	80.00	0.23		18.40			
	Pillar Side	64	0.12		1.65	12.67			
	Pillar faces	3	1.40		2.20	9.24			
	Top of pillars	3	0.35	0.35		0.37			
						280.68	Sqm	401.90	112805.09
9	2 coat wpc painting over plastered surface		Quantity as Same As plaster			280.68	Sqm	188.72	52969.84
10	Form work								
	Pillar	3	1.40		3.00	12.60			
	Top of wall (capping)	2	80.00		0.05	8.00			
						20.60	Sqm	875.87	18042.92
	Sub Total								975374.28
	Value Added Tax @ 13%								0.00
	Grand Total (NRs.)								975374.28

Government of Sudurpashchim Province
Ministry of Physical Infrastructure and Development
Water Supply and Sanitation Division Office
Kailali

Detailed Quantity Estimate & Costing

##

Name of the Work: R.C.C. Over Head Tank 225 CUM

Sn.	Description of Work	No	L	B	H	Quantity	Unit	Rate	Amount	Remarks
1	Site clearance	1	12	12		144	Sqm	138.92	20004.48	
2	E/W in excavation in OS	1	π	$(5.50)^2$	2.475	235.08				
	Soling Base	1	π	$(4.95)^2$	0.2	15.38				
	Valve Chamber	4	2.15	2.05	1.145	20.19				
						270.6464	Cum	105.95	28674.41	
3	Stone/Boulder soling in foundation with sand	1	π	$(4.95)^2$	0.2	15.38				
	Below OHT Base , Outer Peti And Drain	1	π	$(4.95)^2$	0.2	15.38				
	Deduction Pillar	8	0.40	0.4	0.2	-0.26				
	Valve Chamber	4	2.15	2.05	0.2	3.53				
						34.03	Cum	2761.04	93958.19	
4	P.C.C. (1:3:6) in foundation	1	π	$(4.95)^2$	0.225	17.31				
	Valve Chamber foundation	4	2.15	2.05	0.05	0.88				
						18.19	Cum	10563.56	192167.00	
5	P.C.C. (1:2:4) on OHT Base	1	π	$(3.85)^2$	0.10	4.65				
						4.65	Cum	12212.31	56787.24	
6	Brick masonry (1:4)									
	Pipe Support	4	1.2	0.34	0.34	0.55				
	Outer Drain	2	21	0.23	0.23	2.17				
	For Valve Chamber									
	LW	8	2.15	0.23	1.05	4.15				
	SW	8	1.35	0.23	1.05	2.61				
						9.49	Cum	13981.59	132690.60	
7a	R.C.C. (1:1.5:3) in foundation	1	π	$(4.72)^2$	0.30	20.99				
		2	17.91	1.725	0.075	4.63				
		1	17.91	0.45	0.45	3.63				
		8	0.40	0.40	1.50	1.92				
						31.17				
	Valve Chamber Cover Slab	4	2.15	2.05	0.10	1.76				
	Deduction Opening	1	π	$(0.15)^2$	0.30	(-)0.03				
						32.90	Cum	16173.95	532171.48	
b	Cement concrete work (1:1.5:3) for R.C.C. in cement column, Bracing, and stair case upto 3m.	8	0.40	0.40	3.00	3.84				
		8	1.80	0.30	0.60	2.59				
	Stair	2	1.97	0.75	0.10	0.29				
	Landing	2	0.77	0.75	0.16	0.18				
	Steps	1/2*18	0.75	0.18	0.18	0.21				
						7.11	Cum	16173.95	114996.75	
c	Column and bracing and stair case upto 6m	8	0.40	0.40	3.00	3.84				
		8	1.80	0.30	0.30	1.3				
		2	1.97	0.75	0.10	0.29				
		2	0.77	0.75	0.16	0.18				
		1/2*18	0.75	0.18	0.18	0.21				
						5.82	Cum	17910.45	104238.79	
d	Column and bracing and stair case upto 9m	8	0.40	0.40	3.00	3.84				
		8	1.80	0.30	0.30	1.30				
		2	1.97	0.75	0.10	0.29				
		2	0.77	0.75	0.16	0.18				
		1/2*18	0.75	0.18	0.18	0.21				
						5.82	Cum	19646.95	114345.22	
e	Column bracing and stari case upto 12m	8	0.40	0.40	3.00	3.84				
		8	1.80	0.30	0.60	2.59				
		2	1.97	0.75	0.10	0.29				
		2	0.77	0.75	0.16	0.18				
		1/2*18	0.75	0.18	0.18	0.21				
						7.11	Cum	21383.45	152036.29	
f	Column bracing and stir case upto 15m	8	0.40	0.40	3.00	3.84				

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Prepared by:

Checked by:

Approved by:

Analysis of Rates									
Rates are based on approved district rate of Kailali for 2082/083									
	Name of Project :		Bajinatha siddhanatha Water Supply Project						
	Assumptions								
	Density of sand					=		1450.00	kg/m ³
	Density of rubble stone					=		2000.00	kg/m ³
	Density of aggregate					=		1600.00	kg/m ³
	Name of market :		Jhalahari						
	Name of roadhead :		0.00						
	Distance of project site from market on :								
	- Foot					=		0.00	kosh
	-Earthen road					=		0.00	km
	- Gravel road					=		0.50	km
	-Metalled road					=		2.00	km
	Sand and aggregates are available in		Suda Crusher or other						
	It is far from district headquarters, so district rate is used to calculate the cost of sand and aggregate at site.								
	for head load (road head to Project site)					=		0.00	kosh
	If there is tractor/ truck road distance of haulage for sand/aggregate transportation on :								
		-Earthen road				=		0.00	km
		- Gravel road				=		0.00	km
		-Metalled road				=		0.00	km
	Cost of stone per m ³ at road head							1057.00	Rs.
	Cost of sand per m ³ for purchase/district rate							800.00	Rs.
	District Rates :								
	District rate of transportation by head load for :								
	-convenient load					=		4.16	Rs/kg.kosh
	-inconvenient load as bars, ms angle, etc.					=		4.58	Rs/kg.kosh
	- GI & HDPE pipes					=		9.41	Rs/kg.kosh
	- distribution transformer					=		10.74	Rs/kg.kosh
	- electrical poles					=		7.88	Rs/kg.kosh
	District rate of transportation by truck for convenient load on :								
	-Earthen road					=		11.40	Rs/km.quintal
	- Gravel road					=		6.80	Rs/km.quintal
	-Metalled road					=		3.00	Rs/km.quintal
	District rate of transportation by truck for inconvenient/GI pipe load on :								
	-Earthen road					=		11.97	Rs/km.quintal
	- Gravel road					=		10.20	Rs/km.quintal
	-Metalled road					=		3.33	Rs/km.quintal
	District rate of transportation by truck for HDP pipe load on :								
	-Earthen road					=		42.97	Rs/km.quintal
	- Gravel road					=		36.62	Rs/km.quintal
	-Metalled road					=		11.95	Rs/km.quintal
	Rate of load / unload					=		11.65	Rs/quintal
	District rate of Skilled labour					=		1130.00	Rs/d
	District rate of unskilled labour					=		755.00	Rs/d
	District rate of plumber					=		1040.00	Rs/d

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	Rate of some local materials							
1	Sand for 1 m³							
	Cost of 1 m ³ of sand at (as per district rate)						800.00	Rs.
	Cost of transportation:							
	By tractor: metalled road: 1450 kg/ m ³ @ 3 Rs/km.quintal for 0 km haulage						0.00	Rs.
	By tactor: earthen road: 1450 kg/ m ³ @ 11.4 Rs/km.quintal for 0 km haulage				=		0.00	Rs.
	By manual transportation: 1450 kg/m ³ @ 4.16 Rs/kg.kosh for 0 kosh haulage				=		0.00	Rs.
	Rate 1 m³ of sand at site (transporation cost +collection or purchase cost)				=		800.00	Rs.
2	Stone for 1 m³							
	Cost of stone at road head (per m ³)				=		1057.00	Rs.
	Cost of transportation:							
	by head load	2000 kg/m ³ @ 4.16 Rs/kg.kosh for 0 kosh haulage			=		0.00	Rs.
	Rate of 1 m³ of Stone at site				=		1057.00	Rs.
3	Aggregates 10 mm size for 1m³							
	Cost of 1 m ³ of aggregate						1400.00	Rs.
	Cost of transportation:							
	By tractor metalled road: 1600 kg/m ³ @ 3 Rs/km.quintal for 0 km haulage						0.00	Rs.
	By tractor earthen road: 1600 kg/m ³ @ 11.4 Rs/km.quintal for 0 km haulage						0.00	Rs.
	By manual transportation: 1600 kg/ m ³ @ 4.16 Rs/kg.kosh for 0 kosh haulage				=		0.00	Rs.
	Rate of 1 m³ aggregate				=		1400.00	Rs.
4	Aggregates 20 mm size for 1m³							
	Cost of 1 m ³ of aggregate						1890.00	Rs.
	Cost of transportation as above for 10 mm size aggregates						0.00	Rs.
	Rate of 1 m³ aggregate				=		1890.00	Rs.
5	Aggregates 40 mm size for 1m³							
	Cost of 1 m ³ of aggregate						1350.00	Rs.
	Cost of transportation as above for 10 mm size aggregates						0.00	Rs.
	Rate of 1 m³ aggregate				=		1350.00	Rs.
6	Rate of Transportation (from Jhalahari to Site):							
	Road type	Distance	Convenient	Inconvenient/GI pipe			HDPE pipe	
	-Earthen road	0.00	0.00	0.00			0.00	Rs/kg
	- Gravel road	0.50	0.03	0.05			0.18	Rs/kg
	-Metalled road	2.00	0.06	0.07			0.24	Rs/kg
	-For head load	0.00	0.00	0.00			0.00	Rs/kg
	Loading - unloading		0.23	0.23			0.23	Rs/kg
	Rate of transportation per kg :		0.33	0.35			0.66	Rs/kg
	Rate of transportation per kg with CO		0.38	0.40			0.76	Rs/kg
7	Local Wood per 1 m³							
	Rate of local wood per 1 m³, as per approved district rates				=		44100.00	Rs/m³
8	Salwood per 1 m³							
	Rate of sal-wood per 1 m³, as per approved district rates						1 77 000.00	Rs/m³
9	Mud [per m³]							
	District rate of mud				=		268.00	Rs/m³
	[If ther is no specific district rate of mud, it will be obtained as "1.5×labour cost]							

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10	Rate of Transportation of electrical pole and transformer (from Bazaar to Site):							
	Road type	Distance	Electrical pole		Transformer			
	-Earthen road	0.00	0.00		0.00			
	- Gravel road	0.50	0.05		0.05			
	-Metalled road	2.00	0.07		0.07			
	-For head load	0.00	0.00		0.00			
	Loading - unloading		0.35		0.70			
	Rate of transportation per kg :		0.47		0.82	Rs/kg		
	Rate of transportation with CO		0.54	Rs/kg	0.94	Rs/kg		
a)	Connection pipe to WC and septic tank & vent pipe(110 mm ø 2.5 kg/cm ² HDP)							
	Cost of 110 mmØ 2.5 kg/cm ² HDPE pipe per metre							
			@ 1.185	kg/m	290.00	Rs/kg	343.65	Rs.
	Cost of transportation up to site		@ 0.660	Rs/kg			0.78	Rs.
		Installation and joining cost					130	Rs.
			Subtotal:				474.43	Rs.
	Add overhead for contractor @ 15%						71.16	Rs.
		Cost of pipe and installation per m at site					545.59	Rs.
	Supply and installation of 40mm GI (medium) poles for shed per m							
	Supply of 40mm GI (medium) poles and truss member				1.1 m @		592.00	Rs/m
					cost:		651.20	Rs
	Cutting in size, welding and fixing in the site				20% of pipe cost		130.24	
		Subtotal					781.44	
	Add contractor overhead			@	15%		117.216	
	Rate/m						898.66	Rs
b)	10 mmØ bar mesh on MS angle (40×40×6) frame to lantern opening sides size- 0.75×1.00 m							
	weight for 1 screen	(cost includes fabrication also)			24	@	126.68	
	Subtotal						3040.32	
	Add contractor overhead			@	0.15		456.048	
	Cost per screen with CO						3496.368	
c)	Transportation cost for gravel filter media and sand filter media							
	Availability of sand/oblique gravel							
	Road type	Distance	Rate	Cost of transportation				
	-Earthen road	0.00	11.40	Rs/km.quintal	0.00			
	- Gravel road	0.00	6.80	Rs/km.quintal	0.00			
	-Metalled road	0.00	3.00	Rs/km.quintal	0.00			
	-For head load	0.00	4.16	Rs/kg.kosh	0.00			
	Loading - unloading		23.30	Rs/quintal	0.23			
	Rate of transportation per kg :				0.23			

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ANALYSIS OF RATES

GI Pipe Pieces for Fitting Connections

Preparation of pipe pieces of Medium Duty for the fittings connections/arrangements

Assuming 30% for cutting, threading, sundries and 10% for loss, Rate per piece is calculated as:

S.No.	Pipe size and length	Length	Unit	Rate/m	Amount	Add 40%	Subtotal	Rate/piece (Rs.) with Contractor's overhead
1	GI Pipe piece (25 mmØ) 0.30 m long	0.3	m	396.89	119.1	47.6	166.7	191.7
2	GI Pipe piece (25 mmØ) 0.45 m long	0.45	m	396.89	178.6	71.4	250.0	287.6
3	GI Pipe piece (25 mmØ) 0.60 m long	0.6	m	396.89	238.1	95.3	333.4	383.4
4	GI Pipe piece (25 mmØ) 1.0 m long	1	m	396.89	396.9	158.8	555.7	639.0
5	GI Pipe piece (25 mmØ) 1.85 m long	1.85	m	396.89	734.3	293.7	1028.0	1182.1
6	GI Pipe piece (32 mmØ) 0.45 m long	0.45	m	513.15	230.9	92.4	323.3	371.8
7	GI Pipe piece (32 mmØ) 0.60 m long	0.6	m	513.15	307.9	123.2	431.1	495.7
8	GI Pipe piece (32 mmØ) 1.0 m long	1	m	513.15	513.2	205.3	718.4	826.2
9	GI Pipe piece (40 mmØ) 0.60 m long	0.6	m	593.33	356.0	142.4	498.4	573.2
10	GI Pipe piece (40 mmØ) 1.0 m long	1	m	593.33	593.3	237.3	830.7	955.3
11	GI Pipe piece (50 mmØ) 0.60 m long	0.6	m	807.86	484.7	193.9	678.6	780.4
12	GI Pipe piece (50 mmØ) 0.20 m long	0.2	m	807.86	161.6	64.6	226.2	260.1
13	GI Pipe piece (50 mmØ) 0.30 m long	0.3	m	807.86	242.4	96.9	339.3	390.2
14	GI Pipe piece (50 mmØ) 0.35 m long	0.35	m	807.86	282.8	113.1	395.9	455.2
15	GI Pipe piece (50 mmØ) 0.45 m long	0.45	m	807.86	363.5	145.4	509.0	585.3
16	GI Pipe piece (50 mmØ) 1.0 m long	1	m	807.86	807.9	323.1	1131.0	1300.7
17	GI Pipe piece (50 mmØ) 1.60 m long	1.6	m	807.86	1292.6	517.0	1809.6	2081.1
18	GI Pipe piece (65 mmØ) 0.30 m long	0.3	m	1,033.22	310.0	124.0	434.0	499.1
19	GI Pipe piece (65 mmØ) 0.45 m long	0.45	m	1,033.22	465.0	186.0	650.9	748.6
20	GI Pipe piece (65 mmØ) 0.65 m long	0.65	m	1,033.22	671.6	268.6	940.2	1081.3
21	GI Pipe piece (65 mmØ) 0.75 m long	0.75	m	1,033.22	774.9	310.0	1084.9	1247.6
22	GI Pipe piece (80 mmØ) 0.20 m long	0.2	m	1,277.10	255.4	102.2	357.6	411.2
23	GI Pipe piece (80 mmØ) 0.45 m long	0.45	m	1,277.10	574.7	229.9	804.6	925.3
24	GI Pipe piece (80 mmØ) 0.75 m long	0.75	m	1,277.10	957.8	383.1	1341.0	1542.1
25	GI Pipe piece (80 mmØ) 0.60 m long	0.6	m	1,277.10	766.3	306.5	1072.8	1233.7
26	GI Pipe piece (80 mmØ) 0.65 m long	0.65	m	1,277.10	830.1	332.1	1162.2	1336.5
27	GI Pipe piece (80 mmØ) 0.67 m long	0.67	m	1,277.10	855.7	342.3	1197.9	1377.6
28	GI Pipe piece (80 mmØ) 0.30 m long	0.3	m	1,277.10	383.1	153.3	536.4	616.8
29	GI Pipe piece (80 mmØ) 0.77m long	0.77	m	1,277.10	983.4	393.4	1376.7	1583.2
30	GI Pipe piece (80 mmØ) 0.84 m long	0.84	m	1,277.10	1072.8	429.1	1501.9	1727.1
31	GI Pipe piece (80 mmØ) 0.90 m long	0.9	m	1,277.10	1149.4	459.8	1609.2	1850.5
32	GI Pipe piece (80 mmØ) 1.0 m long	1	m	1,277.10	1277.1	510.8	1787.9	2056.1
33	GI Pipe piece (80 mmØ) 1.60 m long	1.6	m	1,277.10	2043.4	817.3	2860.7	3289.8
34	GI Pipe piece (100 mmØ) 0.30 m long	0.3	m	1,895.55	568.7	227.5	796.1	915.6
35	GI Pipe piece (100 mmØ) 0.45 m long	0.45	m	1,895.55	853.0	341.2	1194.2	1373.3
36	GI Pipe piece (100 mmØ) 0.60 m long	0.6	m	1,895.55	1137.3	454.9	1592.3	1831.1
37	GI Pipe piece (100 mmØ) 0.65 m long	0.65	m	1,895.55	1232.1	492.8	1725.0	1983.7
38	GI Pipe piece (100 mmØ) 0.68 m long	0.68	m	1,895.55	1289.0	515.6	1804.6	2075.2
39	GI Pipe piece (100 mmØ) 0.69 m long	0.69	m	1,895.55	1307.9	523.2	1831.1	2105.8
40	GI Pipe piece (100 mmØ) 0.73 m long	0.73	m	1,895.55	1383.8	553.5	1937.3	2227.8
41	GI Pipe piece (100 mmØ) 0.75 m long	0.75	m	1,895.55	1421.7	568.7	1990.3	2288.9
42	GI Pipe piece (100 mmØ) 1.0 m long	1	m	1,895.55	1895.6	758.2	2653.8	3051.8
43	GI Pipe piece (100 mmØ) 1.60 m long	1.6	m	1,895.55	3032.9	1213.2	4246.0	4882.9
44	GI Pipe piece (125 mmØ) 0.30 m long	0.3	m	2,285.93	685.8	274.3	960.1	1104.1
45	GI Pipe piece (125 mmØ) 0.45 m long	0.45	m	2,285.93	1028.7	411.5	1440.1	1656.2
46	GI Pipe piece (125 mmØ) 0.75 m long	0.75	m	2,285.93	1714.5	685.8	2400.2	2760.3
47	GI Pipe piece (125 mmØ) 0.60 m long	0.6	m	2,285.93	1371.6	548.6	1920.2	2208.2
48	GI Pipe piece (125 mmØ) 1.0 m long	1	m	2,285.93	2285.9	914.4	3200.3	3680.4
49	GI Pipe piece (125 mmØ) 1.6 m long	1.6	m	2,285.93	3657.5	1463.0	5120.5	5888.6
50	GI Pipe piece (150 mmØ) 0.60 m long	0.6	m	2,639.00	1583.4	633.4	2216.8	2549.3
51	GI Pipe piece (150 mmØ) 1.0 m long	1	m	2,639.00	2639.0	1055.6	3694.6	4248.8
52	GI Pipe piece (200 mmØ) 0.45 m long	0.45	m	4,881.62	2196.7	878.7	3075.4	3536.7
53	GI Pipe piece (200 mmØ) 0.60 m long	0.6	m	4,881.62	2929.0	1171.6	4100.6	4715.6
54	GI Pipe piece (200 mmØ) 1.0 m long	1	m	4,881.62	4881.6	1952.7	6834.3	7859.4

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Procurement of Pipeline with Laying

Scheme 1

S.No.	Pipe size and series	Unit	Quantity	Unit weight (kg/m)	Total weight (kg)	Supply Rate	Laying and Joining Rate	Total	Amount
1	HDPE pipes								
	Distrubtion line:								
	225 mmØ PN-6, PE 100 HDPE pipe	m	397	6.759	2683.32	2255.01	246.17	2501.18	9 92 968.46
	32 mmØ PN-10, PE 100 HDPE pipe	m	1413	0.208	293.90	69.4	5.00	74.40	1 05 127.20
	40 mmØ PN-10, PE 100 HDPE pipe	m	1020	0.324	330.48	108.1	6.06	114.16	1 16 443.20
	50 mmØ PN-10, PE 100 HDPE pipe	m	2524	0.501	1264.52	167.15	6.06	173.21	4 37 182.04
	63 mmØ PN-10, PE 100 HDPE pipe	m	2847	0.801	2280.45	267.24	83.21	350.45	9 97 731.15
	75 mmØ PN-10, PE 100 HDPE pipe	m	2127	1.135	2414.15	378.67	83.21	461.88	9 82 418.76
	90 mmØ PN-10, PE 100 HDPE pipe	m	1392	1.631	2270.35	544.15	83.21	627.36	8 73 285.12
	110 mmØ PN-6, PE 100 HDPE pipe	m	1568	1.621	2541.73	540.81	122.16	662.97	10 39 536.96
	125 mmØ PN-6, PE 100 HDPE pipe	m	1961	2.086	4090.65	695.95	122.16	818.11	16 04 313.71
	140 mmØ PN-6, PE 100 HDPE pipe	m	1159	2.609	3023.83	870.44	166.41	1036.85	12 01 709.15
	160 mmØ PN-6, PE 100 HDPE pipe	m	1836	3.426	6290.14	1143.02	166.41	1309.43	24 04 113.48
	180 mmØ PN-6, PE 100 HDPE pipe	m	761	4.307	3277.63	1436.94	166.41	1603.35	12 20 149.35
	200 mmØ PN-6, PE 100 HDPE pipe	m	4905	5.34	26192.70	1781.58	246.17	2027.75	99 46 113.75
	Subtotal		23910		56953.84	Rs.			219 21 092.33
	Value added Tax					@ 13.00 %			28 49 742.00
	Total cost of HDPE pipes					Rs.			247 70 834.33

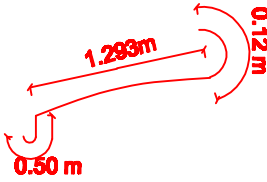
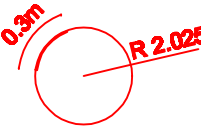
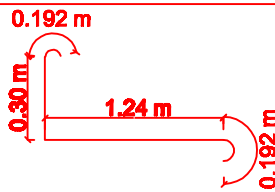
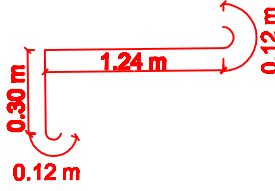
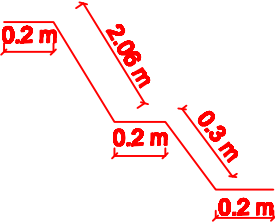
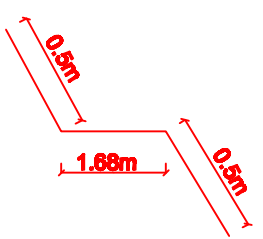
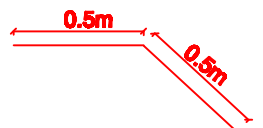
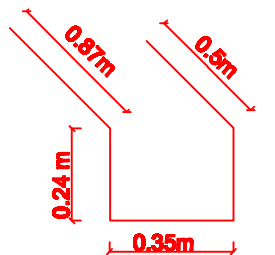
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DETAIL QUANTITY CALCULATION OF OPERATOR/GUARD HOUSE

####

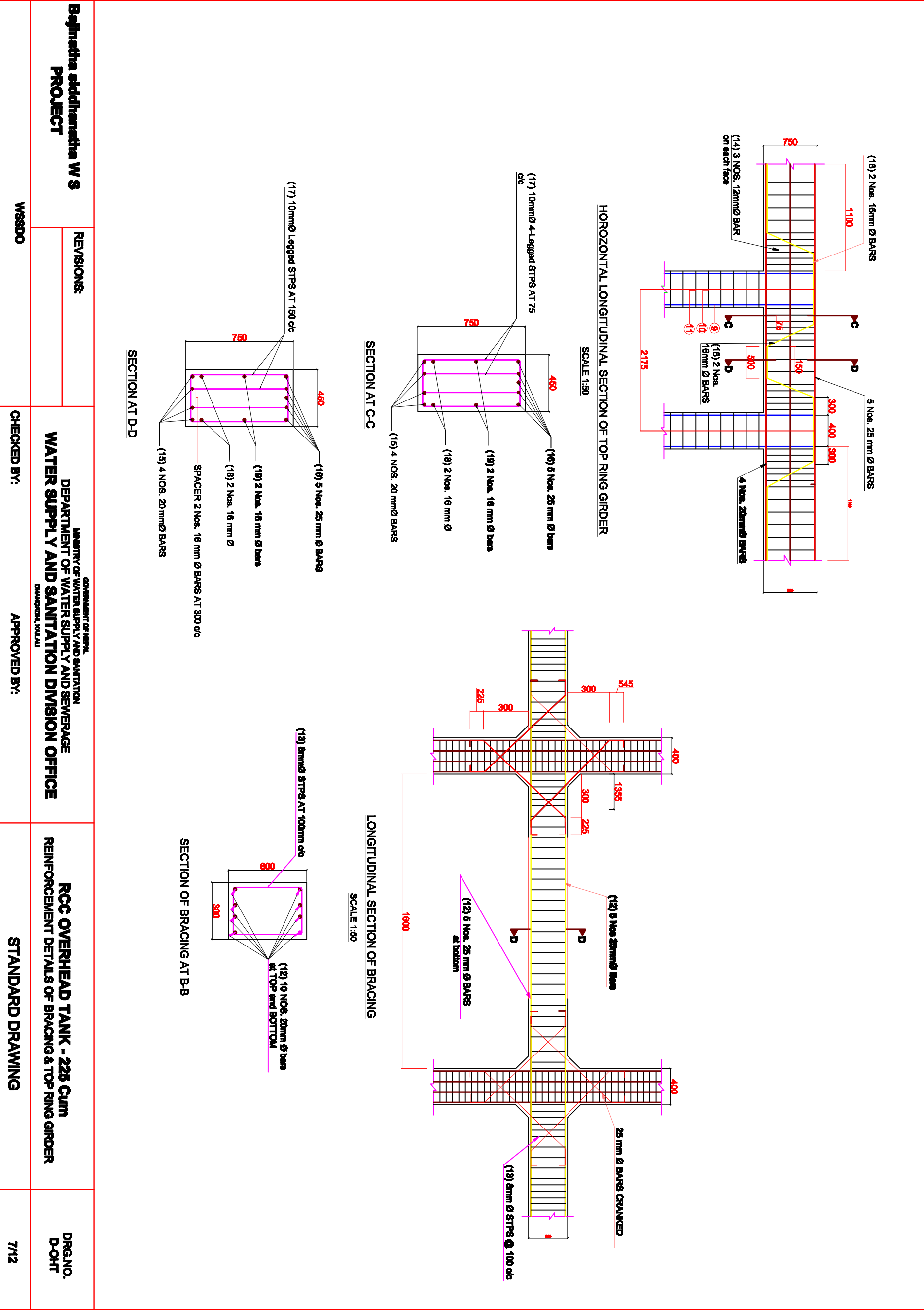
Type of Building :-Office Building/Operator/Guard House										Civil Works
S.No.	Description	No.	Length	Breadth	Height	Quantity	Unit	Rate	Amount	Remarks
1	EARTH WORK									
1.1	Surface dressing and Top soil cutting with 150 to 200 mm depth including disposing the surplus excvated earth within the premises for layout & setting out of building as per drawing & specification.	1	11.00	3.50		38.50				
		1	4.50	1.50		6.75				
					Total	45.25	m ²	138.92	6,286.13	
1.2	Earthwork in excavation in all kinds of soil in foundation trenches, pits, raft, etc. including dressing of sides, ramming of bottom, watering & lift upto 2 meter, stacking of excavated materials at least 4 meter clear from the edge of excavation and backfilling the stacked soil in 150 mm layers where required in plinth under floor, sides of foundation, laying & disposing the layer by properly watering, ramming & then disposing of all surplus excavated soil as directed within 30 mtr. distance from the edge of the building all complete as per drawing, specification and instruction of engineer.									
	Foundation									
	Long Wall	2	11.50	0.75	1.00	17.25				
	Short Wall	4	2.50	0.75	1.00	7.50				
		2	0.75	0.75	1.00	1.13				
	Dwarfwall	1	2.50	0.75	1.00	1.88				
	Step	1	1.50	0.60	0.14	0.13				
		1	0.75	0.60	0.14	0.06				
	Ramp	1	1.75	2.00	0.30	1.05				
					Total	28.99	m ³	626.01	18,147.40	
1.3	Filling with sand brought from outside in 150 mm thick layers under floors foundation tec. Including the cost of watering, ramming, dressing etc. all complete as per drawing, specification & approval of Engineer.									
	Operator Room	1	3.50	3.00	0.10	1.05				
	Toilet	1	2.50	1.08	0.10	0.27				
	Store Room	1	2.50	1.80	0.10	0.45				
	Generator Room	1	4.00	4.50	0.10	1.80				
					Total	3.57	m ³	223.57	797.87	
1.4	Earthwork in filling in floor in perfect line and level with materials from associated excavation including machine compaction in 150 mm layers with sprinkling water required to obtain 90 % density as per drawing, specification and approval of engineer.									
	Operator Room	1	3.50	3.00	0.60	6.30				
	Toilet	1	2.50	1.08	0.60	1.61				
	Store Room	1	2.50	1.80	0.60	2.70				
	Generator Room	1	4.00	4.50	0.60	10.80				
	Staircase Lobby	1	6.50	0.90	0.60	3.51				
		1	2.15	1.49	0.60	1.92				
					Total	26.84	m ³	223.57	6,001.65	

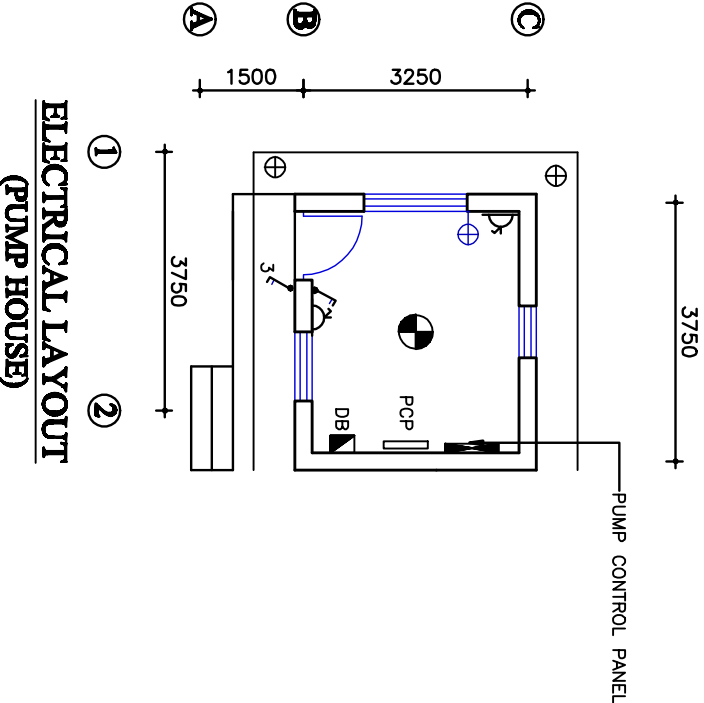
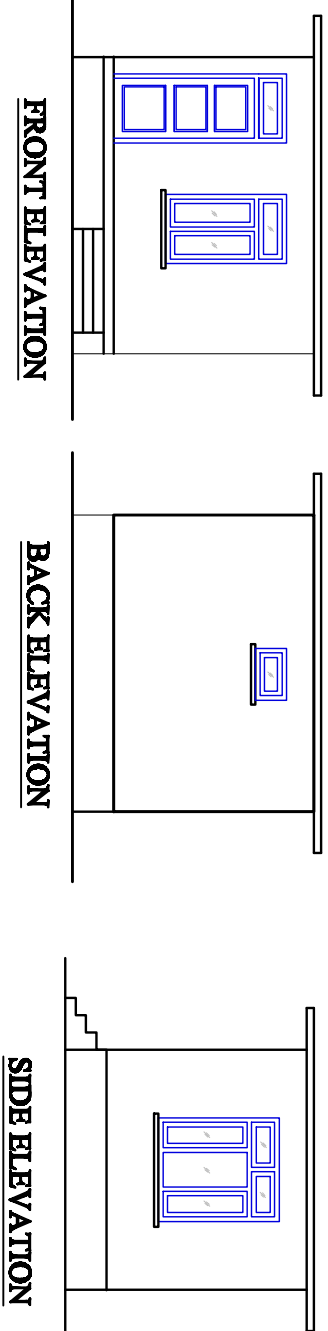
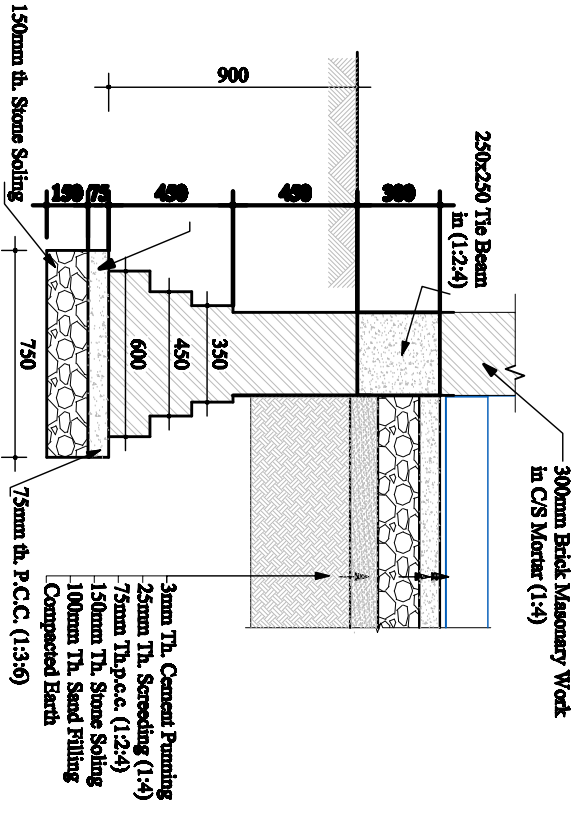
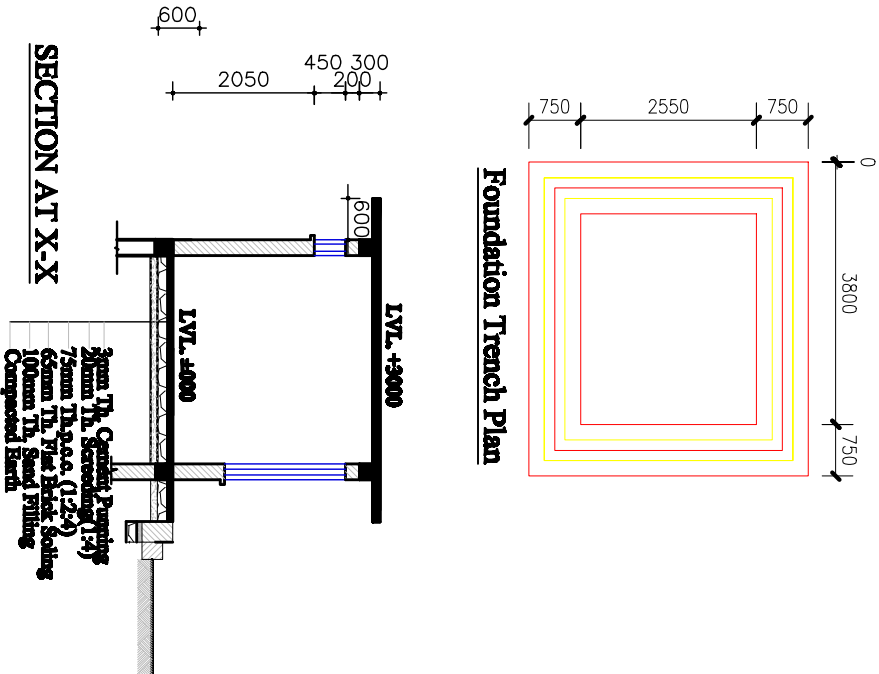
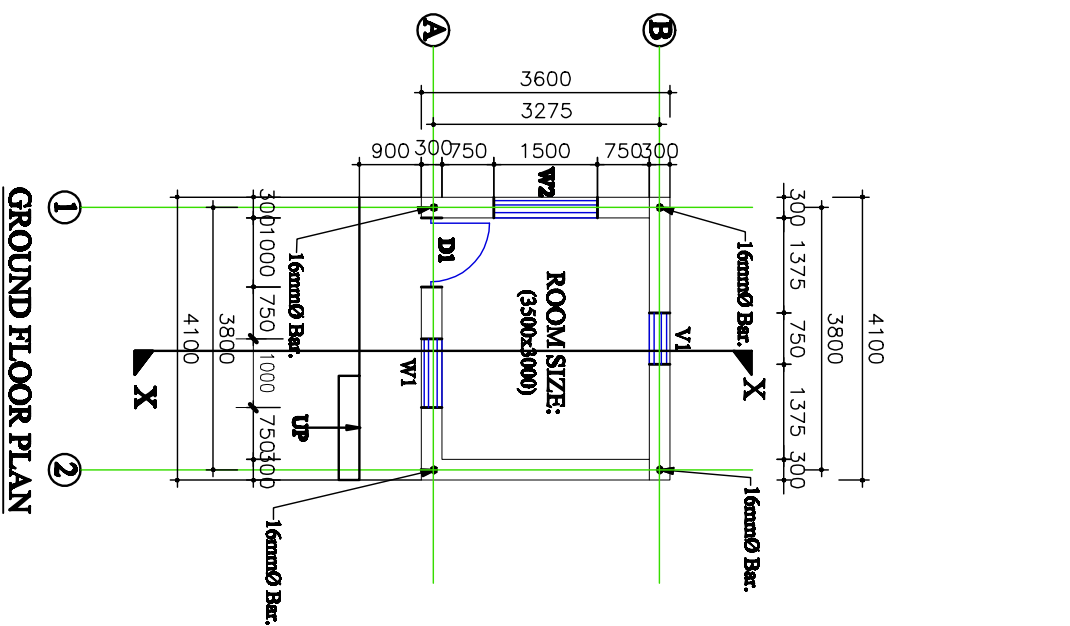
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										DRG.NO. D-OHT		12/12
										RCC OVERHEAD TANK - 225 Cum BAR BENDING SCHEDULE		STANDARD DRAWING
Bar Code	Location of Member	Sketch	Ø (mm)	Number of Bar	No of Member	Weight/m kg	Total Length (m)	Total weight(kg)	Remarks	GOVERNMENT OF KERALA MINISTRY OF WATER SUPPLY AND SANITATION DEPARTMENT OF WATER SUPPLY AND SEWERAGE WATER SUPPLY AND SANITATION DIVISION OFFICE THIRUVANANTHAPURAM, KERALA		
4	Top Dome		10 mm	88	88	0.62	1.91	104.20				
35	Top Dome		10 mm	29	29	0.62	20.35	365.89	0.75m≤R≤2.025m			
36	Manhole		10 mm	4	4	0.62	3.80	9.42				
37	Manhole		8 mm	7	7	0.395	3.70	10.23				
38	Manhole		10 mm	10	10	0.62	2.38	14.75				
39	Manhole		6 mm	18	18	0.222	2.20	8.79				
40	Staircase		16 mm	4 x 11	44	1.578	1.93	134.00				
41	Staircase		10 mm	3 x11	33	0.62	1.93	39.48				
42	Staircase		10 mm	6 x 10	60	0.62	3.27	121.64				
43	Staircase		8 mm	16 x 10	160	0.395	0.72	45.50				
44	Staircase		10 mm	6 *8	48	0.62	2.68	79.76				
45	Staircase		10 mm	6 *8	48	0.62	1.0	29.76				
46	Staircase		12 mm	6	6	0.887	2.20	11.71				
47	Staircase		8 mm	6	6	0.395	0.84	20.00				
NOTES:										REVISIONS:		
										CHECKED BY:		APPROVED BY:

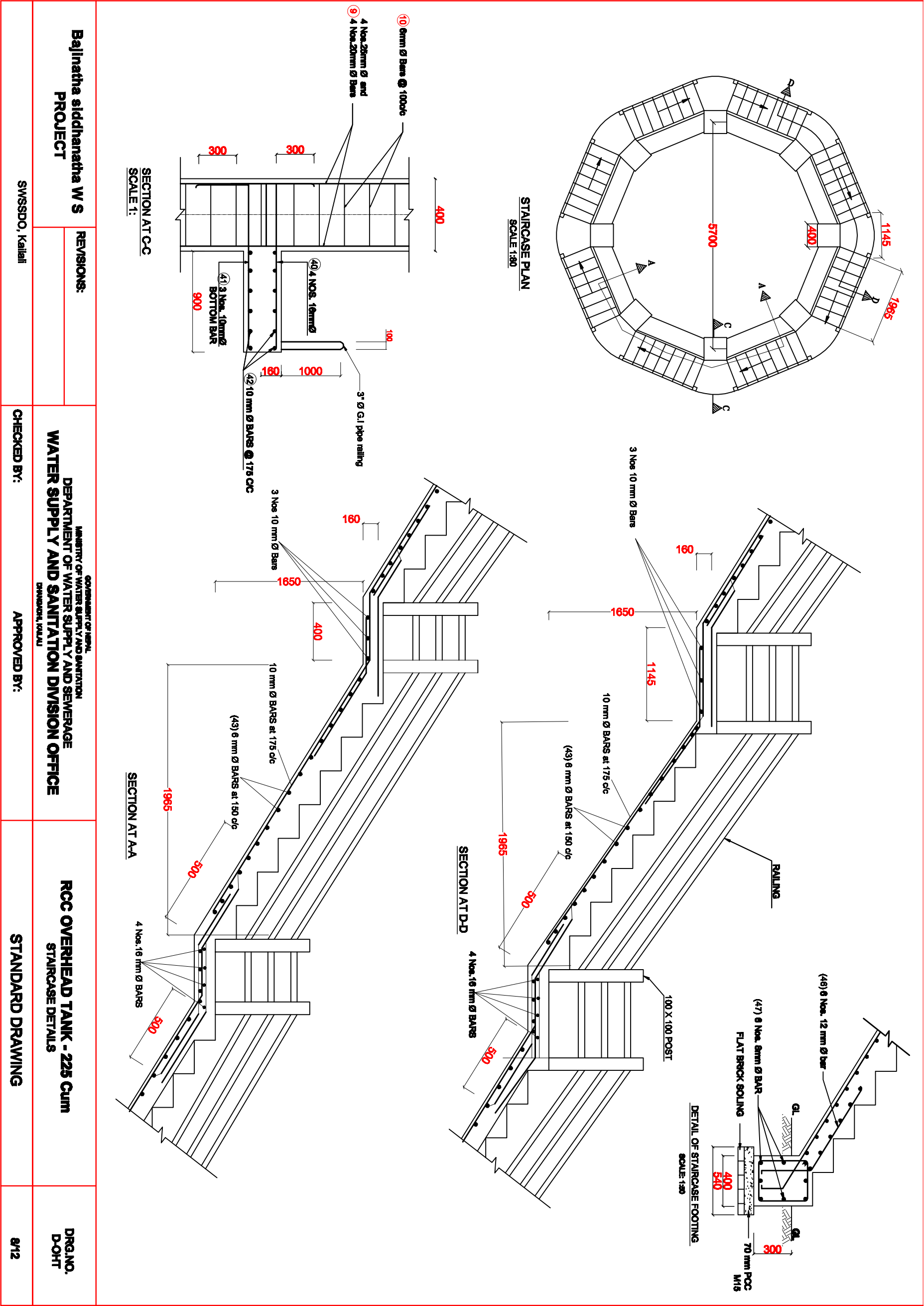
Bar Code	Location of Member	Sketch	Ø (mm)	Number of Bar	No of Member	Weight/m kg	Total Length (m)	Total weight(kg)	Remarks	DRG.NO. D-OHT		1/12
④	Top Dome		10 mm	2 x 28	56	0.62	2.72	94.43		RCC OVERHEAD TANK - 225 Cum BAR BENDING SCHEDULE		
⑤	Top Dome		10 mm	2 x 56	112	0.62	2.02	140.26				
②⑥	Bottom Ring Beam with Balcony		10 mm	194	196	0.62	2.03	246.68				
②⑦	Bottom Ring Beam with Balcony		20 mm	8	8	2.465	31.80	627.09	4.07m≤R≤4.49m			
②⑧	Bottom Ring Beam with Balcony		8 mm	6	6	0.395	31.40	74.41	4.63m≤R≤4.85m			
②⑨	Cylindrical Wall		12 mm	2 x 17	34	0.887	27.70	835.37	4.075m≤R≤4.195m	APPROVED BY:		
③⑩	Cylindrical Wall		10 mm	2 x 6	12	0.62	27.70	206.08	4.075m≤R≤4.195m			
③①	Cylindrical Wall		8 mm	2 x 209	418	0.395	4.7	776.01				
③②	Top Ring Beam		12 mm	6	6	0.887	28.25	150.34	4.075m≤R≤4.275m			
③③	Top Ring Beam		8 mm	189	189	0.395	0.85	63.45				
③④	Top Dome		10 mm	22	22	0.62	5.85	79.79		REVISIONS:		
②	Top Dome		10 mm	22	22	0.62	4.16	56.74				
③	Top Dome		10 mm	44	44	0.62	3.46	94.38		NOTES:		*****

Note:Official project drawing included for study and academic reference during internship. © Water Supply and Sanitation Division Office (WSSDO), Kailali.





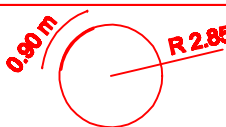
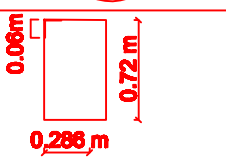
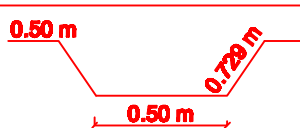
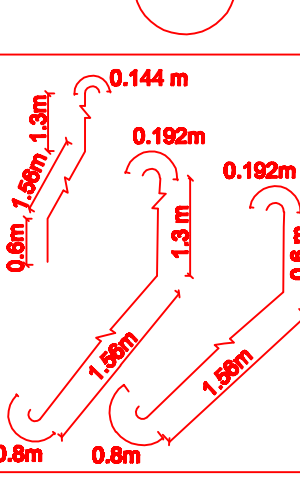
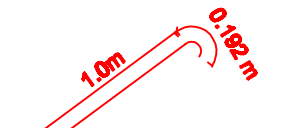
Submitted by: Water Supply and Sanitation Division Office. Dhangadhi, Kailali		PROJECT NAME : Bajinatha Siddhanathaw S Project		Scale: 1:150		Surveyed by : Designed by : Recommended by : Approved by :		PUMP HOUSE (PLAN, ELEVATION, SECTION) FISCAL YEAR 2078/079	
Submitted to: Province Government Suburbanization Province Ministry of Physical Infrastructure Development Water Supply and Sanitation Division Office Dhangadhi, Kailali									



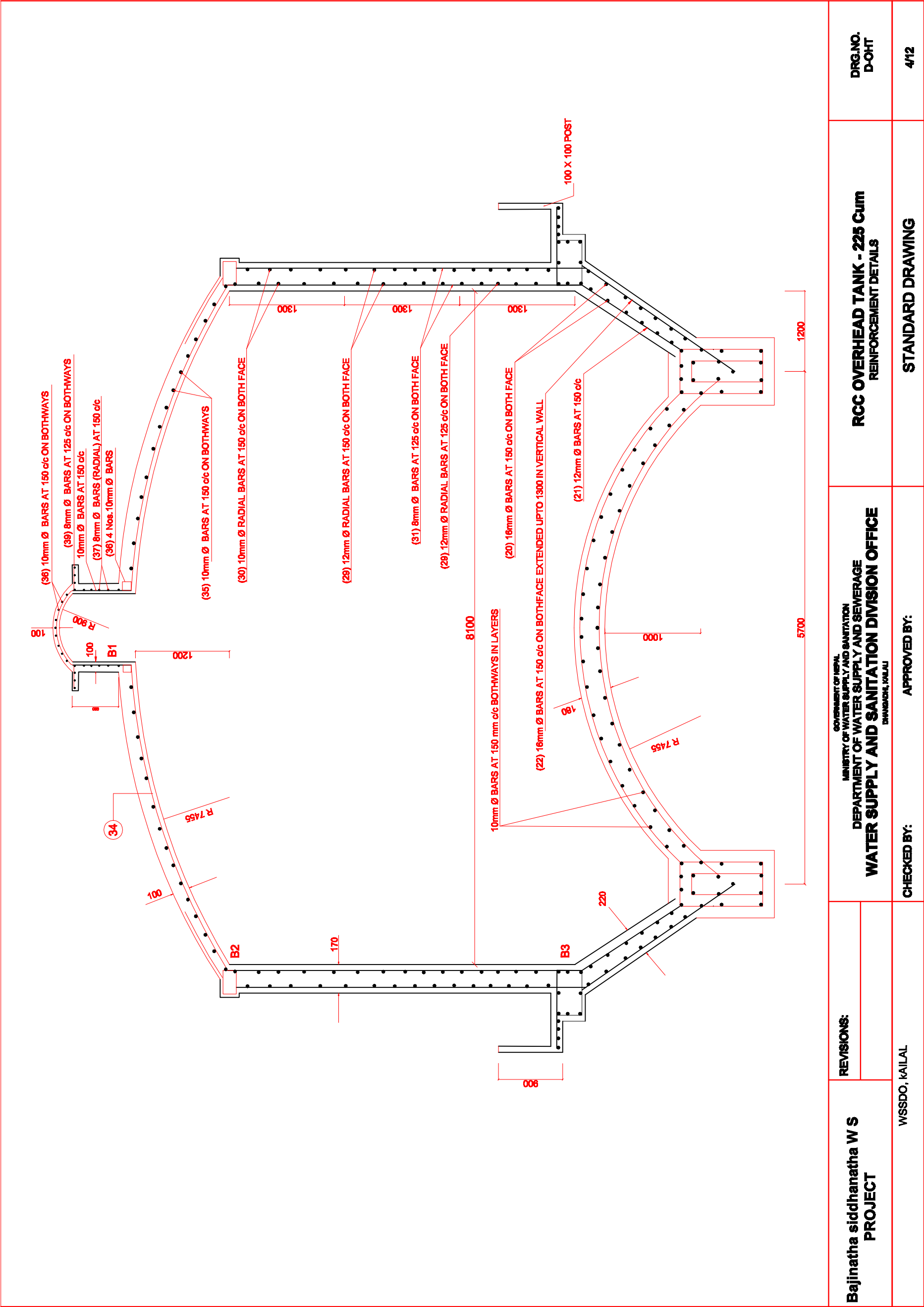
BAR BENDING SCHEDULE -225 CUM TANK

Bar Code	Location of Member	Sketch	Ø (Bar of Bar Per Size)	Number of Bar Per Member	Total No of Bar	Weight/m kg	Total Length (m)	Total Weight (kg)	Remarks	DRG.NO. D-OHT	9/12
①	Foundation		25 mm	50	50	3.854	4.90	944.23		RCC OVERHEAD TANK - 225 Cum BAR BENDING SCHEDULE	STANDARD DRAWING
②	Foundation		25 mm	50	50	3.854	4.10	790.07			
③	Foundation		25 mm	100	100	3.854	3.00	1156.20			
④	Foundation		12 mm	24	24	0.887	21.50	457.69	0.94m≤R≤4.76m Mean radius has been taken	MINISTRY OF WATER SUPPLY AND SANITATION DEPARTMENT OF WATER SUPPLY AND SEWERAGE WATER SUPPLY AND SANITATION DIVISION OFFICE DHANRAJ, KAILALI	APPROVED BY:
⑤	Foundation Beam		25 mm	8	8	3.854	22.00	678.30	2.66m≤R≤3.03m		
⑥	Foundation Beam		16 mm	2	2	1.578	21.60	68.16	2.66m≤R≤3.03m		
⑦	Foundation Beam		8 mm	240	240	0.395	1.15	109.02		REVISIONS:	CHECKED BY:
⑧	Foundation Beam		12 mm	277	277	0.887	2.35	577.39	4 Legged Stirrups		
⑨	Column		25 mm 20 mm	(4) *(8) (4) *(8)	32 32	3.854 2.465	27.85 27.85	3434.68 2196.80			
⑩	Column Stirrups		6 mm	(165) *(8)	1320	0.222	1.50	439.56		NOTES:	*****
⑪	Column Stirrups		6 mm	(165) *(8)	1320	0.222	1.10	322.34			
⑫	Bracing		20 mm	(10)*(4)	40	2.465	21.40	2110.04			
⑬	Bracing Stirrups		8 mm	(4)*(8) *(19)	608	0.395	1.76	422.68		NOTES:	*****
⑭	Bracing Crank		25 mm	(4)*(4) *(8)	128	3.584	2.13	1050.75			
⑮	Bottom Ring Girder		20 mm	4	4	2.465	21.60	212.97	2.67m≤R≤3.02m		

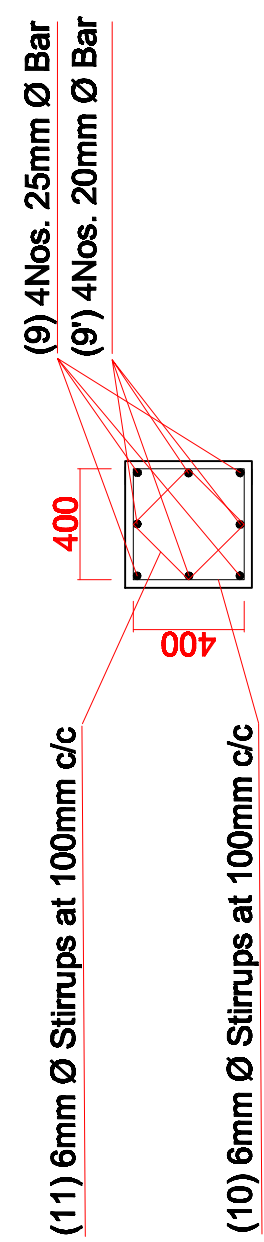
NOTE:Official project drawing studied during internship at WSSDO, Kailali. Included for academic reference only. Not prepared by the author. © WSSDO office, Kailali.

Bar Code	Location of Member	Sketch	Ø (mm)	Number of Bar	No of Member	Weight/m kg	Total Length (m)	Total Weight (kg)	Remarks	DRG.NO. D-OHT	10/12
16	Bottom Ring Girder		25 mm	5	5	3.584	21.60	416.23	2.67m≤R≤3.02m		
17	Bottom Ring Girder		10 mm	20 x 8 20 x 8	160 160	0.62	2.14 1.85	212.28 183.52			
18	Bottom Ring Girder		16 mm	2 x 8	16	1.578	3.33	84.07		RCC OVERHEAD TANK - 225 Cum BAR BENDING SCHEDULE	STANDARD DRAWING
19	Bottom Ring Girder		16 mm	6	6	1.578	21.60	204.50			
20	Conical Slab		16 mm	2 x 13	26	1.578	26.20	1074.93	3.378m≤R≤3.522m		
21	Conical Slab		12 mm	170	170	0.887	3.60	542.84		GOVERNMENT OF NEPAL MINISTRY OF WATER SUPPLY AND SEWERAGE DEPARTMENT OF WATER SUPPLY AND SEWERAGE WATER SUPPLY AND SANITATION DIVISION OFFICE DANGAUCHI, JULAI	APPROVED BY:
22	Conical Slab		16 mm	85	85	1.578	3.85	516.40			
			16 mm	85	85	1.578	3.16	423.85			
23	Bottom Dome		16 mm	164	164	1.578	1.99	514.99		GOVERNMENT OF NEPAL MINISTRY OF WATER SUPPLY AND SEWERAGE DEPARTMENT OF WATER SUPPLY AND SEWERAGE WATER SUPPLY AND SANITATION DIVISION OFFICE DANGAUCHI, JULAI	APPROVED BY:
24	Bottom Dome		10 mm	21*2	42	0.62	10.30	268.21	0.07m≤R≤2.62m		
25	Top Dome		10 mm	2 x 7	14	0.62	3.56	30.90			
1	Top Dome		10 mm	2 x 7	14	0.62	3.56	30.90		REVISIONS:	
2	Top Dome		10 mm	2 x 7	14	0.62	3.42	29.68			
3	Top Dome		10 mm	2 x 14	28	0.62	3.14	54.51			

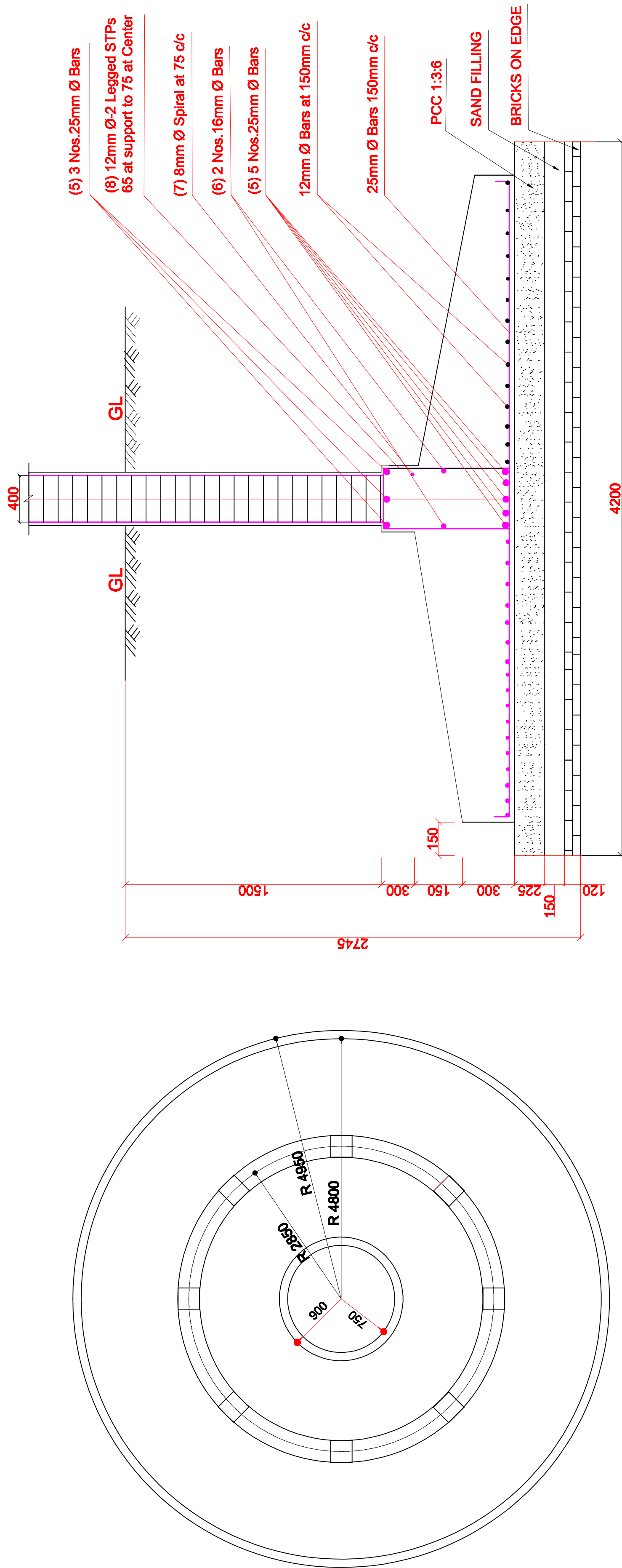
Official project drawing included for study and academic reference during internship under the guidance of engineer. © Water Supply and Sanitation Division Office (WSSDO), Kailali.



Note:Official project drawing included for study and academic reference during internship. © Water Supply and Sanitation Division Office (WSSDO), Kailali.



DETAIL OF COLUMN



FOUNDATION PLAN
scale 1:200

SECTION A - A

Bajinatha siddhanatha W S PROJECT	REVISIONS:	GOVERNMENT OF NEPAL MINISTRY OF WATER SUPPLY AND SANITATION DEPARTMENT OF WATER SUPPLY AND SEWERAGE WATER SUPPLY AND SANITATION DIVISION OFFICE DHANGAUHI, KULAU	RCC OVERHEAD TANK - 225 Cum FOUNDATION DETAILS	DRG.NO. D-OHT
WSSDO		CHECKED BY:	STANDARD DRAWING	5/12

*Note: Official project drawing included for study and academic reference during internship. © Water Supply and Sanitation Division Office (WSSDO), Kailali.